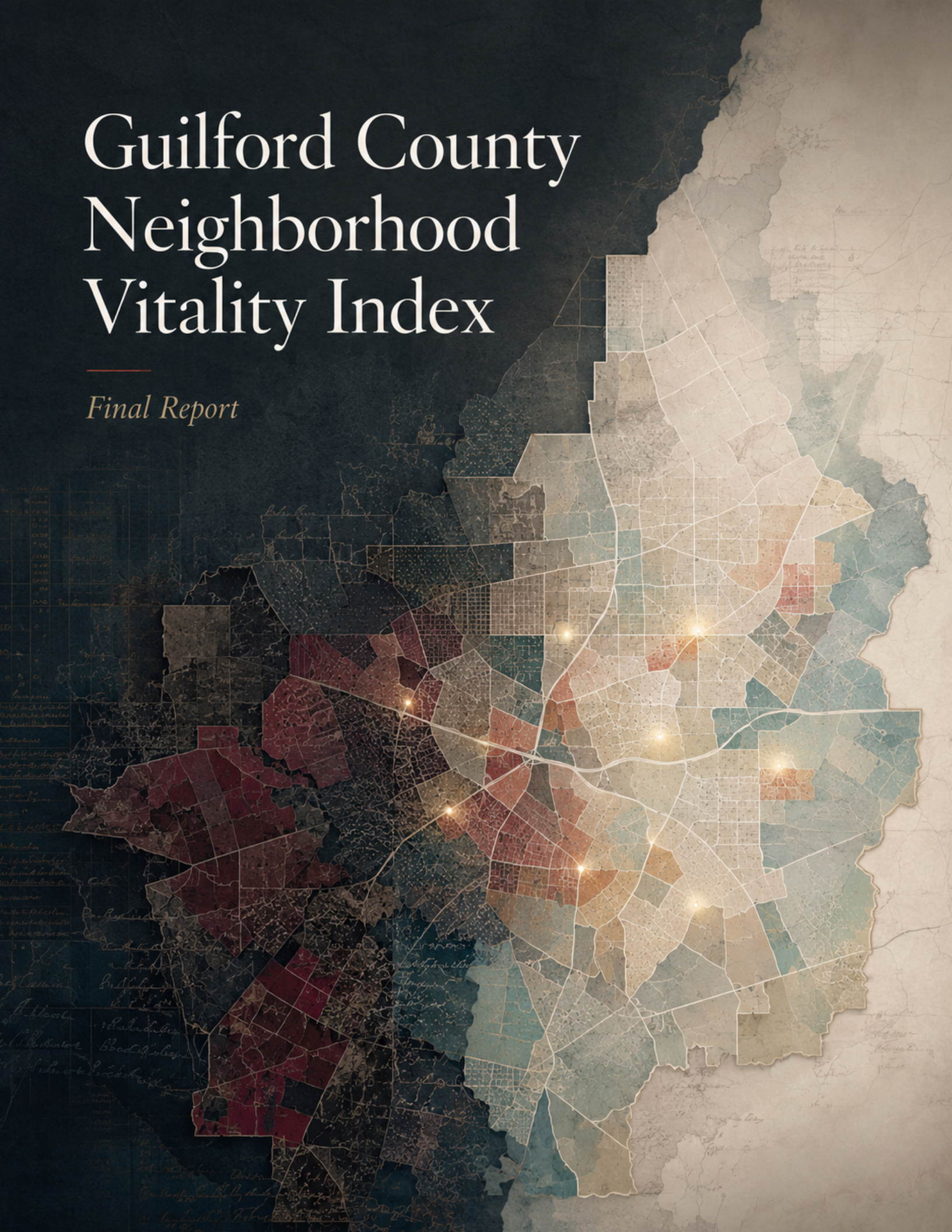


Guilford County Neighborhood Vitality Index

Final Report



GUILFORD COUNTY NEIGHBORHOOD VITALITY INDEX

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1. INTRODUCTION

1.1 PURPOSE AND COMMISSION

This report delivers the *Guilford County Neighborhood Vitality Index*: a block-group-level measure of neighborhood conditions for all 338 block groups of Guilford County, North Carolina, built from 220 verified variables, distilled by criterion validation to ten transparent components, and classified into five categories of neighborhood vitality. The index was commissioned within the Guilford County Community Data Atlas initiative, under the joint sponsorship of Jeff Thigpen, Guilford County Register of Deeds, and Shaunne Thomas, Director of the Guilford County Small Business Entrepreneurship Department, to serve a purpose both sponsors have framed in the same terms: to make the county’s “invisible economy,” the neighborhoods, households, and enterprises that a century of disinvestment has hidden from conventional measurement, visible enough to invest in.

The commission carried three intended uses, and the report is organized to serve them. As an **instrument of geographic decision support**, the index identifies, at block-group resolution and on documented thresholds, where measured neighborhood conditions may warrant concentrated attention. Protected characteristics, including race and ethnicity, are not used to calculate scores or classify block groups. As an **instrument of diagnosis**, its ten-component architecture tells program designers which conditions produce each neighborhood’s classification, so that the response can match the mechanism. And as an **instrument of accountability**, its fully replicable pipeline and annual refresh convert this report from a snapshot into a baseline against which the county’s coming investments can be honestly judged. A companion volume, *An Atlas of Neighborhood Conditions in Guilford County* (Innovative Research Insights, 2026), publishes the complete 213-plate map series underlying this analysis, and the index itself is destined for the county’s public ArcGIS platform under the governance arrangements of Section 8.3.

1.2 THE NINE PILOT TRACTS: A PROOF OF CONCEPT

The work is anchored by nine census tracts selected in advance, as a proof-of-concept geography, by the Coalition of Anchors, the partnership of public, institutional, and community organizations guiding place-based investment in the county: tracts 112.01, 112.02, 113, 114, and 115 in east and central Greensboro, and tracts 138, 139, 142, and 143 in south-central High Point, the latter encompassing the Washington Street corridor and its surrounding communities. These are not arbitrary study areas. Their selection does not itself establish eligibility for any public benefit or confer programmatic priority; those decisions remain subject to lawful, program-specific criteria (Section 3.6). They are the geography of the county’s historically Black neighborhoods, platted under segregation, graded hazardous by the federal Home Owner’s Loan Corporation security maps (aka “red lining maps”), designated for clearance by mid-twentieth-century renewal plans, and severed by the highway construction that followed, a history Section 2 documents from the county’s own archival record. Roughly 34,000

residents live in the nine tracts today, at poverty rates between 28 and 50 percent against the county's 17.

Methodologically, the nine tracts serve this report as a pre-specified proof-of-concept validation geography. They were selected before the index existed, on historical and community knowledge; if the index measures what that community knowledge describes, they should emerge at the bottom of the county's distribution. Section 6.5 reports the test in full: all nine classify Very Low or Low Vitality, and the corridor's tract 139 records the lowest score in the county. The result supports their use as a proof-of-concept validation geography; it does not convert the pilot selection into an eligibility rule. The same methodology applies to all 338 block groups and can be applied to any neighborhood in Guilford County.

Table 1 The nine pilot tracts in brief.

Tract	Population	Poverty rate	GCNVI category (score)
Greensboro 112.01	3,296	44%	Very Low (-1.27)
Greensboro 112.02	3,363	50%	Low (-0.98)
Greensboro 113	3,541	28%	Low (-1.07)
Greensboro 114	5,359	43%	Very Low (-1.47)
Greensboro 115	3,276	50%	Low (-0.90)
High Point 138	4,269	38%	Low (-1.26)
High Point 139	4,187	49%	Very Low (-1.65)
High Point 142	4,550	31%	Low (-1.08)
High Point 143	2,504	28%	Low (-0.99)

1.3 WHAT THE INDEX IS AND IS NOT

One sentence governs every use of this instrument: the index measures the conditions of places, never the worth of people. Its ten components are conditions a county can change, home values and evictions, lending and food access, birth outcomes and civic connection, and its demographic-neutrality safeguards (Section 3.6) enforce the distinction by design: demographic composition is excluded from scoring entirely, and protected characteristics are not used to determine classification, eligibility, or allocation; modeled data are capped and labeled, producer suppression is honored rather than overridden, and the safety component is defined, and must be read, as residents' exposure to harm rather than their propensity for it. The index is county-relative: it describes each block group's position within Guilford County in the mid-2020s and supports no comparison beyond the county line, a role the national benchmark indices retained in the file continue to serve. And it is a measure of association with opportunity, validated against what became of the children each neighborhood raised, not a set of causal guarantees about any single intervention; Section 7 states the boundary between what the measurement can inform and what lawful, program-specific decisions must separately determine.

1.4 ROAD MAP OF THE REPORT

Section 2 grounds the numbers in history and literature: the archival record of how the county’s geography of disadvantage was produced, the theoretical tradition from Du Bois to the Opportunity Atlas that explains its persistence, and the national index lineage against which this instrument’s design is a considered position. Section 3 details the methodology: the two-phase design, the criterion-validation framework, the verification discipline, the screening and selection protocol, the demographic-neutrality safeguards, and the limitations. Section 4 documents the data: five source families, the geography and vintage protocol, the acquisitions of note, and the registered gaps. Section 5 presents the ten components individually, each with its selection evidence and its county and pilot-area maps. Section 6 assembles and validates the composite: distribution, psychometrics, criterion and benchmark validation, the geography of the five categories, the pilot-tract proof-of-concept test, and the block-group evidence for reading beneath the score. Section 7 shows how measurement can inform place-based decision support, including comparison of the index with the county’s existing investment geographies, while program-specific eligibility and allocation remain subject to governing law and responsible agency authority, and Section 8 states the recommendations: eight recommended place-based actions, their sequencing, the governance of the instrument, and the measurement agenda. Section 9 concludes. The appendices carry the complete variable catalog, codebook, statistical results, model documentation, scored dataset, map suite, provenance record, verification summary, the employment-criterion methods note, and the glossary.

2. BACKGROUND AND LITERATURE

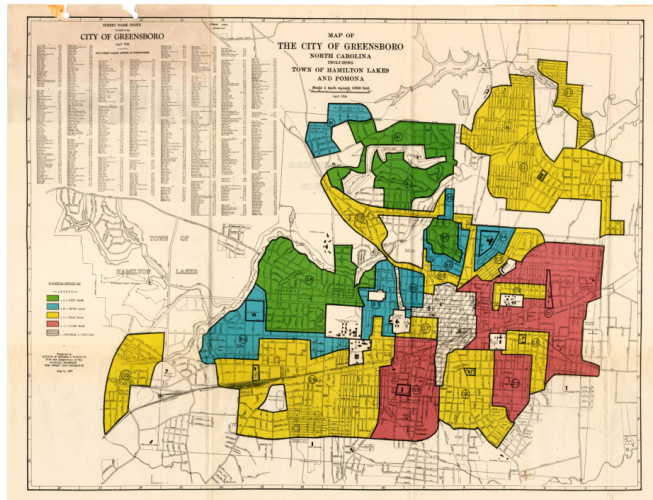
2.1 THE STRUCTURAL ROOTS OF CONCENTRATED DISADVANTAGE

The conditions this report measures were produced, not found, and the production is documented. In both of Guilford County’s principal cities, the neighborhoods that anchor the index’s lowest categories were built by and for Black residents under segregation, systematically devalued by public credit policy, and then physically dismantled by public works. The Guilford County Neighborhood Vitality Index’s maps document the present-day pattern those transactions left behind, read here alongside the historical record that produced it.

Greensboro’s pattern was a dual market by design. As the city grew from railroad village to the state’s third-largest city, its Black neighborhoods grew with it as planned, segregated developments: Warnersville, platted in 1867 when the Quaker agent Yardley Warner subdivided thirty-five acres south of downtown into lots sold to Black families expressly so they could build equity; Jonesboro, near the present campus of North Carolina A&T; and later Scott Park, College Heights, and Clinton Hills (Baylin, 1968; Preservation Greensboro). Even the city’s premier Black neighborhood obeyed the logic of containment: Nocho Park, platted in 1923 as an exclusive development for middle- and upper-income Black residents, complete with planned park, hospital, and school, was created by white developers operating within, never outside, the segregated market. The city had enacted an explicit racial zoning ordinance in 1914; when the Supreme Court struck such ordinances down in 1917, restrictive covenants and exclusionary zoning carried the boundary forward.

Public policy then converted containment into extraction. The 1937 Home Owners’ Loan Corporation residential security map of Greensboro shaded Nocho Park, Clinton Hills, and the surrounding Black neighborhoods yellow and red, “definitely declining” and “hazardous,” on the explicit basis of racial composition, and the denial of mortgage credit that followed left East Greensboro’s land values defenseless long before the bulldozers arrived (Rothstein, 2017; Sills, 2021). Urban renewal completed the work: Warnersville and Jonesboro, neighborhoods of Black homeowners, were cut through and cleared for U.S. 29, public housing, and allied renewal projects, part of a national program that demolished more than 425,000 housing units in over 2,100 projects whose displaced residents were 80 percent Black. Figure 1 places the 1937 security map beside the 2026 vitality classification for central Greensboro; the two maps are separated by ninety years and agree in outline, which is the report’s historical thesis stated cartographically (Sills, 2021). The reader can trace the correspondence district by district: the D-graded red of the map’s east side (the “hazardous” designation laid over Nocho Park, Scott Park, and College Heights) and the D3 grade south of the rail junction sit under the Very Low and Low block groups of panel b, while the A-graded green of Sunset Hills and Irving Park anchors today’s High and Very High arc.

a. HOLC residential security map, surveyed 1936, issued 1937



b. GCNVI vitality classification, 2026

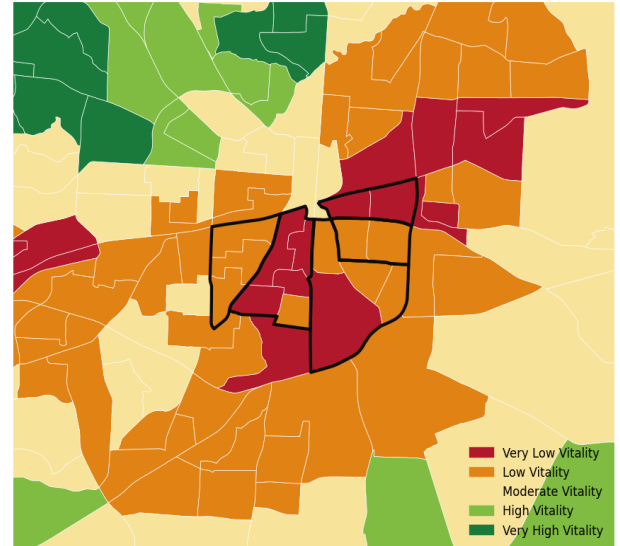


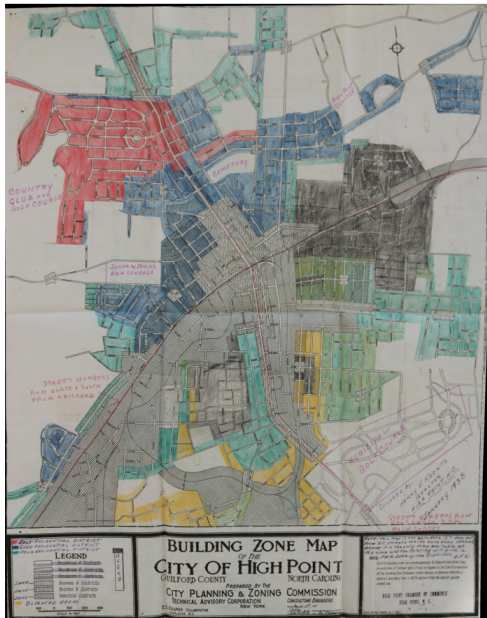
Figure 1 Greensboro 1937 HOLC “Redline” Map vs. GCNVI Vitality Scores 2026

High Point ran the same sequence with a furniture-town accent, and its case is documented in this project’s own prior work, *From Disparity to Parity* (Sills, 2023). Founded in 1859 at the highest point of the North Carolina Railroad survey, the city concentrated furniture and textile manufacturing, and the Black workforce that ran it, in its core; its first city plan followed in 1928, two years after *Euclid v. Ambler* legalized zoning. The East Central area, which corresponds to today’s census tract 139, the most distressed tract in this report’s entire analysis, was repeatedly targeted for redevelopment in part on the basis of its racial composition: a 1962 urban renewal plan classified 77 percent of the area’s structures, 1,061 buildings, as deficient and proposed acquisition and clearance, and a 1986 East Central Small Area Plan renewed the ambition. What the plans did not deliver was investment; between 1985 and 1987 the city committed only \$331,179 in Community Development Block Grant funds to the area, a fraction of the rehabilitation the plans themselves envisioned.

The paper trail begins even earlier than the renewal plans, and Figure 2 reproduces it: the city’s own 1935 Building Zone Map, prepared by its Planning and Zoning Commission with New York consulting engineers and hand-colored by a local realtor, carries in its printed legend, alongside “Best,” “Good,” and “Poor Residential” districts and “Blighted Areas,” an explicit district designation reading “NEGRO.” Race was not an unfortunate correlate of High Point’s zoning; it was a category of it, printed in the legend. The black-hatched districts that legend entry marks, east and south of the rail junction, are today’s tracts 138, 139, 142, and 143, and panel b of the figure shows what ninety years of that designation produced: the same geography, rendered now in the index’s Very Low and Low categories. The Washington Street corridor, once known as High Point’s Black Wall Street, carries the memory in the words of its own leaders: “That started with urban renewal back in the 60s, when they decided to run the [major roadway] on through a black community; that was the downfall of our community” (community leader, Washington Street Needs Assessment, Innovative Research Insights, 2025). Six

decades later, tract 139's median household income sits at roughly a quarter of the county's and prime-age labor-force participation has fallen below one third (Section 6.6). The history is not merely background to that condition; it documents the public policies and investment patterns most strongly associated with the geography observed today. The GCNVI itself measures present conditions and does not establish that any one of those policies is the proximate cause of any individual contemporary outcome.

a. Building Zone Map of the City of High Point, 1935



b. GCNVI vitality classification, 2026

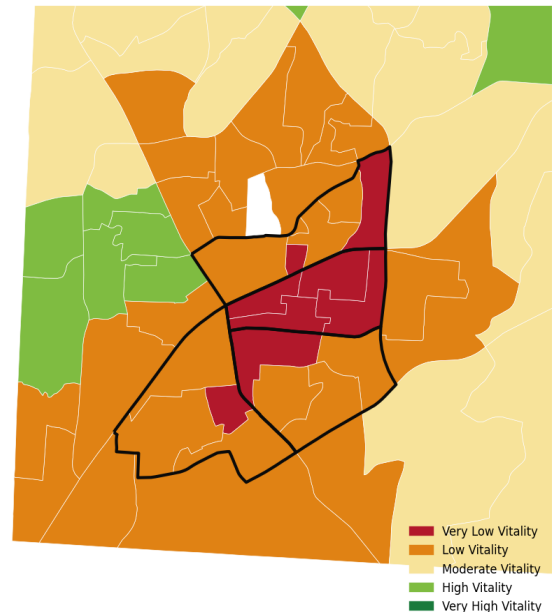


Figure 2 High Point 1935 Zoning Map vs. GCNVI Vitality Scores 2026

2.2 THEORETICAL FRAMEWORK

The intellectual tradition this report works in begins where its method does: with W. E. B. Du Bois, whose block-by-block survey of Philadelphia's Seventh Ward (Du Bois, 1899) founded the practice of mapping the conditions imposed on Black neighborhoods while refusing to mistake those conditions for the character of residents. Every map in this report and its companion Atlas is a continuation of that act.

The century that followed supplied the mechanism, piece by piece. Wilson (1987) showed that spatially concentrated poverty is self-reinforcing: when employment, institutions, and economically stable households are withdrawn from a neighborhood together, the isolation that remains erodes the networks and norms through which striving proceeds. Massey and Denton (1993) identified enforced segregation as the machine that does the concentrating, the institutional arrangement that translates any economic downturn into geographically compounded disadvantage for Black communities specifically. Sampson (2012), from the Chicago project that operationalized neighborhood measurement, demonstrated that neighborhoods possess durable social properties, collective efficacy above all, that shape safety, health, and civic life independently of the individuals who move through them; the social-support and self-response components of the present index are this insight

rendered measurable. Sharkey (2013) established the intergenerational clause: neighborhood disadvantage is inherited, with families “stuck in place” across generations such that a neighborhood’s past ranks among the strongest predictors of its children’s futures. Rothstein (2017) assembled the legal history proving the arrangement was governmental at every level, *de jure* rather than *de facto*. And Chetty and colleagues’ Opportunity Atlas converted Sharkey’s clause into census-scale measurement, reporting, for every neighborhood in America, what actually became of the children raised there; this project adopts those measurements as its validation criteria, so that the index is tested against independent long-run outcome measures consistent with that theoretical tradition. The mechanisms, finally, remain active: Raymond and colleagues (2021) document investor acquisition of single-family housing as the contemporary instrument by which value is extracted from the same neighborhoods the earlier instruments emptied, a channel the index watches through its eviction, ownership, and absentee measures. These theories supply interpretive context; they do not determine the GCNVI score, which is produced by the empirical procedures of Section 3 and excludes protected demographic characteristics from composite construction.

2.3 THE AREA-INDEX LINEAGE

The GCNVI also stands in a measurement tradition, and its design is a considered response to that tradition’s strengths and gaps (the full review is the project’s Neighborhood Indicators and Indices Literature Review, Sills, 2026). The modern line begins with Townsend’s four-variable deprivation score (unemployment, non-car ownership, non-home ownership, overcrowding, standardized and summed unweighted), whose economy and transparency the GCNVI deliberately echoes at ten components; Carstairs and Morris extended the form, and American public health imported it.

The American ecosystem the county could adopt off the shelf comprises a half dozen instruments, each excellent for its purpose and each misfit, in a specific way, to county-scale place-based decision support. The Area Deprivation Index (17 factor-weighted census variables) offers national block-group rankings whose weights and scaling have drawn sustained methodological critique, and whose national norming compresses local variation. The Social Deprivation Index and CDC’s Social Vulnerability Index are built partly from demographic composition itself, minority status among the inputs, which makes them accurate descriptions and inappropriate scoring instruments for this project, whose demographic-neutrality rule excludes protected characteristics from the calculation of neighborhood vitality. The Distressed Communities Index operates at the ZIP scale that business data favor and neighborhoods do not respect. The Child Opportunity Index 3.0 is the field’s most sophisticated child-focused composite, and its 44 indicators include enough modeled and survey inputs that this project uses it as a criterion rather than a component source.

The Home Owners Loan Corporation (HOLC, “Redlining”) map historical literature, finally, has matured into quantitative evidence that the 1930s grades caused durable reductions in ownership, value, and mobility (Aaronson, Hartley, & Mazumder, 2021; Faber, 2020; Nardone et al., 2020; with the cautions of Fishback et al., 2022, and Michney & Winling, 2020, duly noted), and that roughly three quarters of D-graded neighborhoods remain low-to-moderate income today. Table 2.1 summarizes the comparison. What none of the national instruments provides, and what the GCNVI was commissioned to provide, is

the combination this county's work requires: block-group resolution on a current local frame, components a county government can actually move, validation against independent long-run outcomes rather than construct assertion, and measurement safeguards that exclude protected demographic characteristics from scoring and measure conditions of place rather than people.

Table 2 The GCNVI against the national area indices in county use.

Instrument	Inputs	Geography	Method	Demographics in score	Validation	Local actionability
ADI (2022/2024)	17 ACS variables	Block group, national norm	Factor-weighted	Indirect	Construct	Low: survey composites
SDI (2019)	7 ACS variables	Tract	Factor-weighted	Yes (direct)	Construct	Low
SVI (2022)	16 ACS variables, 4 themes	Tract	Percentile rank sum	Yes (theme 3)	Construct	Low
DCI (2019-2023)	7 indicators	ZIP	Rank average	No	Construct	Moderate
COI 3.0 (2024)	44 indicators	Tract, national norm	Weighted composite	No	Outcome-informed	Low: many modeled inputs
GCNVI (2026)	10 components, 8 administrative or measured	Block group, county frame	Equal-weight aligned z	Excluded by rule	Criterion-validated (r = .62 to .87)	High: movable local measures

2.4 THE LOCAL EVIDENCE BASE

The index arrives, finally, into a county that has already documented itself. Across a decade of studies conducted by this firm and its principal in High Point alone, the 2015 Market Segmentation Study, the 2016 Health Report Card, the 2017 Behavioral Health Study, the 2019 Resilience High Point Needs Assessment, the 2020 Residential Drinking Water Quality Study, the 2022 Assessment of Social Health Drivers, the 2022 HealthySteps Needs Assessment, the 2023 Guilford County Workforce Needs Assessment, and the 2025 Washington Street Needs Assessment, a recurring pattern appears: concentrated poverty and persistent, geographically concentrated differences in health, housing, transportation, education, and economic opportunity. Those studies interpret many of those conditions in light of documented racial segregation, discriminatory housing and credit practices, urban renewal, infrastructure decisions, and uneven public and private investment, and several document disproportionate burdens on Black and other non-white residents. Parallel work in Greensboro and countywide (the racialized-planning policy analysis, the 2025 county food-security study, the criminal-justice reentry research) completes the qualitative record on the Greensboro side of the frame.

That record matters to this report in two ways. Methodologically, it supplies the triangulation that the measurement literature counsels: resident accounts, archival records, prior mixed-methods fieldwork, administrative data, and the GCNVI are treated as distinct evidence streams. The prior studies provide context and independent evidence; they do not enter the GCNVI score. Where independently generated quantitative results converge with prior research and resident knowledge, as they do throughout this report, that convergence strengthens confidence that the index is detecting genuine place-based conditions rather than demographic composition.

Substantively, the prior studies establish that the county's problem has never been the identification of need. Many of the neighborhoods that prior studies identified as experiencing the most severe conditions appear at the deprived extreme of every credible national index, on the HOLC maps of the 1930s, and in the testimony of their own residents; they now appear, with block-group precision and criterion validation, in the pages of this report. What has been missing is a countywide, block-group-level, transparent, and reproducible instrument for measuring present conditions under uniform rules. The historical and qualitative record helps explain the patterns; the GCNVI measures them. Neither historical nor current racial composition determines the score, and any later decision about services, funding, contracts, or investment remains subject to lawful, program-specific authority and criteria.

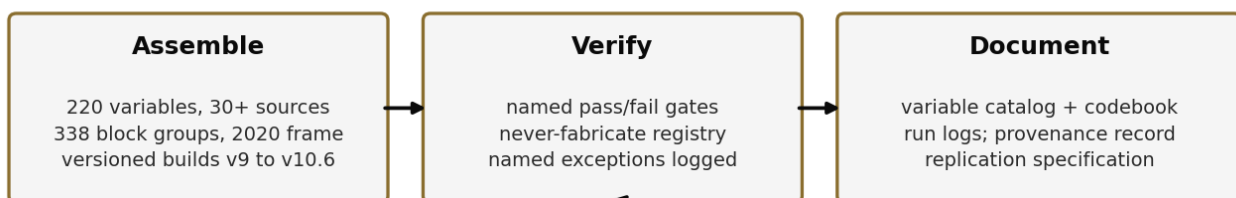
3. METHODOLOGY

3.1 DESIGN OVERVIEW: TWO PHASES

The Guilford County Neighborhood Vitality Index was produced in two phases that correspond to the two obligations of any public measurement instrument: to assemble evidence worth trusting, and to combine it in a way that can be defended. Phase A, acquisition and verification, built a master analytic file of 220 variables for all 338 census block groups in Guilford County on the 2020 census geography, drawing on more than thirty distinct sources and proceeding through versioned pipeline builds (v1 through v10.6) between June and August 2026. Phase B, empirical selection and composite construction, subjected the assembled measures to criterion-based screening, regression and collinearity diagnosis, and a documented selection protocol, yielding a ten-component composite index classified into five vitality categories. Figure 3 summarizes the flow; the subsections that follow describe each stage in the order the work was performed.

Two commitments governed both phases and are worth stating before the details. First, nothing in the master file is fabricated: every value traces to an authoritative source through a documented acquisition route, and where a source could not be obtained, was suppressed by its producer, or failed a quality rule, the gap is registered openly rather than filled by estimation. Second, nothing shipped without verification: every pipeline build concluded with a block of named pass or fail assertions (frame integrity, coverage floors, bounds, internal-consistency inequalities, and source-supplied reconciliations) and was refused publication on any failure. These are methodological choices, but they are also governance choices; a public index invites scrutiny, and the surest response to scrutiny is a file whose every cell can be explained.

PHASE A: ACQUISITION AND VERIFICATION



PHASE B: EMPIRICAL SELECTION AND COMPOSITE CONSTRUCTION

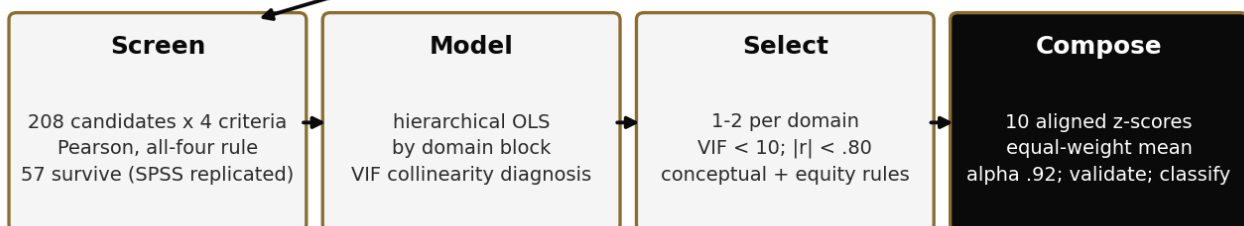


Figure 3 Design overview of the GCNVI

3.2 THE CRITERION-VALIDATION FRAMEWORK

The most consequential methodological decision of the project was the turn, between the first and second design iterations, from a theory-weighted composite to an outcome-anchored one. The original (v1.0) design specified domains and weights on conceptual grounds, in the manner of most practitioner indices. The revised (v2.0) design instead asked an empirical question: among all the conditions measurable at the neighborhood scale, which ones actually distinguish the neighborhoods where children go on to thrive from those where they do not? Answering that question requires independent measures of long-run thriving, and the project adopted four.

Three criteria come from the Opportunity Atlas (Chetty, Friedman, Hendren, Jones, and Porter), which follows the census of American children born between 1978 and 1983 into their adult outcomes and reports those outcomes by the neighborhood in which the children grew up.

1. Y1 is the mean household income rank in adulthood of children raised in the block group by parents at the 25th percentile of the national income distribution;
2. Y2 is the probability that such children reached the top quintile;
3. Y3 is their employment rate at age 30 or later.
4. The fourth criterion, Y4, is the Child Opportunity Index 3.0 composite (diversitydatakids.org), a contemporary, nationally normed measure of the conditions children currently experience.

The four criteria are independent of the candidate predictors in construction and, in the case of Y1 through Y3, separated from them by decades of real time: they measure what became of children raised under each neighborhood's conditions, not the conditions themselves.

Screening against the criteria produced one finding about the criteria themselves that shaped everything downstream. Three of the four move together tightly; the fourth does not. Table 3.1 reports the inter-criterion correlations: income rank, top-quintile probability, and the Child Opportunity Index correlate at .73 to .87, while the employment criterion (Y3) is near-orthogonal to all three ($r = .06, -.02,$ and $-.17$). Nearly every strong predictor of the other three criteria reverses sign on Y3: neighborhoods with worse measured conditions show slightly higher employment among children of low-income parents. This mirrors a documented national pattern in the Opportunity Atlas (earlier labor-force entry in low-mobility areas; college enrollment suppressing measured employment at 30 in high-mobility ones) and is a property of the benchmark, not noise in the county data. The consequence, adopted throughout: the index is selected and validated against Y1, Y2, and Y4, and Y3 is reported separately as a differently interpreted benchmark. Appendix J treats the finding in full.

Table 3 Pearson correlations among the four criterion variables (n = 337 to 338 block groups)

	Y1 income rank	Y2 top-quintile	Y3 employment	Y4 COI 3.0
Y1 income rank	1.000			
Y2 top-quintile probability	.873	1.000		
Y3 employment at 30	.055	-.015	1.000	
Y4 Child Opportunity Index	.805	.726	-.171	1.000

3.3 ACQUISITION & VERIFICATION DISCIPLINE

Phase A was executed as a versioned data pipeline rather than a set of ad hoc downloads, and the distinction matters for both reliability and replication. Each build (v9, v10, v10.1 through v10.6) is a named, logged event: a scripted orchestrator pulled or ingested sources, joined them to the 2020 block-group frame, and wrote a branded master file, a variable catalog, a domain status ledger, and a run log recording every retrieval, transformation, and check. Later builds were targeted increments (voter registration and turnout at v10.2; environmental indicators at v10.3; small-business lending at v10.4; the High Point code-enforcement merge at v10.5; county vital statistics at v10.6), each with its own verification block and each superseding the prior master in full.

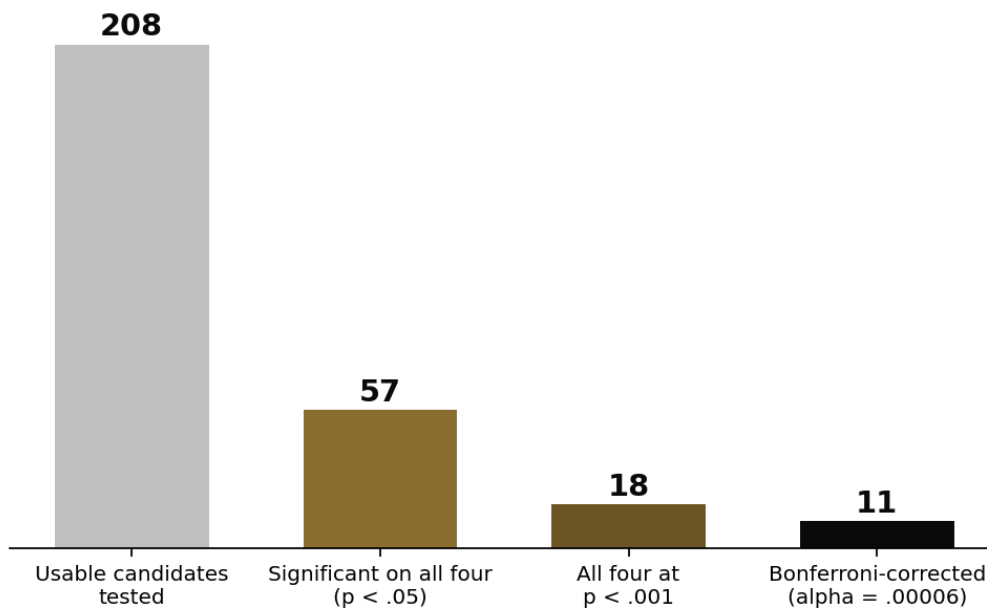
The verification regime deserves specific description because it is the project's principal quality instrument. Every build closes with named assertions that must all pass before outputs are written. The checks encode frame integrity (exactly 338 block groups and 126 tracts, with exact tract-set matches in both directions); coverage floors for every newly added variable; bounds tests (rates and percentiles within their admissible ranges); internal-consistency inequalities (for example, voters cannot exceed registrants, and small-firm lending counts cannot exceed total lending counts); and reconciliation against source-supplied totals wherever the source publishes its own (the county-total row of the CRA disclosure file, for instance, reconciled at 99.7 percent, with the 37 loans filed at unknown tracts registered rather than allocated). Failures functioned as diagnostics rather than nuisances; one build's initial failure surfaced several hundred boundary geocodes that were then excluded explicitly rather than silently absorbed.

The same discipline governs absence. The project maintains a manual-downloads and exceptions registry in which every unobtainable item is recorded with its reason: sources whose publication geography stops above the block group; producer-side suppression (the state's small-population life-expectancy tracts); zero-population geographies (the airport block group, which is carried as a named exception rather than imputed); and items still in negotiation at this writing (the North Carolina Secretary of State business registry). Where sources were produced on 2010-vintage geography, values were carried to the 2020 frame by the population-weighted tract crosswalk, with a minimum-weight rule (at least half of a 2020 tract's weight must be covered by valued 2010 parents) so that crosswalked estimates are never manufactured from fragments; every gap that rule produces was traced to its source suppression and verified as unrecoverable. Appendix H incorporates the full acquisition and provenance record, including the replication specification by which the entire file can be rebuilt.

3.4 SCREENING & REDUCTION

Phase B opened with a deliberately simple, deliberately conservative filter. Of the 220 variables in the master file, 208 were usable candidates (the remainder being identifiers, empty legacy fields, or constants); each was correlated (Pearson, pairwise-complete, $n = 337$ to 338) with each of the four criteria, 832 tests in all. A candidate survived only if significantly associated with all four criteria at once at $p < .05$. Fifty-seven measures met the standard. Because significance is a permissive bar at this sample size, the result is reported as a robustness ladder rather than a single verdict (Figure 4): of the 57 survivors, 18 remain significant on all four criteria at $p < .001$, and 11 survive full Bonferroni correction¹ across all 832 tests ($\alpha = .00006$), a core of associations that cannot plausibly be attributed to multiple-comparison chance. The complete screen (coefficients, exact p values, and pairwise n for all survivors) is Appendix C.

Candidate measures surviving successive significance thresholds



832 tests (208 candidates x 4 criteria); pairwise-complete $n = 337$ - 338 per test · Innovative Research Insights, LLC

Figure 4 The robustness ladder: 208 candidates tested, 57 significant on all four criteria

The screen was independently replicated. The full correlation matrix was recomputed in SPSS from the analysis file, by hand and without reference to the pipeline's code, and the two computations agree exactly: identical pairwise n on every test, coefficients matching to display precision (maximum discrepancy .0005, attributable to rounding), the identical set of 57 all-four survivors, and independent confirmation by SPSS's own engine that the five excluded fields were uncomputable. A measurement

¹ See Armstrong RA. When to use the Bonferroni correction. *Ophthalmic Physiol Opt.* 2014 Sep;34(5):502-8. <https://pubmed.ncbi.nlm.nih.gov/24697967/>.

instrument intended for public governance should not rest on a single implementation and this one does not.

The 57 survivors were then examined jointly. Hierarchical ordinary least squares models were estimated for each criterion, entering the survivors in eleven domain blocks in catalog order, on a listwise analytic sample of 312 block groups after principled exclusions (five candidates removed for margin-of-error missingness that would have collapsed the sample; two removed as exact linear aliases of other candidates). Table 4 summarizes the increments to explained variance. Read jointly with the correlation screen, the models delivered three lessons. First, the survivors are massively redundant: the median variance-inflation factor across the design was 35, with the worst offenders in the thousands, meaning the 57 measures carry far fewer than 57 distinct signals. Second, the national composite indices added almost nothing (an R-squared increment of .01 to .04) once the county's own primary measures were in the model, which is both a validation of the primary data and the empirical warrant for excluding composites from the index itself. Third, explanatory power against the aligned criteria is high (final-model adjusted R-squared of .81 for income rank and .96 for the Child Opportunity Index, the latter inflated by shared inputs) while the employment criterion again stood apart (.35), consistent with the criterion finding of Section 3.2. Blocks entered in catalog order; increments reflect entry position as well as content. The demographic block's large increment is a measure of how completely disadvantage has been racialized in the county's geography; it is reported here as descriptive evidence and excluded from scoring under the demographic-neutrality safeguard of Section 3.6. Full model output, coefficients, and the collinearity diagnostics are Appendix D.

Table 4 Increments to R-squared by domain block, hierarchical OLS (n = 312).

Domain block (entry order)	Y1	Y2	Y3	Y4
A. Household economic position	.237	.154	.034	.351
C. Demographic composition	.338	.239	.080	.493
D. Housing conditions	.001	.001	.001	.000
E. Parcel and court records	.054	.068	.064	.029
F. Built environment	.005	.006	.028	.000
G. Health and environment	.150	.197	.162	.085
H. Food environment	.004	.006	.000	.000
I. Crime and safety	.027	.020	.019	.001
J. Civic and social capital	.004	.007	.003	.001
K. Economic activity	.010	.004	.017	.001
N. External composites	.012	.021	.044	.004
Full model R-squared	.842	.721	.452	.966
Full model adjusted R-squared	.812	.668	.347	.959

3.5 COMPOSITE CONSTRUCTION & CLASSIFICATION

Selection from the 57 survivors to the final components proceeded under written rules applied domain by domain. The statistical rules: a component's selection score was its mean absolute correlation with Y1, Y2, and Y4 (the employment criterion excluded per Section 3.2); components had to hold variance inflation below 10 within the final set; and no two selected components could correlate above .80 in absolute value. The structural rules: one component per qualifying domain, two where a domain's administrative depth justified it; national composites excluded categorically (an index built from indices forfeits both transparency and independence); and modeled small-area estimates capped at a single component, both to limit dependence on model-based data and because the modeled health estimates share inputs with the Child Opportunity Index criterion. The conceptual rules are the subject of Section 3.6. Every domain-level decision, including what was passed over and why, is recorded in the selection matrix (Appendix E), which the report reproduces as Table 5.1.

The ten selected components (vehicle access, homeownership, median home value, eviction judgments, social and emotional support, birth outcomes, food access, personal-crime risk, census self-response, and small-business lending to firms under one million dollars in revenue) are treated individually in Section 5. Construction from them is deliberately plain. Each component was standardized on the county distribution (z-scores across the 338 block groups), sign-aligned so that higher values always indicate greater vitality, and averaged with equal weights. Equal weighting is a considered choice, not a default: with ten components held under a variance-inflation ceiling and loading on a single factor, empirical weights (factor scores, regression weights) reallocate influence toward whichever measures happen to share the most variance, add estimation noise, and cost the index its most valuable public property, which is that any reader can recompute it. A block group is scored when at least nine of its ten components are present; 335 of 338 qualify, and the three that do not (the zero-population airport block group and two data-sparse fragments) are reported as unscored rather than approximated.

The composite is internally coherent and externally valid. Cronbach's alpha (α) across the ten aligned components is .922 and McDonald's omega (Ω) is .935; all ten load between .62 and .91 on a single principal factor, and no component's variance-inflation factor exceeds 6.2. Against the criteria, the composite correlates at .762 with adult income rank, .622 with top-quintile probability, and .873 with the Child Opportunity Index (all $p < .001$), with the employment criterion reported separately at -.224. Benchmarked against the national deprivation indices on the county's own ground, the composite outperforms the Area Deprivation Index and the Social Deprivation Index on income mobility and sits within .03 of the Social Vulnerability Index on child opportunity, while remaining fully transparent, locally derived, and composed of measures a county government can actually move. Section 6 develops the validation in full.

Classification, finally, converts the continuous score into the five categories used throughout this report. Cut points are set at one half and one and one half standard deviations from the county mean, yielding Very Low (24 block groups), Low (94), Moderate (92), High (105), and Very High (20) vitality. The

banding rule is symmetric, documented, and reproducible from the published scores; it involves no judgment about any particular place.

3.6 MEASUREMENT ETHICS & DEMOGRAPHIC-NEUTRALITY SAFEGUARDS

An index designed to measure neighborhood conditions and inform place-based public decision-making carries methodological and ethical obligations beyond statistical performance alone. Guilford County's documented history, including racialized planning and disinvestment, provides essential context for interpreting the geographic patterns the index reveals; that history does not determine any block group's score, classification, or eligibility for a public benefit. The project therefore codified these obligations as explicit measurement safeguards rather than leaving them to interpretation.

First, demographic composition is excluded from scoring by design. Race, ethnicity, nativity, age structure, and their correlates appear in the master file and in this report as descriptive and analytical context, and Table 3.2 shows exactly how much variance they would capture; they are nonetheless barred from the index because who lives in a neighborhood is not a deficit of the neighborhood. Neighborhood conditions may be statistically associated with demographic composition, but that association does not make protected or demographic characteristics appropriate scoring variables. The index instead measures observable conditions of place directly: home values, eviction exposure, lending availability, health burden, food access, transportation access, and other independently defined conditions. This rule extended to quasi-compositional measures: the strongest single screener in the entire candidate set, an index of racial and income polarization, was assigned to the benchmark set rather than the components on precisely these grounds. It remains available for descriptive comparison and validation, but it contributes nothing to the GCNVI score or classification.

Second, modeled data are capped and labeled. One component (social and emotional support) is a model-based small-area estimate, retained because it carries a construct, the collective-efficacy dimension of neighborhood life, that no administrative source measures, and capped at one by rule. The personal-crime component is likewise a modeled vendor index; it was validated against the county's measured 2014 to 2018 compilation ($r = .68$ with the verified violent-crime rate), is cleared for the analysis reported here, and carries a standing condition: its license must be resolved, or the component substituted with the measured series, before the index is republished on a public platform.

Third, suppression is honored, not overridden. Where producers suppress small-population estimates (the state's life-expectancy tracts, the project's own twenty-birth reliability floor for birth-outcome rates), the suppressed cells remain empty in every product of this project, including its maps, which render them as hatched absence. The alternative, imputation, would manufacture precision exactly where the data are weakest and the stakes of error are highest.

Fourth, scores are county-relative and say so. Every z-score, category, and map in this report describes a block group's position within Guilford County in the mid-2020s. The index does not support comparisons to other counties, and its categories are not national designations; the external composites retained in the benchmark set exist for readers who need that national frame. The five GCNVI categories are analytical classifications, not statutory designations and not automatic eligibility

categories for grants, contracts, employment, services, or other public benefits. Any County department or partner using the GCNVI to inform a program or expenditure remains responsible for establishing lawful, program-specific criteria under its own authority.

Fifth, demographic neutrality applies to use as well as construction. Historical and demographic information may be collected, mapped, analyzed, and reported for lawful research, descriptive analysis, public understanding, validation, and compliance purposes. Protected characteristics are not used to calculate a GCNVI score, determine a vitality category, establish eligibility for a County program, or provide differential treatment or special benefits, and no protected characteristic should be reintroduced indirectly through the intentional use of a geographic or statistical proxy. The GCNVI provides evidence about conditions of place; it does not itself authorize an expenditure, establish program eligibility, or determine the award of a contract or public benefit. Later sections refer to this rule as the use safeguard of Section 3.6.

3.7 LIMITATIONS

Six limitations bound the appropriate use of the index, and stating them plainly is part of the method. First, several components and many catalog variables are tract-level measures broadcast to their constituent block groups; for those measures the effective sample is closer to 126 than 338, within-tract variation is invisible by construction, and significance tests are correspondingly optimistic. Second, the American Community Survey estimates that underlie several components carry sampling error that is proportionally largest in exactly the small, high-poverty block groups the index is most concerned with; the project's coefficient-of-variation screening and reliability floors mitigate but do not eliminate the issue. Third, two components are model-based rather than measured, with the caps, validation, and licensing condition described above. Fourth, administrative coverage is not uniform: code-enforcement data cover the two principal cities' jurisdictions, so zeros elsewhere in the county are coverage artifacts, flagged as such wherever the variable appears. Fifth, the file mixes vintages by necessity (criterion outcomes reflect children raised decades ago; the newest administrative data are current-year), and crosswalked 2010-vintage measures carry boundary-recomposition uncertainty that is called out wherever it bites. Sixth, the design is cross-sectional: the index describes conditions, and its correlations with later-life outcomes, however strong, are not by themselves causal estimates of what any specific intervention will change. The measurement agenda of Section 8.4 addresses the gaps that can be closed; the others are inherent to neighborhood measurement and are managed here by disclosure.

4. DATA & SOURCES

4.1 SOURCE FAMILIES

The 220-variable master file draws on five families of sources, each contributing evidence of a different kind, and the index's credibility rests on their triangulation: survey estimates, administrative records, geocoded public files, open commercial data, and national composites cross-check one another throughout. Table 5 summarizes the families; Appendices A and B carry the complete per-variable catalog and analysis codebook.

Table 5 The five source families of the GCNVI master file

Source family	Principal sources	What they contribute	Geography	Vintage
Federal statistical	ACS 2019-2023; LODES; HMDA; CRA; FDIC; HUD/USPS; CDC PLACES; CDC/ATSDR SVI; USDA FARA; EPA EJScreen; Census self-response archive	Demographics, economics, education, housing; employment; mortgage and small-business capital; banking; address vacancy; modeled health; vulnerability; food and environmental exposure; civic response	Block group and tract	2019 to 2025
State	NC State Board of Elections; State Center for Health Statistics (via county); NC HUB certification	Registration and turnout; vital-records inputs; certified small businesses	Address-level, geocoded; tract	2024 to 2026
County and municipal administrative	Guilford County parcels; county courts; Greensboro and High Point code enforcement; Guilford County Health Department	Assessed values, ownership, foreclosure; eviction judgments; violations casework; birth outcomes and life expectancy	Parcel and case level, geocoded; tract	2023 to 2026
Open and commercial	Overture Maps; AGS CrimeRisk (licensed); Social Capital Atlas	Establishments and independence; modeled crime risk; network measures	Point, geocoded; block group; ZIP	2018 to 2026
National composites (benchmarks only)	ADI; SDI; SVI; DCI; COI 3.0	External validation and triangulation; excluded from the index by rule	Block group and tract	2019 to 2024

4.2 GEOGRAPHY & VINTAGE

All 220 variables are expressed on a single frame: the 338 block groups and 126 census tracts of Guilford County as defined in the 2020 census, constructed natively from the Census Bureau's

published geography. Sources arriving at other resolutions were brought to the frame by documented rules. Point and address data (parcels, court records, code cases, establishments, registered voters) were geocoded and assigned by point-in-polygon against the 2020 boundaries, with the Census Bureau’s batch geocoder as the fallback for records lacking coordinates. Match rates and exclusions are logged per source. Tract-level sources were broadcast uniformly to their constituent block groups and are flagged as broadcast in the codebook, with the analytic consequences stated in Section 3.7. Sources produced on 2010-vintage geography (the Opportunity Atlas criteria, the food-access designations, the life-expectancy series, the measured crime compilation) were carried forward by the population-weighted 2010-to-2020 tract crosswalk under the minimum-weight rule described in Section 3.3. ZIP-level measures (the social-capital series) were allocated by residential correspondence and are so labeled. The frame itself, and every join to it, was re-verified at every build.

4.3 ACQUISITIONS OF NOTE

Most of the file arrived through routine, scripted retrieval; four acquisitions were not routine, and each materially raised the index’s quality. They are recorded here with the credit they are owed.

The Guilford County Health Department’s vital-statistics delivery resolved the project’s largest measurement gap. The state publishes births and deaths at the county level only, and the project had prepared a second-best substitution through a city-limits data-use agreement when the county health department delivered full-county tract data outright: birth outcomes for 2022 and 2023 on native 2020 tracts (11,065 pooled births, with low-birthweight, preterm, and prenatal-care detail) and period life expectancy for three windows (2009-2013, 2012-2016, 2016-2020). The delivery gave the index a measured, current, whole-county birth-outcomes component and the county’s authoritative life-expectancy series, and it retired the substitution plan before it was needed.

The City of High Point’s Community Development and Housing Department provided the city’s complete code-enforcement casework, 33,068 cases spanning 1997 to 2026. Merged with Greensboro’s open-data equivalent at build v10.5 (9,543 High Point cases in the harmonized three-year window, 98.3 percent successfully placed), the delivery converted the code-violations measure from a one-city variable with false zeros into genuine two-city coverage, and populated all four High Point pilot tracts with an administrative housing-stress signal they previously lacked.

The environmental-justice indicators survive in the file because of preservation work outside government. With the EPA’s EJScreen tool and data withdrawn from federal servers, the project sourced the complete 2024 national block-group file from the Public Environmental Data Partners’ preservation copy on the Harvard Dataverse, verified it, and extracted the county’s 338 records with all fields intact. Public measurement increasingly depends on such stewardship, and the report notes it deliberately.

Finally, the civic-participation measures were built, not downloaded. The registration and turnout variables rest on a geocoding operation that assigned 210,862 unique registered-voter addresses from the State Board of Elections files to 2020 block groups at a 95.0 percent match rate, accumulated across sessions with a persistent cache and verified against the frame; turnout in the November 2024 general election was then computed from the voter-history file per block group. The census self-

response measure, ultimately one of the ten index components, came from the Bureau's 2020 self-response archive by direct retrieval.

4.4 DATA GAPS & OPENNESS

The file's honesty about what it does not contain is as much a feature as its coverage. Three gaps are material and registered. The North Carolina Secretary of State business registry, which would add new-business formation and a definitive investor-ownership flag, remained in agreement negotiations at this writing; the establishment measures in Domain K therefore rest on the open Overture data, and the registry is the first item on the measurement agenda (Section 8.4). A current-vintage refresh of measured crime incidents could not be obtained on acceptable terms, so the measured series reflects 2014 to 2018 conditions (crosswalked, and labeled with its vintage wherever used), with the modeled vendor index carrying the current-period construct under the conditions stated in Section 3.6. And the Census Bureau's Multidimensional Deprivation Index proved to be published at county scale and above only, and was resolved as unavailable at the index's geography rather than approximated from coarser data.

Beyond these, the registry records every producer-side suppression the file inherits, every named exception, and every variable that screening retired, each with its reason. The complete accounting (the variable catalog of Appendix A, the codebook of Appendix B with its analysis notes, and the acquisition and provenance record of Appendix H) is published with the report so that any reader, including a skeptical one, can trace any number in this document to its source. The project regards that traceability, more than any single statistic, as the foundation of the index's public authority.

5. THE TEN COMPONENTS

5.0 HOW THE TEN WERE CHOSEN

The ten components of the index were not nominated; they survived statistical scrutiny. Every measure in this section entered a field of 208 candidates, passed the all-four criterion screen of Section 3.4, and then prevailed within its thematic domain under the written rules of Section 3.5: the strongest available association with the three aligned criteria (the mean absolute correlation with Y1, Y2, and Y4, reported below as the selection score), variance inflation below 10 within the final set, no pairwise correlation above .80 with any other selected component, and the conceptual and demographic-neutrality safeguards of Section 3.6. Protected demographic characteristics were ineligible for inclusion in the composite regardless of their statistical association with the criterion variables. Table 6 reproduces the complete selection matrix, one decision per domain, including the domains that contributed nothing and why.

The exclusions deserve as much attention as the selections, because they define what the index is. Domain C, demographic composition, was excluded by design: its measures screen powerfully, and the racial-income polarization index was the single strongest screener in the candidate set, but statistical performance alone was not sufficient for inclusion. Race, ethnicity, nativity, age structure, and the other measures of demographic composition describe characteristics of the population rather than the independently defined neighborhood conditions the GCNVI is intended to score, and they were excluded from composite construction under the demographic-neutrality safeguard of Section 3.6. Where analytically useful, these measures are retained for descriptive analysis, historical interpretation, or benchmark validation; they contribute nothing to any block group's score or classification.

Domain N, the national composites, was excluded categorically as the hierarchical models showed they add almost nothing once the county's own measures are present, and an index of indices would forfeit the transparency this project exists to provide. Domain B, education, produced no component because its one all-four survivor (preschool enrollment) was the weakest and least complete qualifier in the field, and Domain F, the built environment, because its sole survivor (a housing-unit count) is conceptually thin as a vitality measure. Education and built-form measures remain correlated with several selected components, including home values, vehicle access, and self-response, but they do not enter the composite because they did not satisfy the project's selection rules; each selected component is defined and interpreted according to the condition it directly measures. Two domains, by contrast, earned double representation: the parcel-and-court records of Domain E and the health records of Domain G each contribute two components, on the strength of administrative depth (E) and the pairing of one modeled construct with one measured outcome (G).

Each subsection below follows one structure: what the component measures and where it comes from; why it was selected; the county pattern; and the pattern in the two pilot areas identified for proof-of-concept validation. Each closes with an evidence box reporting the component's selection score, its

variance-inflation factor within the final ten, its loading on the index's single principal factor, and its coverage of the 338 block groups.

Table 6 The per-domain selection matrix, condensed

Domain	Selection	Score	Decision in brief
A. Household economic position	A12 vehicle access	.413	Selected; income measures scored higher but carry 27 suppressed block groups; A12 is near-complete and measures a distinct mobility barrier
B. Education	none		Excluded: sole qualifier (preschool enrollment) weakest and least complete in field
C. Demographic composition	none		Excluded under demographic-neutrality safeguard; protected demographic characteristics are not scoring inputs; strongest screeners retained for descriptive and benchmark analysis only
D. Housing conditions	D1 homeownership	.442	Selected; value measure duplicates E4; renter share is D1's complement
E. Parcel and court records	E4 home value; E10 evictions	.669; .557	Two selected: largest domain, both complete administrative measures, conceptually distinct ($r = -.44$)
F. Built environment	none		Excluded: sole qualifier conceptually thin (housing-unit count)
G. Health and environment	Social support; G8 birthweight	.819; .475	Two selected: strongest screener in domain (modeled, capped at one) plus measured vital records
H. Food environment	FARA low-income low-access flag	.606	Selected over its aliased single-criterion twin; proper food-desert construct
I. Crime and safety	I6b personal-crime index	.671	Selected with licensing condition; validated against measured 2014-2018 series
J. Civic and social capital	J8 census self-response	.703	Selected over turnout ($r = .72$ between them); broader construct, stronger score
K. Economic activity	K10b small-firm lending	.361	Selected; capital access to firms under \$1M revenue is the domain's core construct
N. External composites	none		Excluded categorically; retained as validation benchmarks

5.1 VEHICLE ACCESS (A12)

What it measures. The share of households with no vehicle available, drawn from the American Community Survey 2019-2023 five-year estimates at the block-group level and reversed in the index so that higher scores mean better access. In a county whose employment is dispersed across suburban corridors and whose transit network is thin, a household without a car is a household whose labor market has been shrunk to walking and bus distance; the measure operationalizes what the spatial-mismatch literature has documented for half a century.

Why it was selected. Domain A's income and poverty measures screened somewhat more strongly, but each carries 27 block groups suppressed for unacceptable margins of error; vehicle access delivers nearly the same criterion signal (selection score .413) with near-complete coverage, and it contributes a construct, physical reach, that the income measures proxy only indirectly. Its variance-inflation factor of 1.53 is the lowest in the final set: the component brings information the other nine do not.

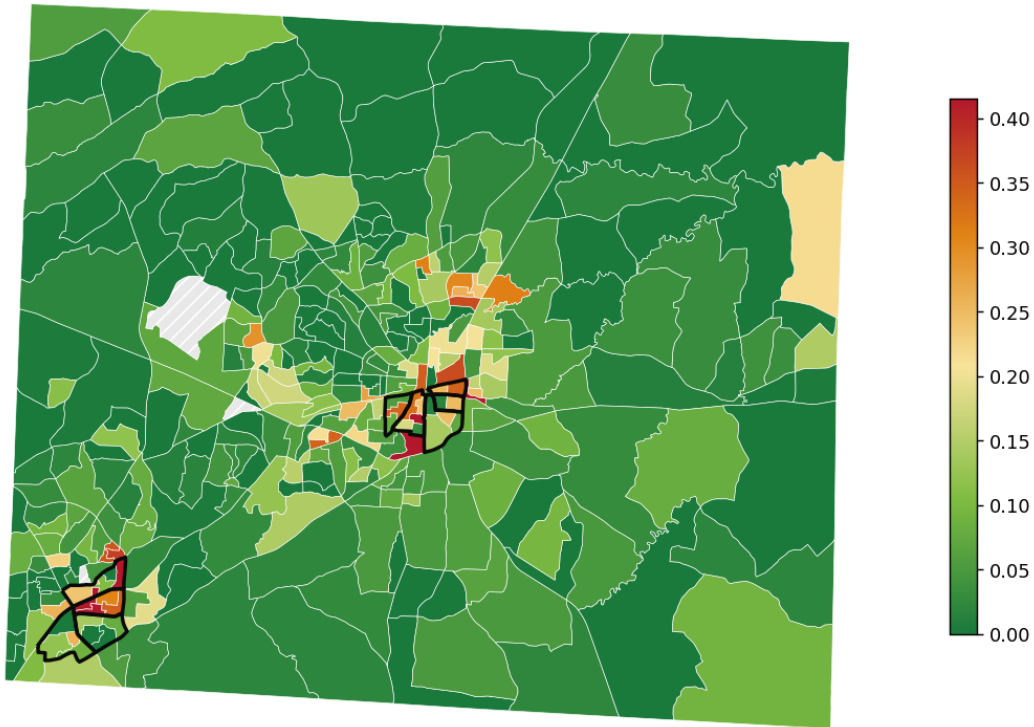
The county pattern. Countywide, 8 percent of households lack a vehicle, but the county map shows how misleading that average is: access approaches universality across the suburban and rural periphery and collapses in the urban cores, where the measure reaches its maximum of 76 percent of households in a single block group.

The pilot areas. The pilot tracts average 20 percent of households without a vehicle in the Greensboro cluster and 24 percent in High Point, two and a half to three times the county rate, with block-group peaks near one household in two. The Greensboro map shows the deficit hugging the historically redlined east side; in High Point it is deepest in tract 139, where, as Section 6.6 details, half of all households have no car in a city built for them.

Evidence: selection score .413; VIF 1.53; factor loading .62; coverage 335 of 338.

Vehicle access: households without a vehicle (share): Guilford County

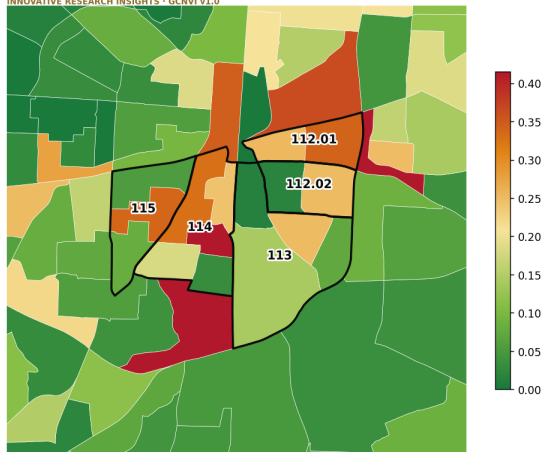
INNOVATIVE RESEARCH INSIGHTS · GCNVI v1.0



Green = direction of greater vitality · scaled p2-p98 · black outlines = priority tracts · hatched = no data · 2020 block groups
Innovative Research Insights, LLC

Vehicle access: households without a vehicle (share): Greensboro target area

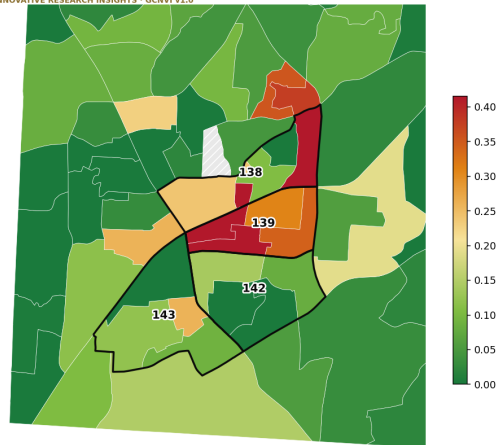
INNOVATIVE RESEARCH INSIGHTS · GCNVI v1.0



Green = direction of greater vitality · scaled p2-p98 · black outlines = priority tracts · hatched = no data · 2020 block groups
Innovative Research Insights, LLC

Vehicle access: households without a vehicle (share): High Point target area

INNOVATIVE RESEARCH INSIGHTS · GCNVI v1.0



Green = direction of greater vitality · scaled p2-p98 · black outlines = priority tracts · hatched = no data · 2020 block groups
Innovative Research Insights, LLC

Figure 5 Vehicle Access: Households without a vehicle

5.2 HOMEOWNERSHIP (D1)

What it measures. The owner-occupied share of occupied housing units, from the same ACS release. Ownership is the tenure through which American households have historically accumulated wealth, stabilized housing costs, and acquired a durable stake in neighborhood conditions; its local absence is both a consequence of past credit exclusion and a mechanism of present instability.

Why it was selected. Within Domain D, the median-value measure duplicates the stronger parcel-record component of Section 5.3 and carries 31 missing block groups, and the renter share is arithmetically D1's complement; homeownership was the domain's clean, complete, conceptually primary survivor (selection score .442, VIF 2.15).

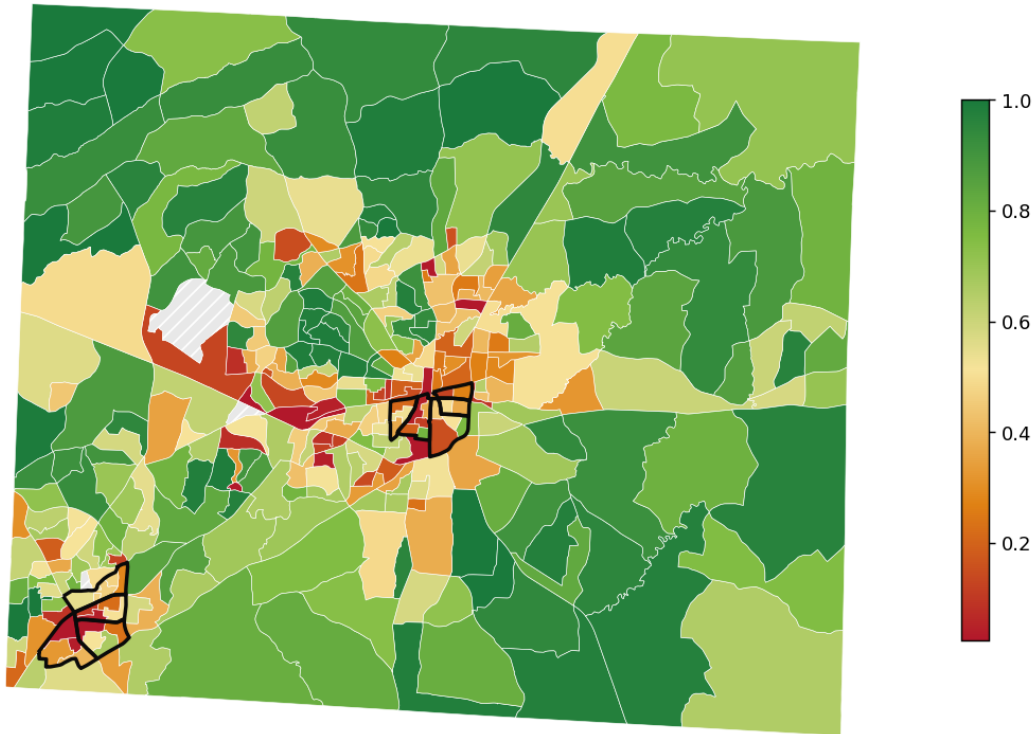
The county pattern. The county averages 59 percent owner occupancy across block groups, spanning the full range from under 5 percent to essentially 100. The map's green belt of high ownership wraps the county's periphery; the orange-red core traces the rental concentrations of the two cities, which are simultaneously the geography of the county's lowest values and highest eviction exposure.

The pilot areas. Ownership averages 36 percent across the Greensboro pilot tracts and 31 percent in High Point's, roughly half the county norm. The maps show the gradient sharply: green owner-occupied block groups sit immediately across the tract boundaries from pilot-tract block groups in the teens and twenties, a juxtaposition that is the residue of where mortgage credit was and was not extended.

Evidence: selection score .442; VIF 2.15; factor loading .74; coverage 335 of 338.

Homeownership rate (%): Guilford County

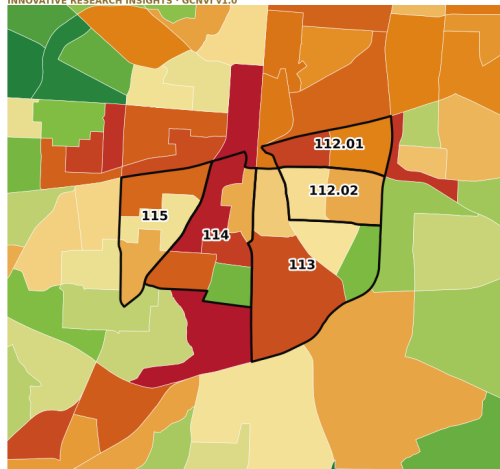
INNOVATIVE RESEARCH INSIGHTS · GCNVI v1.0



Green = direction of greater vitality · scaled p2-p98 · black outlines = priority tracts · hatched = no data · 2020 block groups
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Homeownership rate (%): Greensboro target area

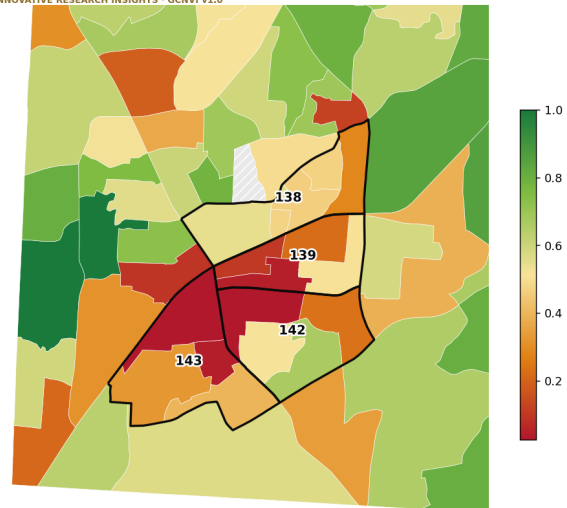
INNOVATIVE RESEARCH INSIGHTS · GCNVI v1.0



Green = direction of greater vitality · scaled p2-p98 · black outlines = priority tracts · hatched = no data · 2020 block groups
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Homeownership rate (%): High Point target area

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Green = direction of greater vitality · scaled p2-p98 · black outlines = priority tracts · hatched = no data · 2020 block groups
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Figure 6 Homeownership Rate

5.3 MEDIAN HOME VALUE (E4)

What it measures. The median assessed value of residential parcels in each block group, computed from the Guilford County parcel file at current assessment vintage. Unlike survey-reported values, the parcel record is a census of the housing stock itself; the measure captures, in a single number, what a century of investment, appraisal, and credit allocation has deposited in each neighborhood's principal asset.

Why it was selected. E4 is the strongest administrative measure in the entire candidate set (selection score .669) and was the only variable in the full hierarchical models significant for three of four criteria while holding variance inflation below 10; it anchors the index's wealth dimension. Its court-record companion (Section 5.4) measures displacement pressure; the two correlate at only -.44 and were both retained under the two-per-domain allowance.

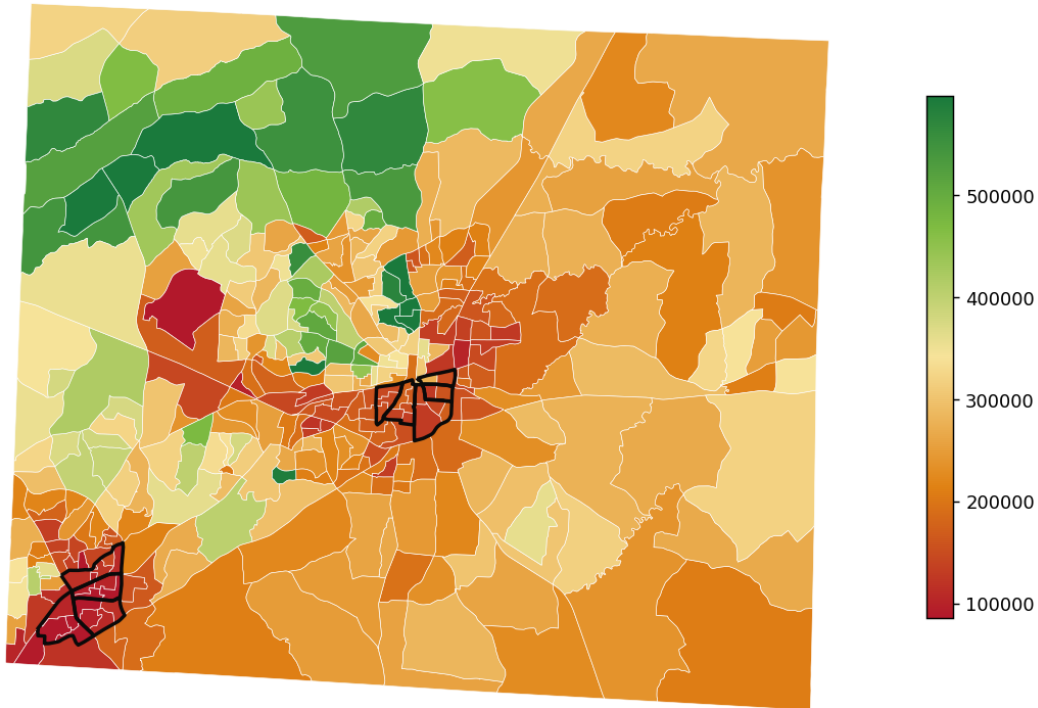
The county pattern. The county's block-group medians average \$268,348 and range past one million dollars; the map is the report's clearest picture of accumulated advantage, with the northwest quadrant glowing green and the two urban cores, and only they, rendered in red.

The pilot areas. The pilot-tract averages are \$156,150 in the Greensboro cluster and \$99,600 in High Point's, roughly forty and sixty percent below the county mean respectively, with individual block groups in the corridor under \$70,000. Read jointly with Section 5.2, the maps say precisely what blocked wealth accumulation means: fewer owners, owning assets worth a third of the county norm.

Evidence: selection score .669; VIF 2.45; factor loading .78; coverage 338 of 338.

Median home value (\$): Guilford County

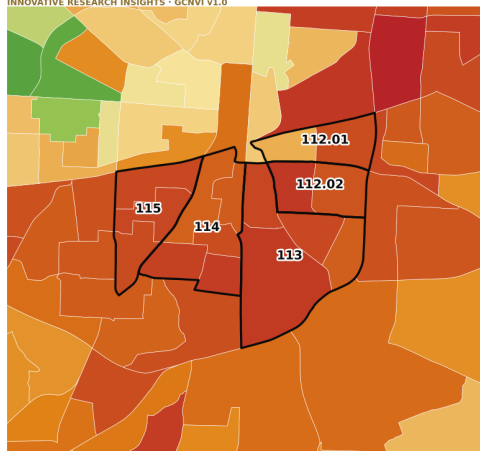
INNOVATIVE RESEARCH INSIGHTS · GCNVI v1.0



Green = direction of greater vitality · scaled p2-p98 · black outlines = priority tracts · hatched = no data · 2020 block groups
Innovative Research Insights, LLC

Median home value (\$): Greensboro target area

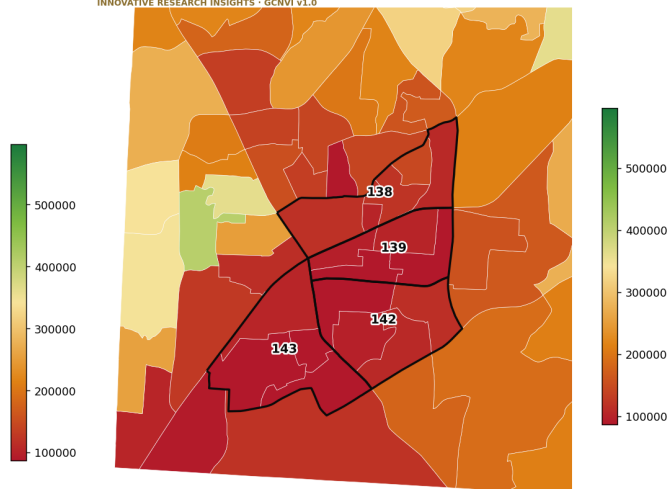
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Green = direction of greater vitality · scaled p2-p98 · black outlines = priority tracts · hatched = no data · 2020 block groups
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Median home value (\$): High Point target area

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Green = direction of greater vitality · scaled p2-p98 · black outlines = priority tracts · hatched = no data · 2020 block groups
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Figure 7 Median Home Value

5.4 EVICTION JUDGMENTS (E10)

What it measures. Eviction judgments per hundred renter households over a three-year window, computed from Guilford County court records geocoded to the 2020 frame, reversed in the index. Eviction is displacement recorded one household at a time: each judgment is a forced move, with the school disruption, possession loss, credit scarring, and health consequences the literature has established.

Why it was selected. The measure is complete, current, administrative, and conceptually distinct from the stock measures beside it (selection score .557, VIF 1.72); it carries the index's instability dimension, the difference between a neighborhood where low-income renters can stay and one where they cannot.

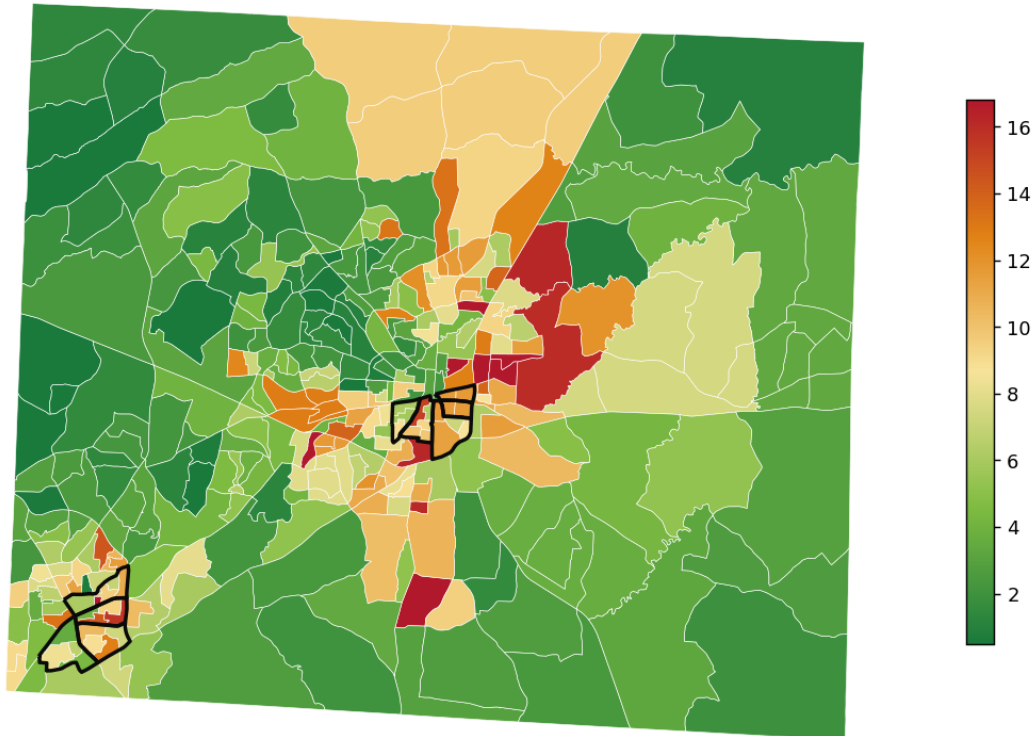
The county pattern. The county averages 6.2 judgments per hundred renter households; the map shows the burden concentrated in the same urban cores as the value deficit, with peaks approaching 24 per hundred, a rate at which a typical renter block turns over by court order within a few years.

The pilot areas. The pilot tracts average 9.8 (Greensboro) and 9.2 (High Point) judgments per hundred renter households, roughly fifty percent above the county rate; corridor block groups reach 10 to 11. The maps make one further point the composite cannot: several otherwise moderate block groups adjacent to the pilot tracts light up on this component alone, an early-warning signature discussed in Section 7.3.

Evidence: selection score .557; VIF 1.72; factor loading .68; coverage 338 of 338.

Eviction judgment rate: Guilford County

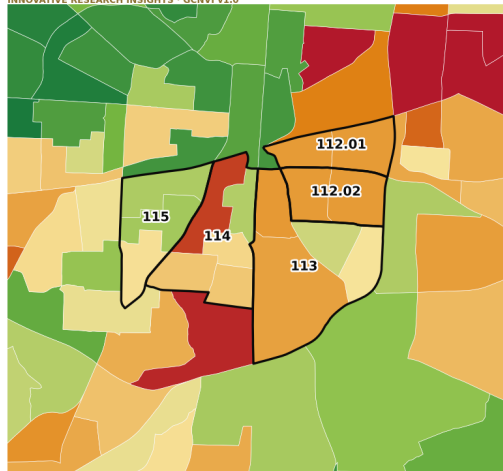
INNOVATIVE RESEARCH INSIGHTS · GCNVI v1.0



Green = direction of greater vitality · scaled p2-p98 · black outlines = priority tracts · hatched = no data · 2020 block groups
Innovative Research Insights, LLC

Eviction judgment rate: Greensboro target area

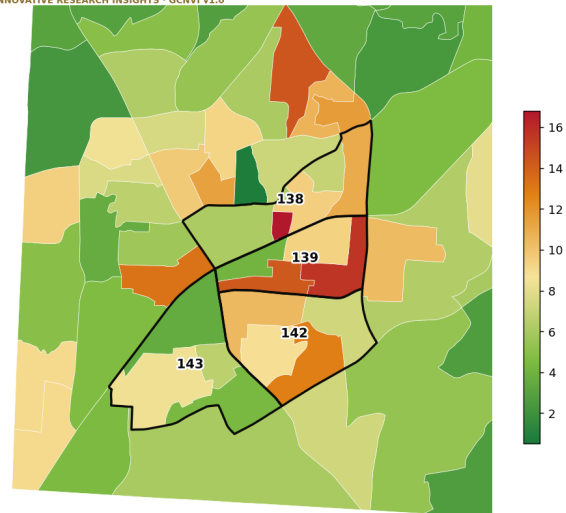
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Green = direction of greater vitality · scaled p2-p98 · black outlines = priority tracts · hatched = no data · 2020 block groups
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Eviction judgment rate: High Point target area

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Green = direction of greater vitality · scaled p2-p98 · black outlines = priority tracts · hatched = no data · 2020 block groups
Innovative Research Insights, LLC

Figure 8 Eviction Judgement Rate

5.5 SOCIAL & EMOTIONAL SUPPORT (PLACES)

What it measures. The modeled share of adults reporting that they lack the social and emotional support they need, from the CDC PLACES 2024 small-area estimates, reversed in the index. The measure is the file's closest approach to the construct the neighborhood-effects literature calls collective efficacy: the density of relationships through which neighbors sustain one another and enforce shared expectations.

Why it was selected. It is the strongest screener in the entire candidate field (selection score .819) and loads at .91 on the index's single factor; no administrative source measures its construct at all. It is also a model-based estimate, and the index therefore admits it under the cap of Section 3.6: one modeled PLACES component, this one, chosen over its equally strong siblings because its construct overlaps least with the components already present. Its VIF of 6.14 is the highest in the final set and sits comfortably under the ceiling.

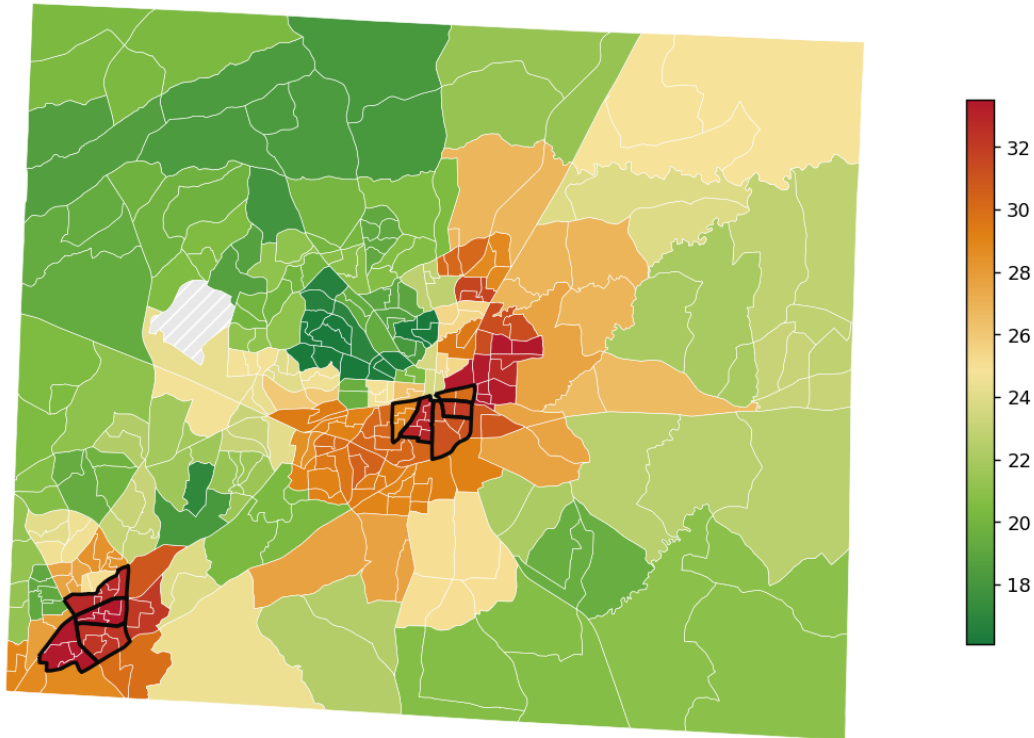
The county pattern. The county averages 25 percent of adults lacking support, ranging from 15 percent in the most connected block groups to 35 at the maximum; the county map's red geography coincides, block for block, with the administrative deficits of the preceding subsections, which is the empirical content of the claim that disinvestment erodes social infrastructure along with physical.

The pilot areas. The pilot tracts average 31 percent (Greensboro) and 33 percent (High Point), against the county's 25; the High Point cluster contains the county maximum. The reader should hold the modeled character of the estimate in mind at block-group resolution; its tract-scale pattern, however, is corroborated by every measured component beside it.

Evidence: selection score .819; VIF 6.14; factor loading .91; coverage 337 of 338.

Lacking social and emotional support (%): Guilford County

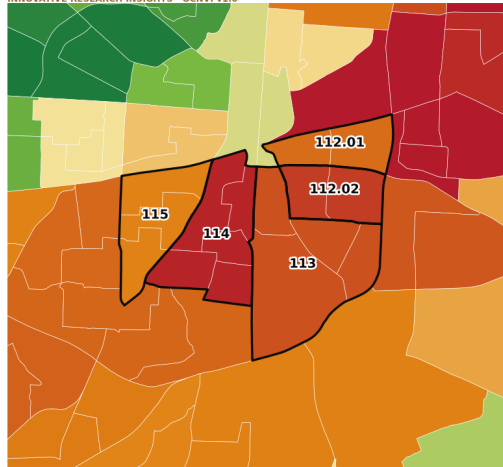
INNOVATIVE RESEARCH INSIGHTS · GCNVI v1.0



Green = direction of greater vitality · scaled p2-p98 · black outlines = priority tracts · hatched = no data · 2020 block groups
Innovative Research Insights, LLC

Lacking social and emotional support (%): Greensboro target area

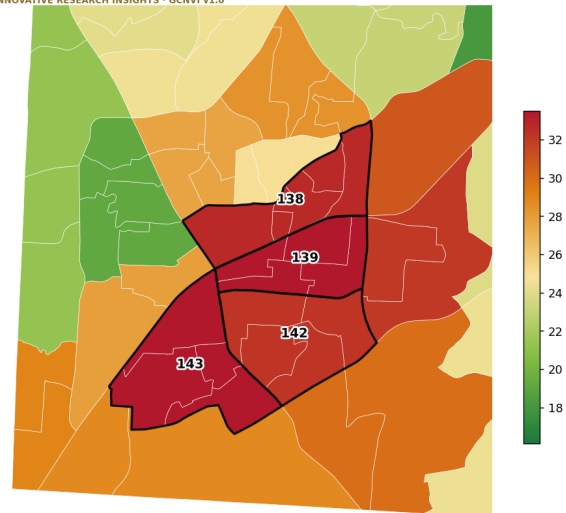
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Green = direction of greater vitality · scaled p2-p98 · black outlines = priority tracts · hatched = no data · 2020 block groups
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Lacking social and emotional support (%): High Point target area

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Green = direction of greater vitality · scaled p2-p98 · black outlines = priority tracts · hatched = no data · 2020 block groups
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Figure 9 Lacking Social and Emotional Support

5.6 BIRTH OUTCOMES (G8)

What it measures. The share of births at low birthweight (under 2,500 grams), pooled over 2022 and 2023, computed from the Guilford County Health Department’s tract-level vital records on native 2020 tracts (11,065 births countywide) and reversed in the index. Low birthweight is among epidemiology’s most sensitive summary measures of chronic community stress: it integrates maternal health, environmental exposure, health-care access, and the physiological weathering of sustained hardship into a single, measured outcome at the very start of life.

Why it was selected. Against the modeled estimates that dominate Domain G’s candidates, G8 is measured, current, and whole-county, delivered by the county’s own health authority (Section 4.3); it pairs the domain’s modeled social construct with a recorded biological outcome (selection score .475, VIF 1.59). Rates rest on a twenty-birth reliability floor, and the handful of suppressed small-tract cells are carried as suppressed.

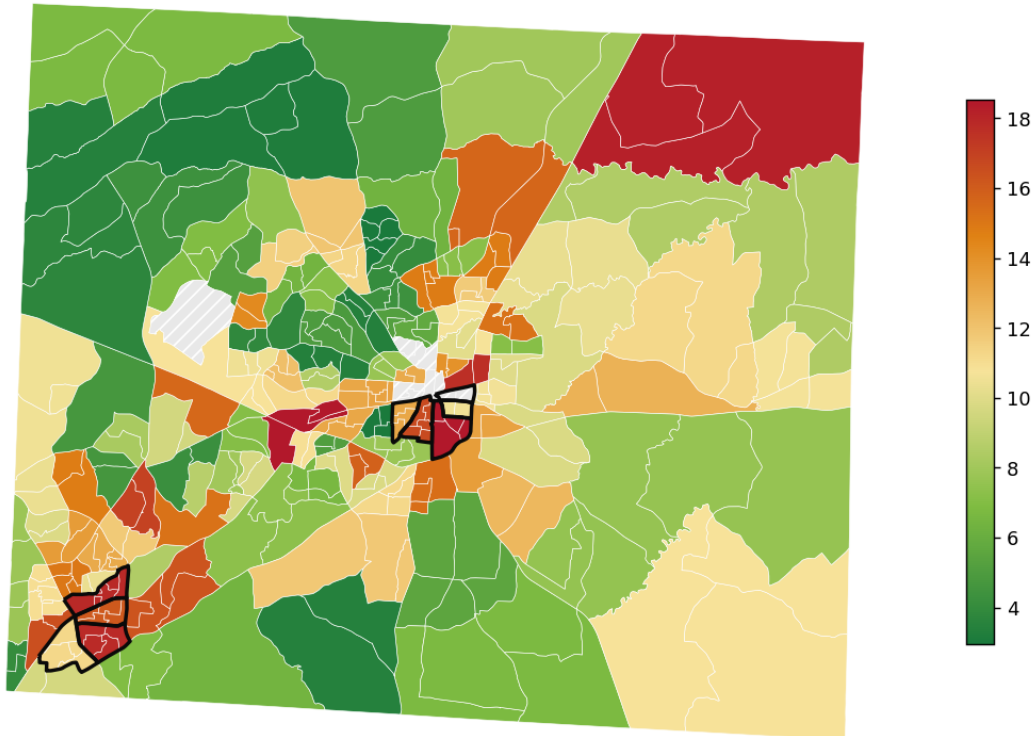
The county pattern. The county rate is 10.1 percent of births; the map’s healthy green periphery and red cores reproduce, in newborn weight, the geography every prior component drew in dollars and judgments, from under 3 percent of births in the healthiest block groups to 19 percent at the maximum.

The pilot areas. The pilot tracts average 15.7 percent (Greensboro) and 15.9 percent (High Point), a rate half again to nearly double the county’s, and the corridor tracts of Section 6.6 approach 16 to 18 percent. No statistic in this report states the stakes of neighborhood conditions more directly: the gradient is visible at birth.

Evidence: selection score .475; VIF 1.59; factor loading .65; coverage 328 of 338 (source suppression honored).

Low birthweight, 2022-23 (%): Guilford County

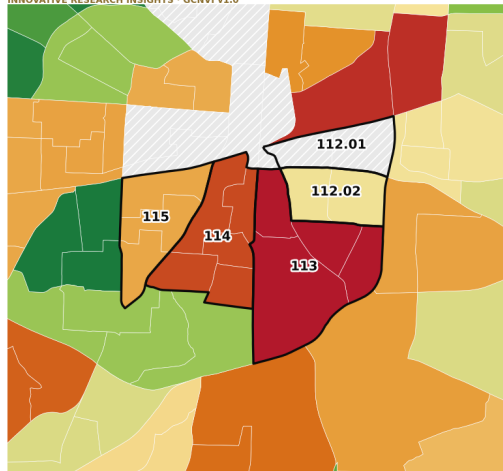
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Green = direction of greater vitality · scaled p2-p98 · black outlines = priority tracts · hatched = no data · 2020 block groups
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Low birthweight, 2022-23 (%): Greensboro target area

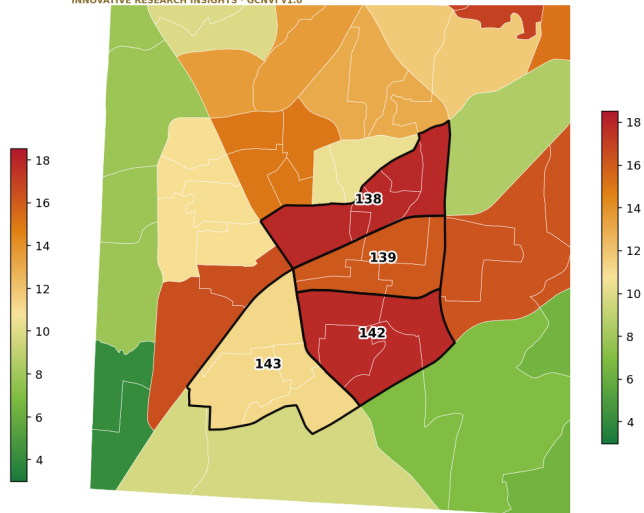
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Green = direction of greater vitality · scaled p2-p98 · black outlines = priority tracts · hatched = no data · 2020 block groups
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Low birthweight, 2022-23 (%): High Point target area

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Green = direction of greater vitality · scaled p2-p98 · black outlines = priority tracts · hatched = no data · 2020 block groups
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Figure 10 Low Birthweight, 2022-2023

5.7 FOOD ACCESS (FARA)

What it measures. The USDA Food Access Research Atlas designation of low-income, low-access areas (the analytic definition of a food desert: low-income tracts whose residents live beyond half a mile urban or ten miles rural from a supermarket), carried from 2010-vintage tracts by the population-weighted crosswalk and reversed in the index. The component is binary by construction and its maps are rendered accordingly.

Why it was selected. The designation combines the two conditions that jointly constrain diet, income and distance, into the domain's proper construct (selection score .606); its aliased single-condition twin was excluded as an exact linear duplicate. As a tract-vintage broadcast measure it carries the resolution caveats of Section 3.7, stated wherever it appears.

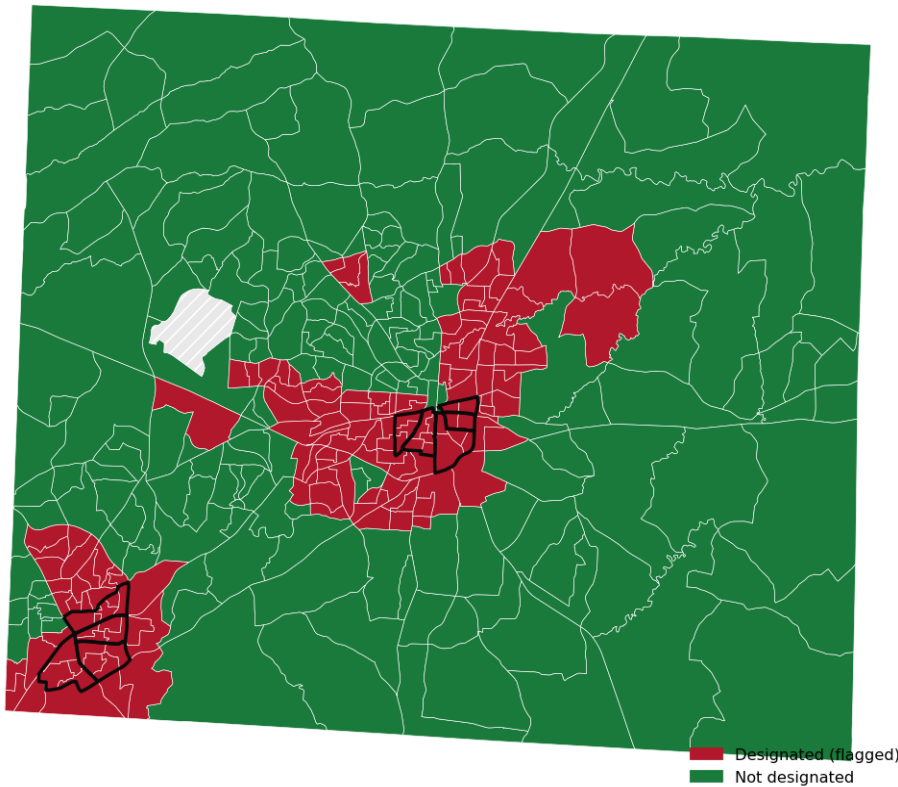
The county pattern. Forty-four percent of the county's block groups sit in designated low-income, low-access territory, a share that surprises readers who equate food deserts with a handful of neighborhoods; the map shows the designation spanning both urban cores and a broad rural east.

The pilot areas. Every pilot-tract block group in both clusters carries the designation: the Greensboro and High Point pilot-area maps are red without exception. The component consequently contributes no within-pilot variation, but its uniformity is itself the finding, and its factor loading of .84 attests how tightly the designation tracks the index's other dimensions countywide.

Evidence: selection score .606; VIF 3.17; factor loading .84; coverage 337 of 338.

Food desert designation (LILA): Guilford County

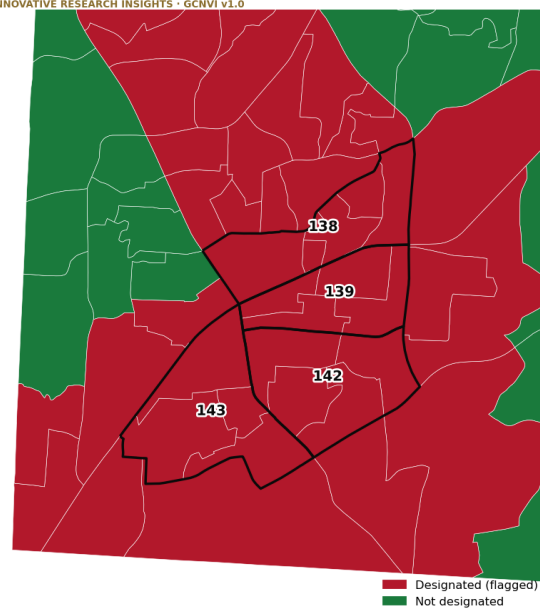
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Green = direction of greater vitality · scaled p2-p98 · black outlines = priority tracts · hatched = no data · 2020 block groups
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Food desert designation (LILA): High Point target area

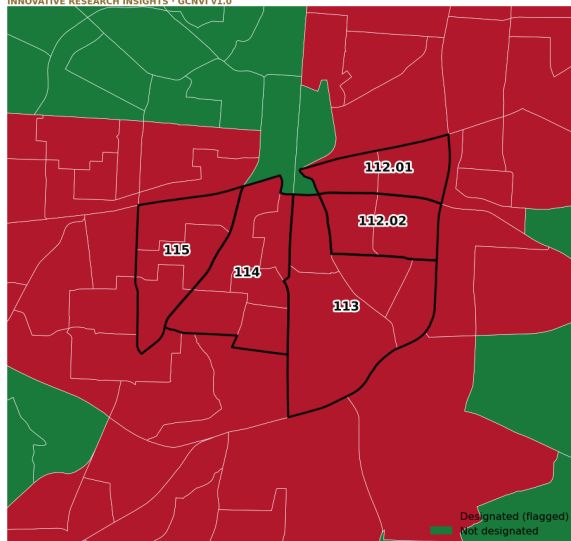
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Green = direction of greater vitality · scaled p2-p98 · black outlines = priority tracts · hatched = no data · 2020 block groups
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Food desert designation (LILA): Greensboro target area

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Green = direction of greater vitality · scaled p2-p98 · black outlines = priority tracts · hatched = no data · 2020 block groups
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Figure 11 Food Desert Designation

5.8 PERSONAL-CRIME RISK (I6B)

What it measures. Esri's modeled index of personal-crime risk (assault, robbery, and related offenses against persons), scaled to a national average of 100 and reversed in the index. The measure estimates residents' exposure to victimization at a location; it is not, and must never be read as, a measure of residents' propensity.

Why it was selected. The county's measured incident compilation is authoritative but reflects 2014 to 2018 conditions on 2010 geography; the modeled index carries the current-period construct, screens strongly (selection score .671), and was validated against the measured series ($r = .68$ with the verified violent-crime rate). In the delivered vintage the index is uniform within tracts, an additional resolution caveat.

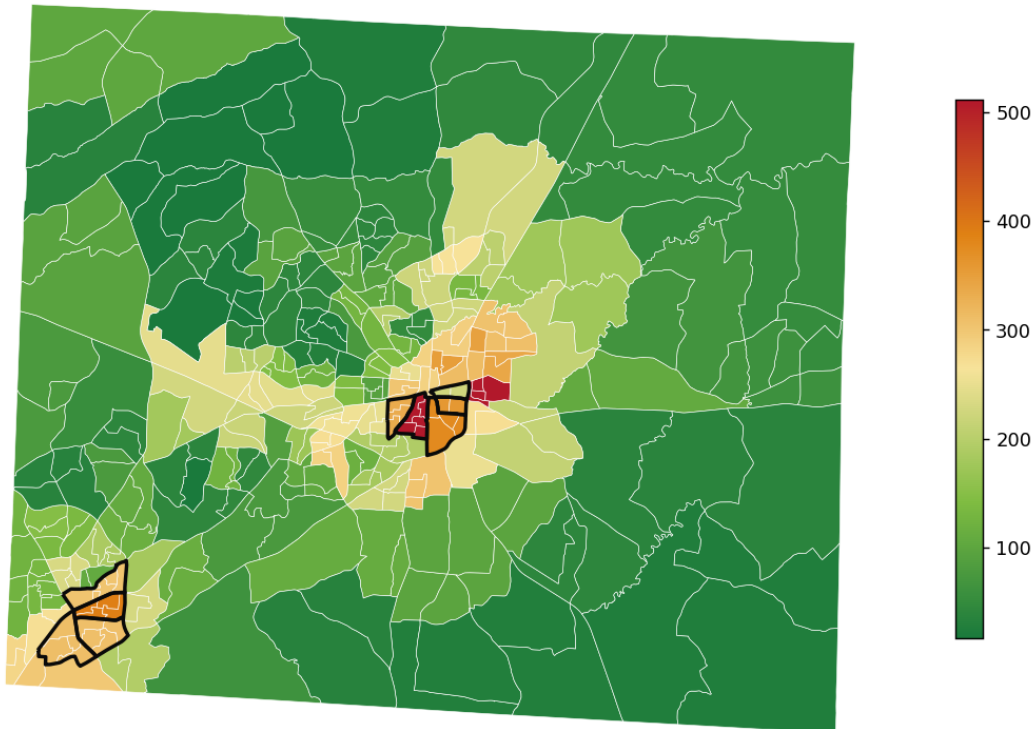
The county pattern. The county averages 161 on the national-100 scale, from essentially zero in the quietest block groups to 554 at the maximum; the map's red geography is the by-now-familiar core, which is the expected signature of eroded collective efficacy, thin commercial activity, and absentee ownership rather than a statement about the people who live there.

The pilot areas. The pilot tracts average 403 (Greensboro) and 327 (High Point), two to two and a half times the county mean and three to four times the national baseline; the Greensboro cluster contains the county's two highest tracts. Section 7's intervention logic reads this component strictly as exposure to be reduced through investment.

Evidence: selection score .671; VIF 3.46; factor loading .87; coverage 338 of 338; modeled, licensing condition standing.

Personal-crime risk index (modeled): Guilford County

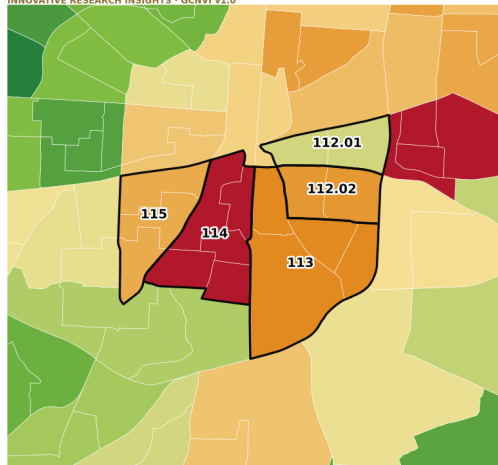
INNOVATIVE RESEARCH INSIGHTS · GCNVI v1.0



Green = direction of greater vitality · scaled p2-p98 · black outlines = priority tracts · hatched = no data · 2020 block groups
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Personal-crime risk index (modeled): Greensboro target area

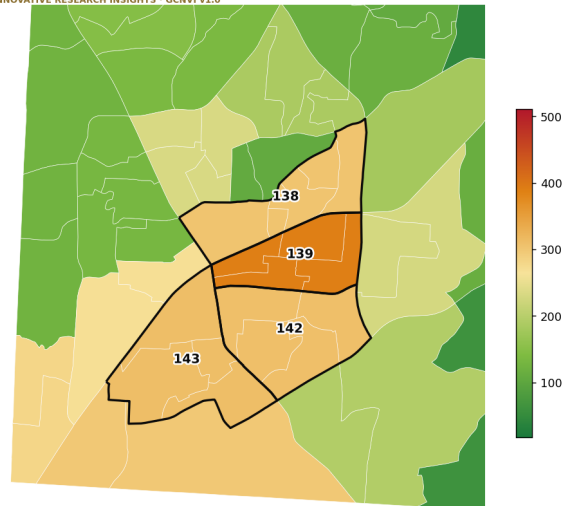
INNOVATIVE RESEARCH INSIGHTS · GCNVI v1.0



Green = direction of greater vitality · scaled p2-p98 · black outlines = priority tracts · hatched = no data · 2020 block groups
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Personal-crime risk index (modeled): High Point target area

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Green = direction of greater vitality · scaled p2-p98 · black outlines = priority tracts · hatched = no data · 2020 block groups
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Figure 12 Personal-Crime Risk Index

5.9 CENSUS SELF-RESPONSE (J8)

What it measures. The share of households that self-responded to the 2020 decennial census, from the Census Bureau’s response-rate archive, taken as the file’s most demanding measure of civic engagement and institutional trust. Unlike turnout, self-response has no partisan valence, no registration prerequisite, and no mobilization industry; it records whether a household, asked once by its government to be counted, answered.

Why it was selected. Among the civic candidates it is the strongest screener (selection score .703) and the broadest construct; the turnout measure, its principal rival, correlates with it at .72 and measures a narrower, registrant-conditioned behavior. Its loading of .91 ties for the highest in the index: where trust in institutions is intact, nearly everything else the index measures is intact with it.

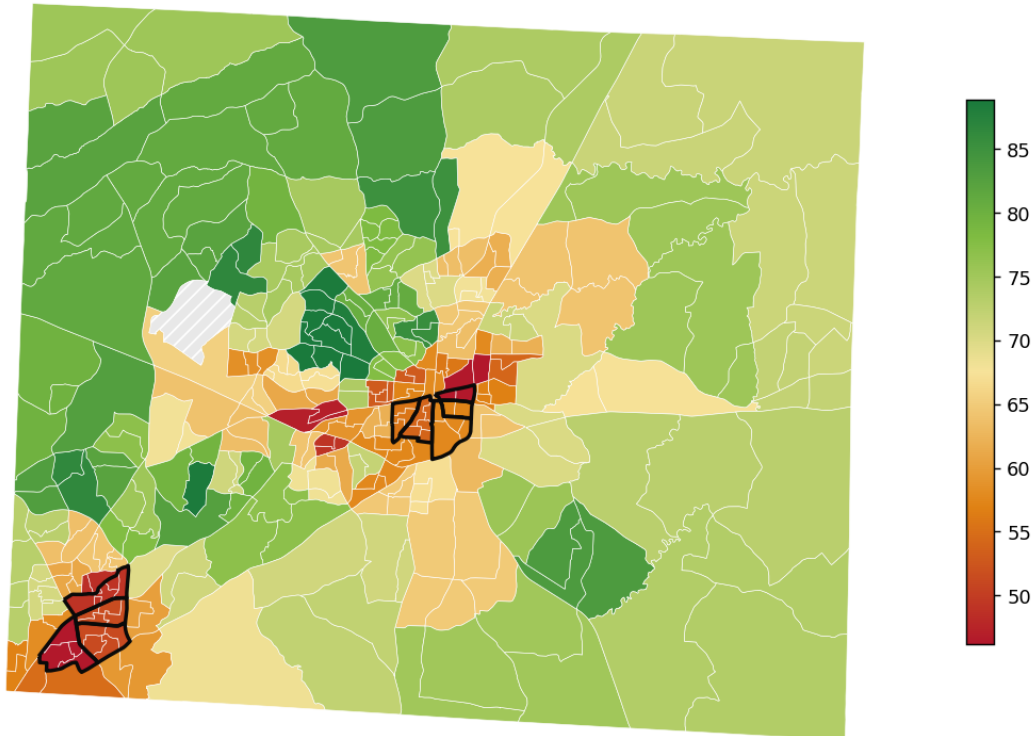
The county pattern. The county averaged 68 percent self-response, from 39 percent at the minimum to 91 at the maximum; the map’s gradient is unusually smooth, declining continuously from the periphery into the cores, which is what an erosion of institutional connection, rather than any single administrative boundary, looks like.

The pilot areas. The pilot clusters averaged 54 percent (Greensboro) and 49 percent (High Point), fourteen to nineteen points under the county; the undercount risk those figures imply is itself a mechanism of disadvantage, since census counts drive a decade of funding formulas. Every enumeration dollar the pilot areas lose is a measured consequence of the condition this component records.

Evidence: selection score .703; VIF 5.11; factor loading .91; coverage 337 of 338.

Census self-response (%): Guilford County

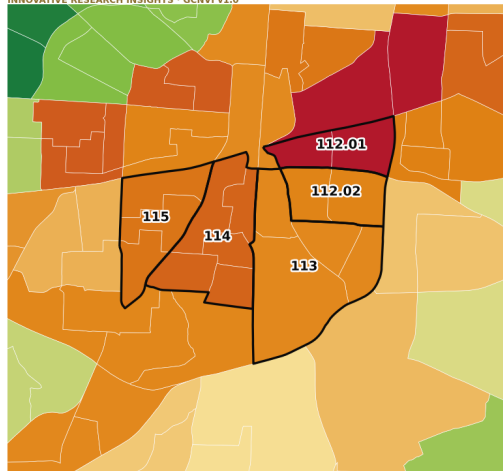
INNOVATIVE RESEARCH INSIGHTS · GCNVI v1.0



Green = direction of greater vitality · scaled p2-p98 · black outlines = priority tracts · hatched = no data · 2020 block groups
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Census self-response (%): Greensboro target area

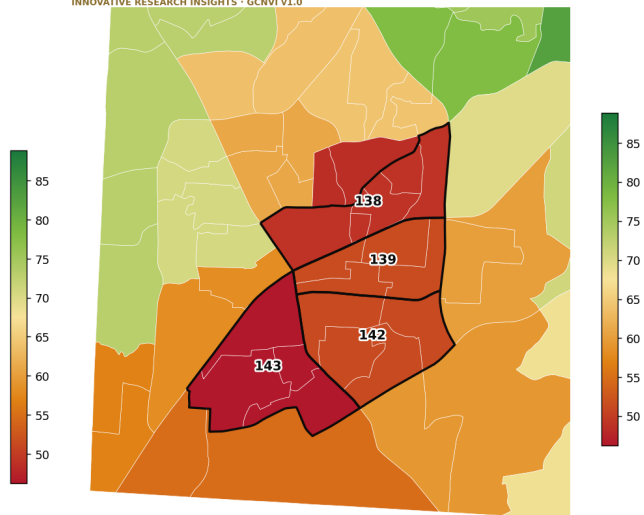
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Green = direction of greater vitality · scaled p2-p98 · black outlines = priority tracts · hatched = no data · 2020 block groups
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Census self-response (%): High Point target area

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Green = direction of greater vitality · scaled p2-p98 · black outlines = priority tracts · hatched = no data · 2020 block groups
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Figure 13 Census Self-Response Rate

5.10 SMALL-BUSINESS LENDING (K10B)

What it measures. The count of Community Reinvestment Act-reported small-business loan originations to firms with revenues under one million dollars, from the 2024 FFIEC disclosure files at the tract level. The measure is the file's most direct record of institutional capital reaching the smallest local firms, the businesses most likely to be neighborhood-owned and neighborhood-serving.

Why it was selected. It is the strongest qualifier in the economic-activity domain (selection score .361) and its construct, capital access, is the domain's core in the Newkirknomics framing that shaped Domain K; its VIF of 1.54 confirms it carries signal the housing and health components do not. Its selection score is the lowest of the ten, and the index accepts that deliberately: a vitality measure without an economic-capital component would be conceptually incomplete, and the component's independence is precisely its value.

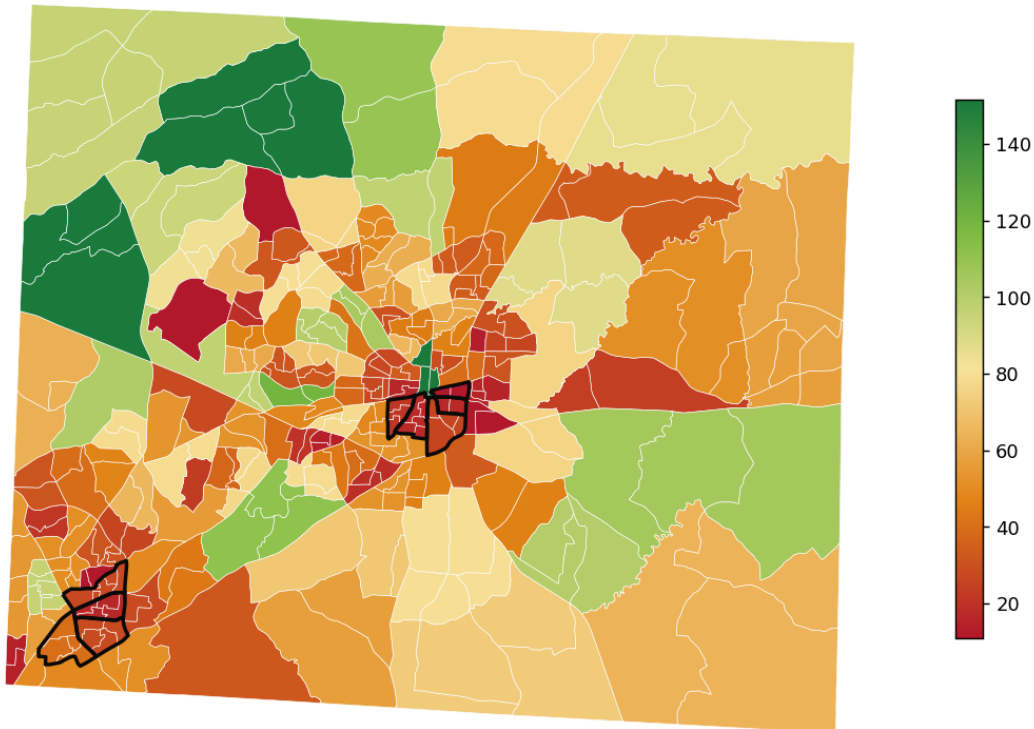
The county pattern. The county averages 56 small-firm originations per block group's tract, with a maximum of 178; the map shows lending following the commercial geography of the county's growth corridors, and thinning, though not vanishing, in the cores.

The pilot areas. The pilot tracts average 19 originations (Greensboro) and 28 (High Point), one third to one half of the county norm, in the very areas where establishment counts (Part I of the companion Atlas) show hundreds of operating firms. The gap between businesses present and capital delivered is the operational definition of the invisible economy this project was commissioned to make visible, and Section 7.3 builds its capital-access recommendations directly on this component.

Evidence: selection score .361; VIF 1.54; factor loading .64; coverage 338 of 338.

Small-business lending to firms under \$1M: Guilford County

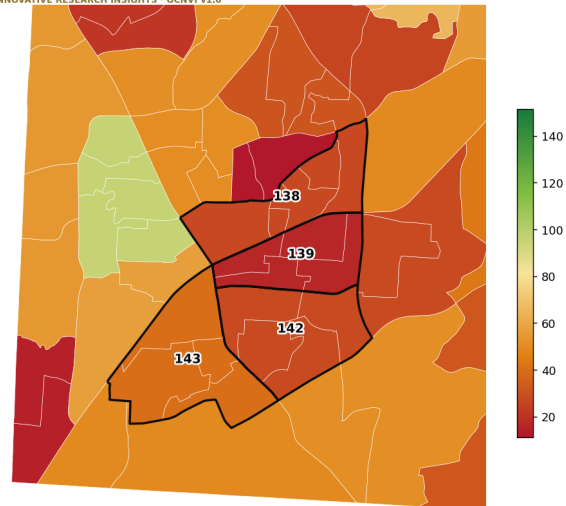
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Green = direction of greater vitality · scaled p2-p98 · black outlines = priority tracts · hatched = no data · 2020 block groups
Innovative Research Insights, LLC

Small-business lending to firms under \$1M: High Point target area

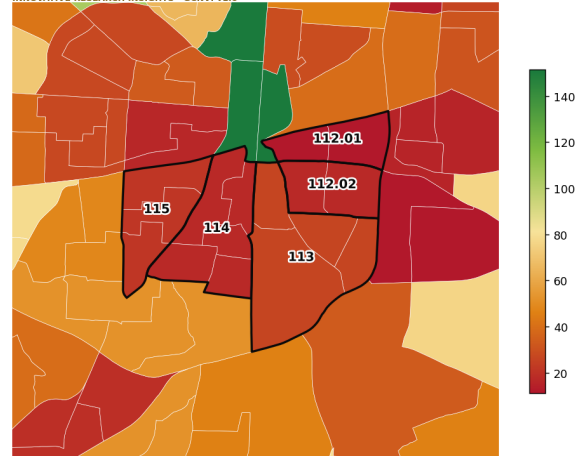
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Green = direction of greater vitality · scaled p2-p98 · black outlines = priority tracts · hatched = no data · 2020 block groups
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Small-business lending to firms under \$1M: Greensboro target area

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Green = direction of greater vitality · scaled p2-p98 · black outlines = priority tracts · hatched = no data · 2020 block groups
Innovative Research Insights, LLC

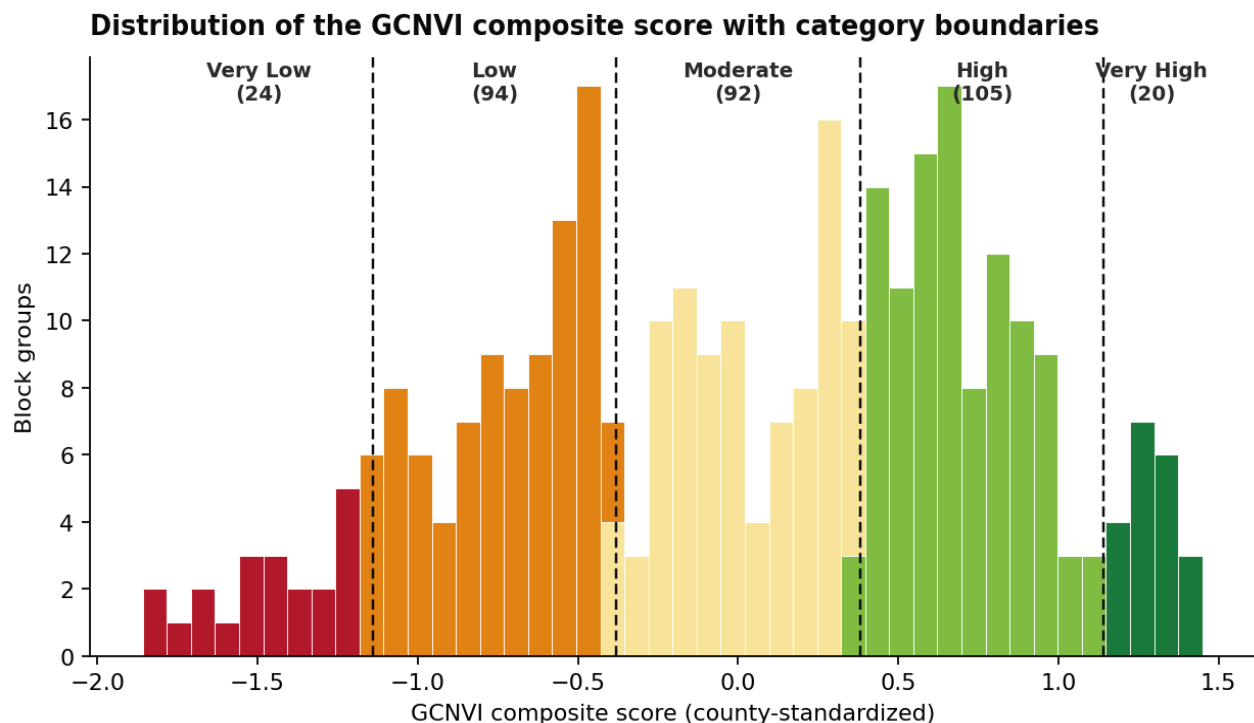
Figure 14 Small-Business Lending to Firms Under \$1m

6. THE COMPOSITE INDEX

6.1 CONSTRUCTION RECAP & SCORE DISTRIBUTION

The composite is the equal-weighted mean of the ten components of Section 5, each standardized on the county's 338 block groups and sign-aligned so that higher always means greater vitality; a block group is scored when at least nine of ten components are present, which 335 of 338 satisfy. The resulting score is centered at zero by construction and spans roughly -2.1 to +1.5 across the county, with a standard deviation of 0.76.

Figure 15 displays the distribution with the five category boundaries in place. Two features deserve note before any map is consulted. First, the distribution is single-peaked and modestly left-skewed: the county has a broad middle, a long tail of deeply disadvantaged block groups, and a shorter reach of advantage, which is what a century of concentrated disinvestment should produce arithmetically. Second, the category boundaries at one half and one and one half standard deviations from the mean are symmetric and mechanical; the categories contain 24, 94, 92, 105, and 20 block groups from Very Low to Very High not because any threshold was tuned to produce those counts but because that is where the county's neighborhoods fall.



335 scored block groups; boundaries at county mean $\pm$ 0.5 and $\pm$ 1.5 SD · Innovative Research Insights, LLC

Figure 15 Distribution of the composite score across the 335 scored block groups.

6.2 RELIABILITY & INTERNAL STRUCTURE

A composite is defensible only if its components cohere, and the index's ten cohere strongly. Internal-consistency reliability is high (Cronbach's $\alpha = .922$; McDonald's $\Omega = .935$), values ordinarily associated with purpose-built psychometric scales rather than assemblages of administrative measures. Principal-factor analysis extracts a single dominant dimension: all ten components load between .62 and .91 (Table 7), no rotation is required, and no second factor approaches interpretability. The index is, in the measurement sense, one thing; the shared dimension the components define is what this report calls neighborhood vitality.

Coherence has not been purchased with redundancy. The selection rules of Section 3.5 held every component's variance-inflation factor below 6.2 within the final set (Table 6.1), and the highest pairwise correlation among components is .78; each of the ten contributes variance the other nine do not. The combination, high shared loading with bounded inflation, is exactly the structure an equal-weighted composite wants: components that measure one construct from ten angles rather than one angle ten times.

Table 7 Psychometrics of the ten components

Component	Loading	VIF	Coverage
Social and emotional support	.91	6.14	337
Census self-response	.91	5.11	337
Personal-crime safety (modeled)	.87	3.46	338
Food access	.84	3.17	337
Median home value	.78	2.45	338
Homeownership	.74	2.15	335
Eviction judgments	.68	1.72	338
Birth outcomes	.65	1.59	328
Small-business lending	.64	1.54	338
Vehicle access	.62	1.53	335

6.3 CRITERION VALIDATION AND BENCHMARKS

The index was built to predict long-run opportunity, and it does. Across scored block groups the composite correlates at +.762 with the adult income rank of children raised in each block group, +.622 with their probability of reaching the top quintile, and +.873 with the Child Opportunity Index (all $p < .001$). Figure 16 displays the first of these relationships with the pilot-tract block groups highlighted: they cluster where the index says they should, in the joint lower-left of measured conditions and realized mobility. The employment criterion, per the finding of Section 3.2, is reported separately ($r = -.224$) and interpreted in Appendix J.

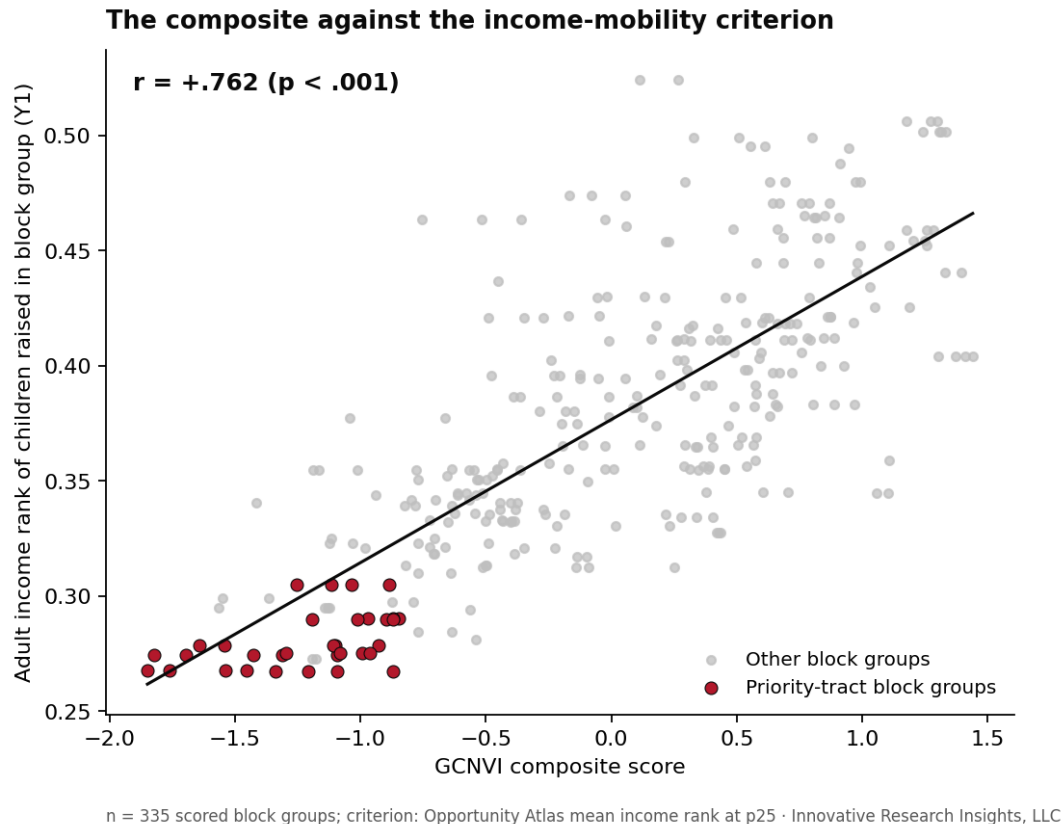


Figure 16 The composite against the income-mobility criterion

The proper test of a local index, however, is comparative: does it earn its place against the national instruments a county could adopt off the shelf? Table 8 reports the absolute correlations of the composite and five national benchmarks with the three aligned criteria, computed on the county's own block groups. The composite outperforms every national index on income mobility (.762 against .675 for the ADI, .736 for the SDI, .711 for the SVI) and matches the field on child opportunity, sitting within .03 of the SVI's .906. That residual edge for the SVI on Y4 deserves its footnote: the SVI and the Child Opportunity Index are both constructed largely from ACS demographic and socioeconomic inputs, so their agreement partly reflects shared ingredients rather than independent confirmation, a circularity the GCNVI deliberately avoids by excluding demographic composition and national composites alike. The practical claim is therefore precise: **on the county's own ground, a transparent ten-component index of locally actionable measures validates as well as or better than the national instruments, while telling the county exactly which movable conditions its score is made of.**

Table 8 Absolute Pearson correlations with the three aligned criteria, Guilford County block groups.

Index	Y1 income rank	Y2 top quintile	Y4 child opportunity
GCNVI (this report)	.762	.622	.873
Area Deprivation Index (standardized, 2022)	.675	.599	.843
Social Deprivation Index (2019)	.736	.603	.872
Social Vulnerability Index (2022)	.711	.651	.906
Distressed Communities Index (2019-2023)	.668	.561	.723
Racial-income polarization (ICE, benchmark)	.753	.618	.906

Table 9 completes the validation by regressing each criterion on all ten components jointly; a brief guide to reading it is warranted, because multiple regression answers a narrower question than validation does. Each coefficient reports a component's unique association with the criterion after the other nine are held fixed. With ten deliberately intercorrelated components, unique contributions concentrate in the measures that span the composite's common factor most broadly: social and emotional support dominates every aligned model (beta = .61 to .73), with home value, evictions, and food access carrying consistent secondary weight.

Several coefficients run small or negative for components whose bivariate validity is unambiguous (small-business lending most visibly); these are suppression effects, the routine arithmetic of correlated predictors, and they say nothing about component validity, which rests on the screening evidence of Section 3.4 and the loadings of Table 8. The models' joint fit is the point: the ten components together account for 73 percent of the variance in income mobility and 92 percent in child opportunity, while the employment model's near-null fit (R-squared = .096) confirms once more that Y3 benchmarks a different construct. The composite, not any single coefficient, is the instrument.

Table 9 OLS regression of the four criterion variables on the ten components

GCNVI component	Model 1: Income rank (Y1)	Model 2: Top-quintile probability (Y2)	Model 3: Employment at 30 (Y3)	Model 4: Child Opportunity Index (Y4)
Vehicle access (share without vehicle, reversed)	+0.016 (.668)	-0.040 (.393)	+0.010 (.883)	+0.013 (.529)
Homeownership rate	-0.104* (.017)	-0.067 (.229)	-0.156 (.050)	-0.008 (.757)
Median home value	+0.094* (.041)	+0.216*** (<.001)	-0.208* (.013)	+0.228*** (<.001)
Eviction judgment rate (reversed)	+0.118** (.002)	+0.141** (.004)	-0.047 (.507)	+0.016 (.466)
Social and emotional support (lack, reversed)	+0.609*** (<.001)	+0.617*** (<.001)	-0.005 (.970)	+0.727*** (<.001)
Healthy birthweight (LBW share, reversed)	+0.041 (.271)	+0.043 (.363)	-0.027 (.685)	-0.001 (.944)
Food access (LILA flag, reversed)	-0.183*** (<.001)	-0.160* (.017)	+0.030 (.757)	-0.035 (.232)
Personal-crime safety (risk index, reversed)	+0.243*** (<.001)	+0.167* (.017)	+0.027 (.786)	-0.077* (.012)
Census self-response rate	+0.128 (.058)	-0.060 (.485)	+0.143 (.247)	+0.153*** (<.001)
Small-business lending (under \$1M firms)	-0.138*** (<.001)	-0.198*** (<.001)	-0.145* (.037)	-0.020 (.339)
R squared	0.730	0.559	0.096	0.915
Adjusted R squared	0.721	0.545	0.067	0.912
df (regression, residual)	10, 315	10, 315	10, 315	10, 315
F	85.0***	40.0***	3.3***	339.5***

* p < .05; ** p < .01; *** p < .001. Intercepts are zero by construction under full standardization and are omitted. Model 3's near-null fit reflects the documented near-orthogonality of the employment criterion to the other three benchmarks and is reported for completeness, not validation.

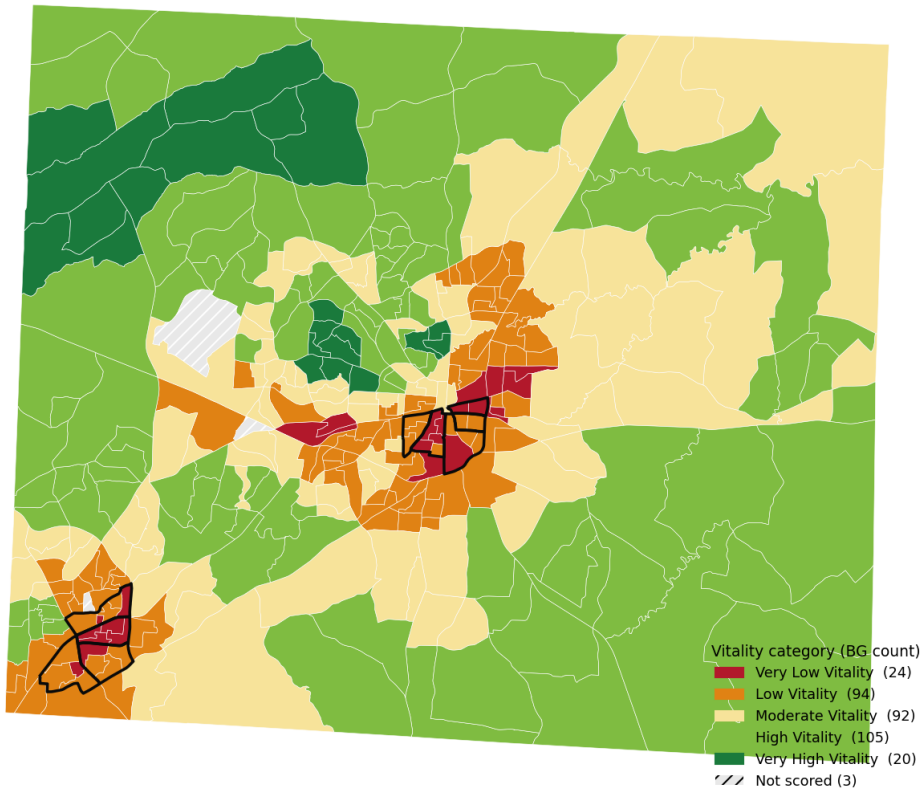
6.4 THE GEOGRAPHY OF VITALITY

Figure 17 presents the five-category classification for the county, and its geography states the report's historical argument cartographically. The Very High and High categories form a continuous arc across the county's north and west, the postwar suburban and exurban growth zone. The Very Low and Low categories concentrate in two compact formations: east and southeast Greensboro, and the south-central quarter of High Point, with a thinner band tracing the mill and rail corridor between them. The county's category map and its 1930s security-rating map (Section 2) are separated by ninety years and agree in outline; the spatial coincidence of designated disinvestment then and measured deficit now is the central empirical fact the index documents.

The additional two maps zoom the continuous score to the two pilot areas. Both show the same internal structure at closer range: scores rise continuously with distance from the historical core, and category boundaries cut through the pilot clusters' edges, where transition block groups (Moderate ratings adjacent to Very Low) mark exactly the frontier where displacement pressure and reinvestment meet. Section 7 returns to those edges.

Guilford County Neighborhood Vitality Index: Five-Category Classification

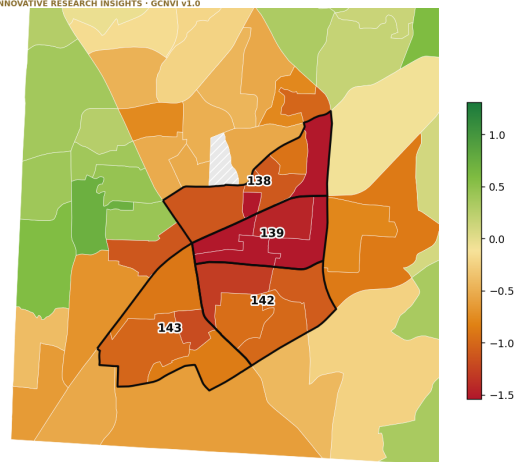
INNOVATIVE RESEARCH INSIGHTS · GCNVI v1.0 MODEL · 10 COMPONENTS · $\alpha = .92$



338 block groups (2020 frame) · categories = county mean $\pm$ 0.5 / 1.5 SD · black outlines = nine Coalition priority tracts
Components: vehicle access · homeownership · home value · evictions · social support · birth outcomes · food access · personal-crime risk · census self-response · small-business lending
Higher scores (green) = greater neighborhood vitality · Innovative Research Insights, LLC

GCNVI v1.0 Composite Score: High Point target area (tracts 138 to 143)

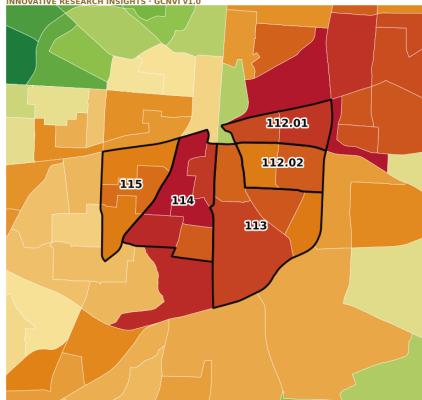
INNOVATIVE RESEARCH INSIGHTS · GCNVI v1.0



Green = greater vitality · scaled p2-p98 · black outlines = priority tracts · hatched = not scored · 2020 block groups
Innovative Research Insights, LLC

GCNVI v1.0 Composite Score: Greensboro target area (tracts 112.01 to 115)

INNOVATIVE RESEARCH INSIGHTS · GCNVI v1.0



Green = greater vitality · scaled p2-p98 · black outlines = priority tracts · hatched = not scored · 2020 block groups
Innovative Research Insights, LLC

Figure 17 The Five-Category Classification, Guilford County, Greensboro, and High Point Pilot Areas

6.5 PILOT-TRACT PROOF-OF-CONCEPT VALIDATION

The nine tracts selected in advance as the project’s pilot geography provide an external proof-of-concept test of the GCNVI. Because prior research, administrative data, archival records, and community knowledge had independently documented severe neighborhood conditions in these areas before the index was constructed, the pilot asks whether a countywide measurement system built under uniform, demographically neutral rules detects those conditions on its own. It does.

Table 10 reports the tract-level results: all nine fall in the Very Low or Low modal categories, with mean scores from -0.90 to -1.65, between 1.2 and 2.2 standard deviations below the county mean. Tract 139 in High Point, the Washington Street corridor’s southern half, records the lowest mean score in the county, with all four of its block groups rated Very Low. Of the 118 block groups countywide in the bottom two categories, 33 (28 percent) lie inside the nine pilot tracts, which contain roughly 8 percent of the county’s block groups. That concentration is evidence that the index detects, without consulting them, conditions that were documented independently before it existed. The result validates the pilot as a proof-of-concept geography; it does not validate the pilot selection as a program eligibility rule or confer priority for any public benefit, and the same methodology applies to every block group in the county.

Table 10 The nine pilot tracts on the composite: proof-of-concept results in the bottom two modal categories

Tract	Block groups	Mean score	Modal category	Very Low / Low count
Greensboro 112.01	2	-1.27	Very Low	2 / 0
Greensboro 112.02	2	-0.98	Low	0 / 2
Greensboro 113	4	-1.07	Low	1 / 3
Greensboro 114	5	-1.47	Very Low	4 / 1
Greensboro 115	3	-0.90	Low	0 / 3
High Point 138	5	-1.26	Low	2 / 3
High Point 139	4	-1.65	Very Low	4 / 0
High Point 142	4	-1.08	Low	1 / 3
High Point 143	4	-0.99	Low	1 / 3

Figure 18 profiles the two clusters component by component, and the profiles justify reading them separately. Both clusters sit below the county average on all ten components, but not identically: the Greensboro cluster’s deepest deficits are safety exposure and small-business lending, while the High Point cluster falls furthest on civic response, vehicle access, and social support. The two pilot clusters reach similar composite categories through different combinations of measured conditions, which is why the composite must be read together with its component profile rather than as a demographic or historical label; Section 7 takes up how those profiles can inform place-based decision support.

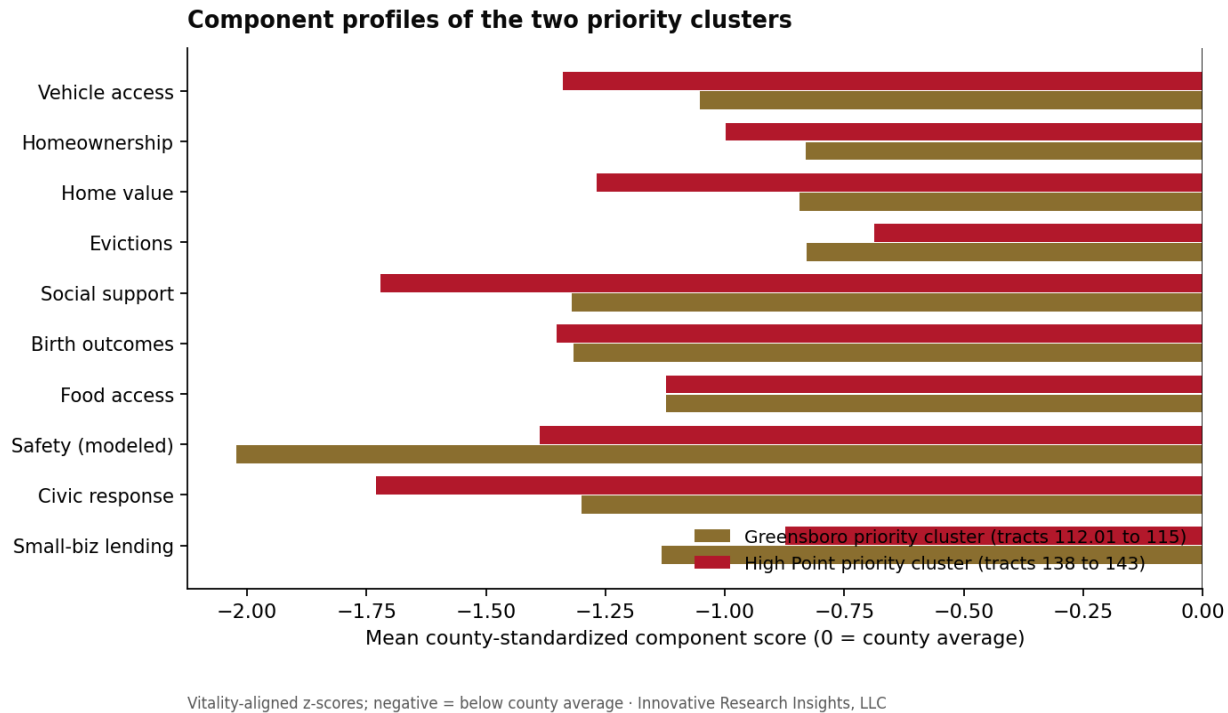


Figure 18 Component profiles of the two pilot clusters: mean vitality-aligned z-scores

6.6 READING BELOW THE COMPOSITE

The composite ranks but it does not diagnose. Two block groups can arrive at the same score by opposite routes, and two adjacent tracts in the same category can require different interventions entirely. The report closes its measurement argument with the case that proves the point, the two tracts of the Washington Street corridor, profiled at block-group resolution in Table 11.

Tracts 138 and 139 sit one boundary apart, share a commercial spine, and land one category apart on the index (Low at -1.26; Very Low at -1.65). The block-group table shows how differently they get there. Tract 139 is uniform deprivation: every block group poor, renter-dominated, and low-valued, with the county's most extreme cell (block group 139.1: median income \$9,436, SNAP receipt 84 percent, ownership 11 percent). Tract 138 is micro-segregated: block groups at 62.7 and 52.8 percent poverty adjoin block groups at 3.9 and 2.6 percent, a heavily Latino renter block group beside an entirely Black homeowner one, so that the tract's averages describe no block group in it. **A program aimed at "tract 138" would miss both of its realities at once; the index's block-group resolution, and the discipline of reading components beneath scores, exist precisely for this.**

The same discipline guards against a subtler error: mistaking symptoms of depletion for signals of health. Within the corridor, measured unemployment is low where labor-force detachment is severe; rents are low where the stock is subsidized and degraded; business-address vacancy is low where the commercial base is thin rather than thriving. Each datum, read alone and naively, flatters; read within the component structure, each indexes the deficit it disguises. Section 7.4 elevates this into a general caution for the index's users.

Table 11 Block-group profile of the Washington Street corridor tracts

Block group	Population	Median HH income	Poverty %	SNAP %	Owner-occupied %	Median value
138.1	1,256	\$23,167	62.7	58.3	28.3	\$107,700
138.2	1,227	\$38,989	3.9	12.1	46.2	\$121,400
138.3	582	\$62,893	2.6	16.7	46.9	\$145,700
138.4	27	NA	70.4	100.0	47.1	\$101,000
138.5	1,177	\$42,987	52.8	36.8	54.5	\$117,650
139.1	1,068	\$9,436	62.1	84.4	10.5	\$94,700
139.2	854	\$23,369	30.6	65.6	4.3	\$68,400
139.3	531	\$12,405	39.9	34.7	50.8	\$81,550
139.4	1,734	\$35,927	62.6	44.8	22.0	\$100,800
County BG median	1,460	\$64,135	11.1	10.3	62.0	\$246,300

7. IMPLICATIONS FOR PLACE-BASED DECISION SUPPORT

7.1 FROM MEASUREMENT TO DECISION SUPPORT

The index was developed as a neighborhood measurement and geographic decision-support instrument, and its design anticipates three distinct uses that should be kept distinct. First, as geographic decision support: the five categories rest on documented, symmetric, reproducible thresholds and can identify the places where measured conditions warrant further review, planning, or consideration for place-based programs, with the published scored dataset as the record of that measurement. The Very Low and Low categories are analytical classifications, not automatic eligibility designations. Any grant, expenditure, contract, service, incentive, or other public benefit must operate under the lawful authority and program-specific eligibility criteria of the responsible agency, and protected characteristics, which are not part of the GCNVI classification, should not be introduced into subsequent eligibility or allocation decisions except where applicable law expressly requires it. Second, as a diagnostic instrument: because the composite is built from ten interpretable components, any block group's classification can be decomposed into the specific conditions that produce it, so that program planning can respond to documented conditions rather than assumption; Section 7.3 systematizes this reading. Third, as a monitoring baseline: the components refresh on annual or better cadence, and the meaningful question three years from now is not simply which block groups fall into the Low and Very Low categories, which this report settles for the baseline year, but whether the places that received lawful, program-specific interventions have begun to move, which only a maintained baseline can answer. What the index is not, and cannot honestly be presented as, is a set of causal effect estimates: it identifies where conditions are most severe and which conditions those are, and it ranks no intervention's efficacy. The pairing of its diagnoses with the intervention evidence base is the work of this section; the recommended actions are the work of the next, and the program decisions that follow remain the work of responsible agencies and partners under applicable law.

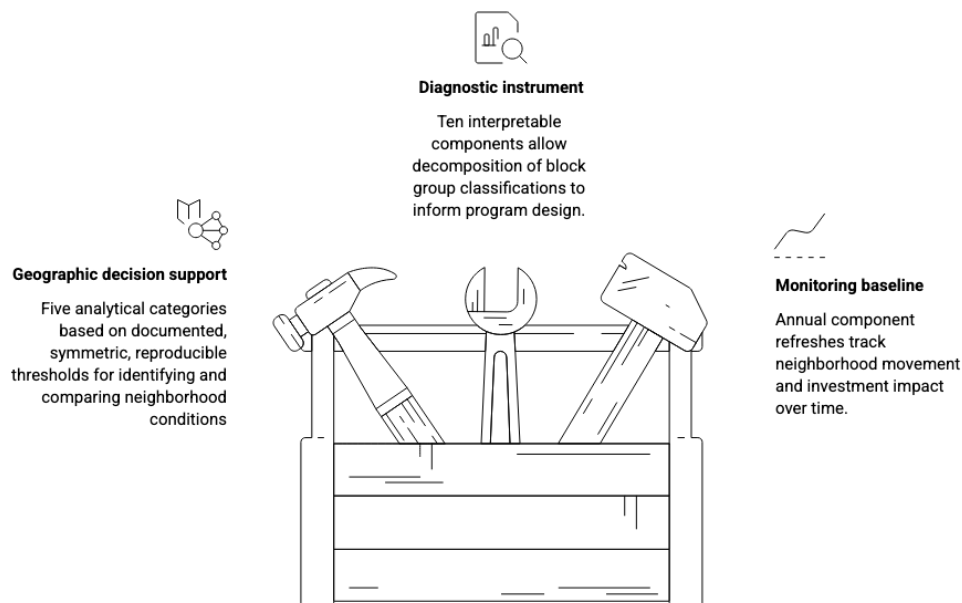


Figure 19 Uses for the GCNVI Scores

7.2 THE LOW & VERY LOW GEOGRAPHY

The bottom two categories comprise 118 block groups housing 166,512 residents, just under a third of the county's population, of whom 30,996 live in the 24 Very Low block groups. Their geography is compact: 74 of the 118 lie in Greensboro's tract range (110,931 residents, concentrated east and southeast of the center), 31 in High Point's (34,704 residents, in the south-central quarter), and 13 in the mill-corridor and rural-fringe remainder. Table 12 summarizes the inventory; the complete block-group listing accompanies the scored dataset (Appendix F).

Table 12 Inventory of the bottom-two-category geography.

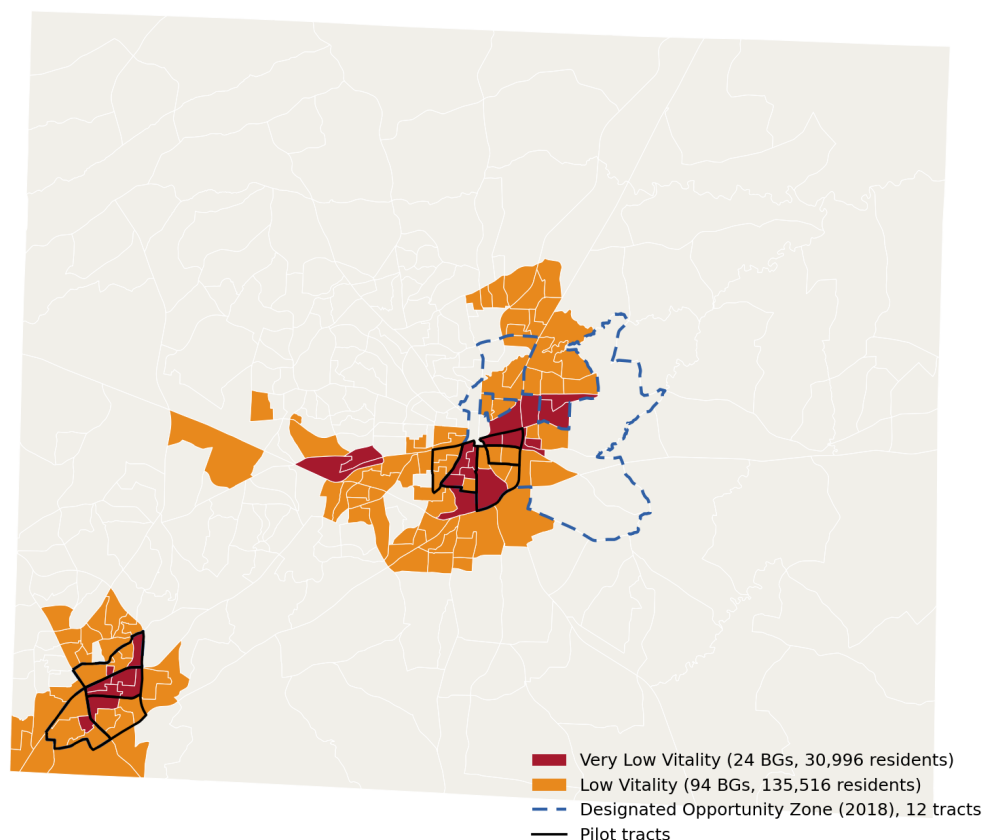
Geography	Very Low BGs	Low BGs	Combined population	Share inside a designated Opportunity Zone
Greensboro core tracts	18	56	110,931	31% of BGs
High Point core tracts	6	25	34,704	32% of BGs
Remainder of county	0	13	20,877	8% of BGs
County total	24	94	166,512	28% of BGs

Figure 20 lays this geography against the county's existing federal investment designations, the twelve census tracts designated as Qualified Opportunity Zones in 2018, and the comparison carries the section's sharpest practical finding. The designations and the measured need overlap far less than either the program's purpose or the county's planning assumptions would suggest: only 28 percent of the bottom-two block groups, containing 38,407 of their 166,512 residents, lie inside a designated zone, so that 128,105 residents of measured Low and Very Low Vitality neighborhoods sit outside the county's principal federal capital-attraction geography.

The misalignment reaches the pilot tracts themselves. Six of the nine are designated (112.01, 112.02, 113, and 114 in Greensboro; 138 and 143 in High Point); three are not, and the exclusions are not marginal cases: tract 139, the lowest-vitality tract in the county by this report's measurement, carries no designation, and neither do 115 or 142. The point is not retrospective criticism of the 2018 selections, which predate this measurement; it is prospective. Where the County or its partners are authorized to participate in developing geographic criteria for state, federal, or local programs (program applications, CDBG target areas, New Markets Tax Credit strategies, any successor to the Opportunity Zone program), the index supplies one transparent empirical input for evaluating how well existing or proposed geographies correspond to measured neighborhood conditions, and Figure 20 shows exactly where the current lines and the measured conditions diverge. The governing law and program requirements of each funding source, including federal eligibility, nondiscrimination, procurement, and geographic rules, remain controlling; Figure 20 identifies correspondence and non-correspondence and does not itself establish a replacement eligibility geography.

The Low and Very Low Vitality geography against the federal Opportunity Zone designations

INNOVATIVE RESEARCH INSIGHTS · GCNVI v1.0 · MAP 7.1



2018 ZOZ designations (US Treasury/CDFI Fund) mapped to 2020 tract successors · only 28% of bottom-two block groups (38,407 of 166,512 residents) lie inside a designated zone; tract 139, the county's lowest-vitality tract, is not designated · Innovative Research Insights, LLC

Figure 20 Low and Very Low Vitality vs. Federal Opportunity Zone Designation

7.3 COMPONENT-GUIDED INTERVENTION LOGIC

The composite identifies where measured conditions warrant closer examination; the components identify which conditions produce that classification. Because each of the ten measures a distinct, movable condition, a depressed reading on any one points to a family of interventions with an evidence base of its own, and a block group's component profile can inform preliminary consideration of those intervention families. Table 13 states the crosswalk; three profile patterns from Section 6 illustrate its use. The crosswalk is illustrative rather than prescriptive: it does not establish program eligibility, require a particular expenditure, or substitute for program-specific evidence, legal authority, community engagement, feasibility analysis, or evaluation of alternatives.

Table 13 The component-to-intervention crosswalk.

Component (depressed)	What it signals	Intervention family
Vehicle access	Labor market shrunk to walking and transit distance	Transit service and frequency; vanpool and guaranteed-ride programs; vehicle-access lending; employer shuttle partnerships
Homeownership	Blocked entry to wealth-building tenure; absentee conversion pressure	Down-payment assistance; heirs-property and title-clearing services; community land trusts for permanent affordability; anti-displacement property-tax relief
Median home value	Suppressed equity; appraisal and reinvestment gap	Owner-occupied rehabilitation grants; appraisal review, homeowner education, fair-housing compliance, and reinvestment strategies; targeted infill and acquisition-rehab; repair-first code enforcement
Eviction judgments	Active displacement; instability cascade into schools, health, employment	Eviction diversion and mediation; right-to-counsel; emergency rental assistance; landlord engagement and habitability enforcement
Social and emotional support	Depleted social infrastructure; eroded collective efficacy	Community centers and third places; congregation-anchored programming; community health workers; violence-interruption staffing
Birth outcomes	Chronic maternal stress; prenatal access failure	Group prenatal care; community doula and home-visiting programs; Medicaid enrollment outreach; the county's early-childhood infrastructure (Ready Ready) extended corridor-first
Food access	Income-and-distance nutrition constraint	Grocery attraction and cooperative development; mobile markets; SNAP incentive programs; corner-store conversion
Personal-crime safety	Exposure produced by disinvestment, not resident propensity	Blight remediation and greening; street lighting and environmental design; violence interruption; place management, never enforcement-first saturation
Census self-response	Institutional distrust; undercount and funding loss	Trusted-messenger civic infrastructure; community navigators; complete-count investment ahead of 2030; registration and participation drives through anchor institutions
Small-business lending	Capital rationing to the smallest firms; invisible economy	CDFI capitalization and presence; CRA advocacy with the county's lenders; character-based and revenue-based lending products; resident-led investment funds; lawful small-business procurement outreach, supplier-readiness assistance, competitive-access initiatives, and anchor-institution procurement strategies, implemented consistent with applicable procurement law

The profiles of Section 6.5 show the crosswalk's point. The Greensboro cluster's deepest deficits are **safety exposure and small-firm lending**: its component profile points first to place management, blight remediation, and CDFI capacity. The High Point cluster falls furthest on **civic response, vehicle access, and social support**: its profile points first to trusted-messenger infrastructure, mobility, and the congregation-anchored social fabric that Section 2's history shows the corridor already possesses. And within a single tract, the block-group resolution matters again as tract 138's internal variation (Section 6.6) demonstrates the value of block-group-level decision support, since tract-wide averages would obscure the materially different conditions of 138.1 and 138.5 and their neighbors.

7.4 WHAT THE EVIDENCE CAUTIONS AGAINST

Four misuses of this index are foreseeable, and naming them is part of delivering it. First, single-indicator targeting: no component alone, however strong, is sufficient evidence for program eligibility or resource allocation, because the corridor evidence shows individual indicators inverting their apparent meaning (low unemployment masking labor-force detachment; low rent masking degraded subsidized stock; low business vacancy masking a thin commercial base). The composite exists because the components discipline one another. Second, misreading exposure as propensity: the safety component measures the victimization risk that disinvestment imposes on residents, and any use of it to justify enforcement saturation rather than investment inverts both the evidence and the project's purpose. Third, demographic proxying: the index was built, at deliberate cost in explained variance, without demographic composition, and any downstream re-introduction of race as a targeting variable, directly or through the intentional use of geographic, statistical, or program criteria as substitutes for protected characteristics, would undo that design and expose the county's programs to exactly the harms the demographic-neutrality safeguards exist to prevent. Geographic targeting should rest on independently meaningful conditions of place and the lawful criteria of the program involved, never on an area's racial or ethnic composition. Fourth, gentrification blindness: a rising composite score is the goal only if the residents who endured the low score are present to experience the high one. The index measures places, and places can improve by exchanging their populations; Section 8's housing recommendations therefore pair every appreciation-oriented investment with an anti-displacement instrument, and the monitoring plan (Section 8.3) tracks eviction, ownership, and absentee measures precisely to catch improvement of the wrong kind. Any anti-displacement assistance or preference must likewise be structured under lawful, program-specific criteria such as income, housing tenure, property condition, or cost burden rather than protected demographic characteristics.

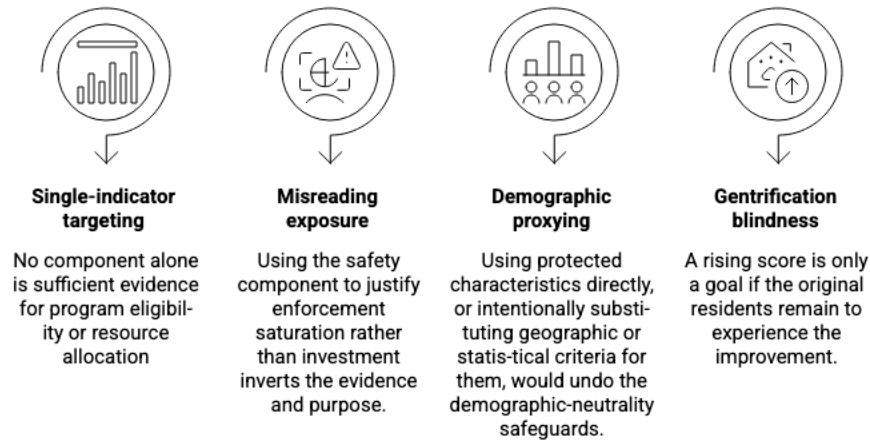


Figure 21 Four Potential Misuses of the GCNVI

8. RECOMMENDATIONS

The recommendations that follow are organized as eight recommended place-based actions for consideration in the areas where the GCNVI identifies Low or Very Low conditions, followed by the sequencing logic, the governance of the instrument itself, and the measurement agenda. Each recommendation names the components it is intended to move and the section that evidences it. They are presented for consideration by the County and relevant partners, subject to each institution's lawful authority, program-specific criteria, funding-source requirements, and independent decision-making; none is within any single institution's gift, which is the reason the Coalition exists.

8.1 RECOMMENDED PLACE-BASED ACTIONS FOR LOW & VERY LOW VITALITY AREAS

1. Stabilize residents in place: home repair, heirs-property resolution, and anti-displacement protection. The bottom-two geography contains the county's oldest housing stock, with corridor median years built ranging from 1944 to 1951, and bears its heaviest eviction burden (Sections 5.3, 5.4, 6.6). A priority policy family for further program development includes owner-occupied home repair scaled to aging and pre-1960 housing conditions; lead-safe remediation where environmental evidence indicates exposure risk; heirs-property and title-clearing assistance so households can qualify for repair investments and retain future equity; expanded eviction-prevention and diversion strategies, including evaluation of legal-assistance models; and lawful property-tax or other anti-displacement protections where rising values may place long-term residents at risk. Any specific program, eligibility standard, tax-relief mechanism, or legal-service model would require separate review of statutory authority, funding sources, program criteria, and responsible-agency capacity, with eligibility based on independently measured housing conditions and household need rather than GCNVI classification alone. Moves: evictions, home value, homeownership, birth outcomes.

2. Build community development capacity: CDCs for the corridors. Sections 2 and 5 together show neighborhoods rich in institutions but lacking sufficient community-development vehicles to carry out acquisition, rehabilitation, redevelopment, and long-term stewardship. The County, municipalities,

philanthropy, and other partners should evaluate whether community development corporations or comparable locally accountable entities could strengthen the pilot corridors—East Greensboro, Washington Street, and Southside in High Point—as well as other areas demonstrating similar measured conditions. Where appropriate, these entities could be supported through capitalization, technical assistance, development-pipeline support, and governance structures that ensure meaningful resident participation and accountability to the neighborhoods they serve. Any public funding, organizational structure, acquisition authority, or governance requirement would need to be established through the responsible institution’s lawful process, program criteria, and capacity. Moves: homeownership, home value, lending, social support.

3. Expand permanently affordable and mixed-income housing, including community land trust models. Where housing programs are authorized and independently measured conditions demonstrate affordability, ownership, or displacement pressure, decision-makers should evaluate permanent-affordability strategies such as community land trusts, deed-restricted housing, shared-equity models, and appropriately designed mixed-income development. These approaches can help preserve the long-term value of public investment for residents and future households while providing an alternative to the investor-acquisition dynamics documented in Section 2.2. Mixed-income development may be particularly appropriate in transition areas where it can broaden housing choice and support neighborhood integration without contributing to displacement. The appropriate tool, geography, affordability period, and eligibility standards should be determined through the governing housing program, funding source, adopted plans, and applicable law rather than by GCNVI classification alone. Moves: homeownership, home value, evictions.

4. Finance local ownership: capital for the invisible economy. Section 5.10’s finding, wherein hundreds of operating establishments alongside roughly one third of the county’s small-firm lending points to a need to strengthen capital access where measured lending gaps are documented, beginning with the pilot corridors and extending to other areas showing similar conditions. The County, municipalities, financial institutions, CDFIs, philanthropy, and anchor institutions should evaluate expanded small-business lending and investment tools; engage deposit-holding banks around Community Reinvestment Act obligations and opportunities using the index as supporting evidence alongside lawful, institution-specific criteria; assess locally governed or resident-led investment vehicles where legally and financially feasible; and expand small-business procurement outreach, supplier-readiness assistance, competitive-access initiatives, and anchor-institution purchasing opportunities consistent with applicable law and institutional authority. The Small Business Entrepreneurship Department’s role should center on entrepreneurship, business formation and survival, supplier readiness, capital access, technical assistance, and lawful procurement pathways rather than protected-class procurement preferences or set-asides. Moves: lending, home value, social support.

5. Invest cradle-to-career, corridor-first. The birth-outcomes component (Section 5.6) and the preschool-enrollment counter-signal in Section 5’s education evidence point to both significant need and existing community assets. The GCNVI should be used as one planning input for sequencing and evaluating prenatal, early-childhood, school-age, and career-pathway investments, beginning with the

pilot corridors where independently measured conditions indicate need and extending to other areas with similar conditions. Illustrative strategies include group prenatal care, doula support, home visiting, expanded pre-K access, Ready for School Ready for Life interventions, and coordinated school-age and career pathways such as shift_ed and the GCS Guilford Guarantee, where the responsible institutions determine that these approaches fit program-specific need, eligibility, service capacity, adopted plans, and lawful authority. “Corridor-first” should therefore be understood as a sequencing preference informed by the index rather than as an eligibility rule. The index’s criterion variables capture many of the longer-term outcomes shaped by childhood neighborhood conditions and can help track whether those conditions and outcomes improve over time, while not by themselves establishing the causal effect of any particular intervention. Moves: birth outcomes and, in the criterion sense, the broader set of long-term opportunity outcomes.

6. Repair infrastructure and environmental conditions. The Low and Very Low geographies include areas with documented infrastructure and environmental burdens, including older housing and pipes, heat exposure, lead risk, and elevated pollution indicators (Sections 5.6, 6.6, and the companion Atlas plates). Capital planning should be informed by the GCNVI and Atlas as an evidence source alongside engineering assessments, adopted capital plans, statutory authority, feasibility, funding requirements, and other lawful criteria. Sidewalks, lighting, tree canopy, stormwater improvements, lead-line replacement, and blight remediation should be considered where the relevant conditions are independently documented, with Very Low areas serving as a reasonable starting point for review rather than as an automatic funding rule. Evidence on greening, remediation, and the built environment can also inform strategies to improve safety and health. Final prioritization should remain with the responsible capital-planning and infrastructure processes. Moves: safety, home value, birth outcomes.

7. Integrate services where the components converge. Where multiple adverse conditions converge, the GCNVI component profile can help agencies and community partners identify places where coordinated or co-located service delivery may improve access and reduce fragmentation. Illustrative services include health care, benefits navigation, housing counseling, workforce assistance, and legal aid, delivered where appropriate through trusted community institutions, including congregations and neighborhood-based nonprofits. Program eligibility and service delivery should remain governed by each program’s lawful criteria, responsible-agency authority, service capacity, and adopted procedures. A direct-cash or guaranteed-income demonstration may also warrant consideration as a complement to service integration where legally and financially feasible, but any such initiative would require separate legal, budgetary, funding-source, program-authority, and equal-protection review before implementation, with participant eligibility based on lawful program-specific criteria such as income and residency. The GCNVI may inform site selection, planning, and evaluation and provide a baseline for tracking change, but it should not by itself establish eligibility geography or entitlement to a public benefit. Moves: social support, self-response, vehicle access, evictions.

8. Review the policy architecture and alignment. Section 2 documents how earlier public instruments including racially explicit zoning, HOLC security maps, clearance plans, and highway routing all shaped the geography reflected in current neighborhood conditions. The historical record supports a systematic review of present-day zoning, code enforcement, capital allocation, appraisal practices,

and related policies to assess how they interact with measured conditions and adopted policy objectives in Low and Very Low areas. Where the County or partner institutions are authorized to consider geographic criteria, the GCNVI may serve as one transparent empirical input for evaluating whether discretionary investment geographies align with documented need. More broadly, the index should be adopted publicly and consistently as a common evidence base for place-based planning, evaluation, and monitoring, while remaining one input rather than an automatic allocation rule. Statutory authority, adopted criteria, funding requirements, program rules, and professional judgment should continue to govern each specific decision. **Moves:** the production function of all ten.

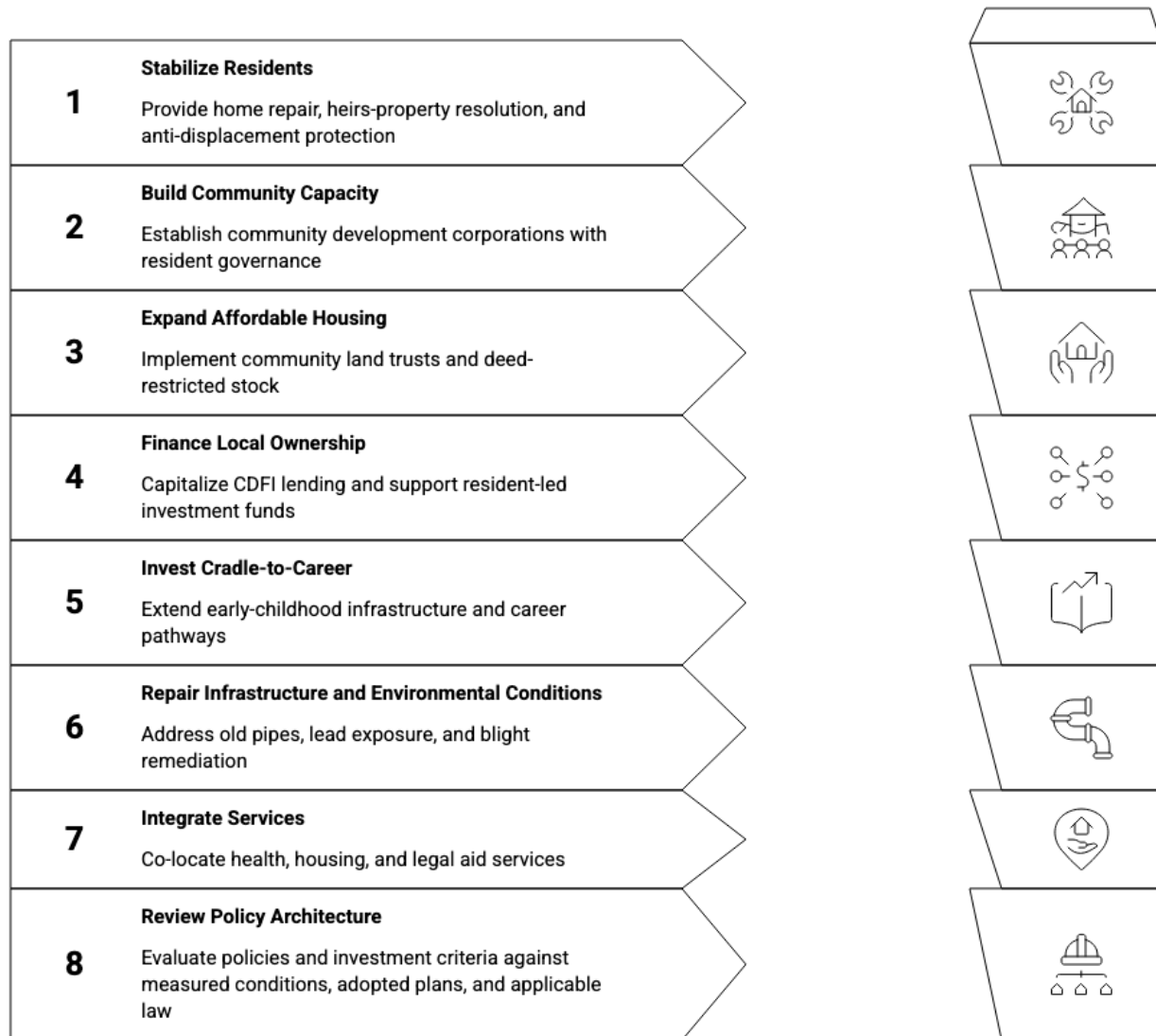


Figure 22 Recommended Place-Based Actions for Areas with Low or Very Low Vitality

8.2 SEQUENCING & THE INSTITUTIONAL LEDGER

The corridor analysis of Section 6.6 generalizes into the agenda’s sequencing principle: build through the assets that are present toward the levers that are absent. The standing assets of the areas classified Low or Very Low are institutional, congregations at the county’s highest density, early-childhood enrollment at county rates, an operating establishment base, and the resident knowledge documented across a decade of fieldwork; its absent levers are financial capacity, ownership opportunity, ownership equity, capital access, and economic attachment. The sequence follows: near-term commitments (years one and two) run through the institutions that already hold trust, repair, diversion, navigation, enrollment, procurement, while the capacity commitments (CDCs, land trusts, investment funds) are chartered and capitalized; medium-term commitments convert trust into assets, ownership pipelines, corridor acquisition, business lending at scale; and the long-horizon commitments, cradle-to-career and policy repair, run continuously underneath. Two disciplines govern throughout: anti-displacement instruments precede appreciation-oriented investment wherever both appear, and every action reports against the index’s own components, so that the ledger of the next review is the same instrument as the diagnosis of this one. The sequencing is illustrative rather than self-executing; each participating institution remains responsible for its own legal authority, program design, eligibility, procurement, funding, and implementation decisions.

8.3 GOVERNANCE, PLATFORM, & REFRESH

The GCNVI is an instrument, and instruments require custody. Three arrangements are recommended. First, platform: the scored dataset, component layers, and map services should transfer to the County GIS Department on the ArcGIS platform already planned for the Community Data Atlas, with the replication specification (Appendix H) as the technical handoff and the public dashboard carrying the how-to-read guidance of this report’s front matter, including the demographic-neutrality, county-relative, non-causal, and non-eligibility safeguards of Sections 1.3, 3.6, and 7. Second, publication rules: suppressed cells remain suppressed on every public surface; and no public view should present block-group scores without their category context and caveats, and public release remains subject to source-specific licensing, privacy, suppression, security, and data-use restrictions. Third, refresh: an annual re-run against the year’s ACS, CRA, HMDA, court, code, and vital-records releases, on the versioned pipeline this project delivers, with the 2030 census self-response measure incorporated when it arrives and each vintage published alongside, never over, its predecessors, so that change is measurable and the baseline of record (this report forming v1.0) is permanent.

8.4 THE MEASUREMENT AGENDA

Five improvements would materially strengthen v2, in priority order. The North Carolina Secretary of State registry integration remains first: it can add business-formation and entity-registration information that may strengthen business analysis and assist entity matching, but Secretary of State records do not by themselves establish current legal title, beneficial ownership, or a definitive investor-ownership classification. Any ownership analysis should retain source provenance and the limits of inference from public records. Second, the High Point permits feed, completing two-city administrative

coverage on the investment side as v10.5 completed it on enforcement. Third, where producer rules permit, research access to the suppressed small-population life-expectancy estimates for the eight affected tracts, including the parent geography associated with pilot tract 112, wherein producer suppression rules, privacy protections, minimum-cell requirements, and data-use restrictions remain controlling. Fourth, a current-vintage measured-crime refresh with the two police departments and sheriff, retiring the UNCG modeled component and its license condition together. Fifth, and of a different kind: a co-produced community-asset layer, measures of the corridor's institutional wealth defined with its residents, so that the next edition of the Atlas records not only the conditions imposed on these neighborhoods but the capacities they have built despite them, on their own terms. That layer should enter as supplemental descriptive evidence first; any future inclusion in the scored index must satisfy the same selection, validation, demographic-neutrality, reproducibility, data-quality, and governance rules applied to the existing components. That final item returns the work to where Section 2.2 began it, with Du Bois: the map of conditions is the beginning of the argument, never the whole of the people.

9. CONCLUSION

This report set out to measure neighborhood vitality in Guilford County and found that the present-day pattern cannot be read apart from a documented history of planning, housing, transportation, credit, and investment decisions. The ten components that survived empirical selection, home values and ownership, evictions and food access, birth outcomes and social support, safety exposure, civic trust, and small-business capital, are distinct measures that load on a single common dimension of neighborhood conditions while retaining separate substantive meaning. Their spatial pattern overlaps in important ways with historic planning and disinvestment geographies documented in Section 2. The 1935 High Point building zone map with a racial category printed in its legend, the 1937 Greensboro security map with its red grades laid over the named Black neighborhoods, the clearance plans and highway routings that followed: the correspondence between those historical geographies and the Very Low and Low block groups of the 2026 vitality map is the report's central descriptive finding. The theoretical tradition from Du Bois through Wilson, Sampson, Sharkey, and Chetty supplies its interpretation: neighborhood conditions are produced and sustained by cumulative public and private decisions, and conditions that were produced can be changed. The GCNVI itself does not establish that any particular historical policy is the proximate cause of any individual contemporary score; what the evidence supports is the use of the index as a transparent, demographically neutral instrument for describing current conditions, monitoring change, and informing lawful place-based decision support.

The nine pilot tracts, selected in advance, provide a proof-of-concept validation geography. All fall without exception in the Low and Very Low categories, consistent with conditions documented before index construction through prior research, administrative data, archival records, and community knowledge. This convergence strengthens confidence that the GCNVI is detecting genuine present-day neighborhood conditions. It does not convert pilot selection into a program eligibility designation, confer priority for a public benefit, or make historical or current demographic composition a scoring criterion. The same methodology applies to all scored block groups countywide. The convergence also supports continued use of resident and institutional knowledge as an independent evidence stream alongside quantitative measurement.

Neither community knowledge nor statistical measurement by itself determines public action. The GCNVI gives Guilford County a transparent, replicable, criterion-validated, block-group-resolution instrument governed by demographic-neutrality and measurement safeguards, aligned with a companion Atlas, scored dataset, versioned pipeline, and recommended place-based actions for consideration. Any subsequent program eligibility, expenditure, contract, procurement action, service, incentive, direct-cash program, or other public benefit must be authorized and governed by applicable law, funding requirements, adopted policies, and the lawful program-specific criteria of the responsible institution. The GCNVI provides a common evidence base for understanding neighborhood conditions, coordinating departmental and community questions, and evaluating change over time. Historical and demographic information may inform interpretation; protected characteristics do not determine GCNVI scores or classifications. The instrument is delivered as a foundation for lawful decision support, Esri/GIS implementation, land-record enhancement, entrepreneurship strategy, interdepartmental collaboration, and continued measurement.

REFERENCES

- Aaronson, D., Hartley, D., & Mazumder, B. (2021). The effects of the 1930s HOLC “redlining” maps. *American Economic Journal: Economic Policy*, 13(4), 355–392. <https://doi.org/10.1257/pol.20190414>
- Acevedo-Garcia, D., McArdle, N., Hardy, E. F., Crisan, U. I., Romano, B., Norris, D., Baek, M., & Reece, J. (2014). The Child Opportunity Index: Improving collaboration between community development and public health. *Health Affairs*, 33(11), 1948–1957. <https://doi.org/10.1377/hlthaff.2014.0679>
- Acevedo-Garcia, D., Noelke, C., McArdle, N., Sofer, N., Hardy, E. F., Weiner, M., Baek, M., Huntington, N., Huber, R., & Reece, J. (2020). Racial and ethnic inequities in children’s neighborhoods: Evidence from the new Child Opportunity Index 2.0. *Health Affairs*, 39(10), 1693–1701. <https://doi.org/10.1377/hlthaff.2020.00735>
- Baylin, J. (1968). [Study of Greensboro’s segregated residential development; full citation to be confirmed against the From Disparity to Parity source list.]
- Branas, C. C., South, E., Kondo, M. C., Hohl, B. C., Bourgois, P., Wiebe, D. J., & MacDonald, J. M. (2018). Citywide cluster randomized trial to restore blighted vacant land and its effects on violence, crime, and fear. *Proceedings of the National Academy of Sciences*, 115(12), 2946–2951. <https://doi.org/10.1073/pnas.1718503115>
- Butler, D. C., Petterson, S., Phillips, R. L., & Bazemore, A. W. (2013). Measures of social deprivation that predict health care access and need within a rational area of primary care service delivery. *Health Services Research*, 48(2), 539–559. <https://doi.org/10.1111/j.1475-6773.2012.01449.x>
- Carstairs, V., & Morris, R. (1989). Deprivation: Explaining differences in mortality between Scotland and England and Wales. *British Medical Journal*, 299(6704), 886–889. <https://doi.org/10.1136/bmj.299.6704.886>
- Chetty, R., Friedman, J. N., Hendren, N., Jones, M. R., & Porter, S. R. (2018). The Opportunity Atlas: Mapping the childhood roots of social mobility (NBER Working Paper No. 25147). National Bureau of Economic Research. <https://doi.org/10.3386/w25147>
- Chetty, R., Jackson, M. O., Kuchler, T., Stroebel, J., Hendren, N., Fluegge, R. B., et al. (2022a). Social capital I: Measurement and associations with economic mobility. *Nature*, 608(7921), 108–121. <https://doi.org/10.1038/s41586-022-04996-4>
- Chetty, R., Jackson, M. O., Kuchler, T., Stroebel, J., Hendren, N., Fluegge, R. B., et al. (2022b). Social capital II: Determinants of economic connectedness. *Nature*, 608(7921), 122–134. <https://doi.org/10.1038/s41586-022-04997-3>

- Connolly, N. D. B. (2014). *A world more concrete: Real estate and the remaking of Jim Crow South Florida*. University of Chicago Press.
- Du Bois, W. E. B. (1899). *The Philadelphia Negro: A social study*. University of Pennsylvania Press.
- Faber, J. W. (2020). We built this: Consequences of New Deal era intervention in America's racial geography. *American Sociological Review*, 85(5), 739–775.
<https://doi.org/10.1177/0003122420948464>
- Fishback, P., Rose, J., Snowden, K. A., & Storrs, T. (2022). New evidence on redlining by federal housing programs in the 1930s. *Journal of Urban Economics*, 141, 103462.
<https://doi.org/10.1016/j.jue.2022.103462>
- Flanagan, B. E., Gregory, E. W., Hallisey, E. J., Heitgerd, J. L., & Lewis, B. (2011). A social vulnerability index for disaster management. *Journal of Homeland Security and Emergency Management*, 8(1), Article 3. <https://doi.org/10.2202/1547-7355.1792>
- Innovative Research Insights. (2025). *Washington Street needs assessment*. Innovative Research Insights, LLC.
- Innovative Research Insights. (2026). *An atlas of neighborhood conditions in Guilford County, North Carolina*. Innovative Research Insights, LLC.
- Jackson, K. T. (1985). *Crabgrass frontier: The suburbanization of the United States*. Oxford University Press.
- Kind, A. J. H., & Buckingham, W. R. (2018). Making neighborhood-disadvantage metrics accessible: The Neighborhood Atlas. *New England Journal of Medicine*, 378(26), 2456–2458.
<https://doi.org/10.1056/NEJMp1802313>
- Krieger, N., Waterman, P. D., Spasojevic, J., Li, W., Maduro, G., & Van Wye, G. (2016). Public health monitoring of privilege and deprivation with the Index of Concentration at the Extremes. *American Journal of Public Health*, 106(2), 256–263.
<https://doi.org/10.2105/AJPH.2015.302955>
- Massey, D. S., & Denton, N. A. (1993). *American apartheid: Segregation and the making of the underclass*. Harvard University Press.
- Messer, L. C., Laraia, B. A., Kaufman, J. S., Eyster, J., Holzman, C., Culhane, J., Elo, I., Burke, J. G., & O'Campo, P. (2006). The development of a standardized neighborhood deprivation index. *Journal of Urban Health*, 83(6), 1041–1062. <https://doi.org/10.1007/s11524-006-9094-x>
- Michney, T. M., & Winling, L. (2020). New perspectives on New Deal housing policy: Explicating and mapping HOLC loans to African Americans. *Journal of Urban History*, 46(1), 150–180.
<https://doi.org/10.1177/0096144218819429>

- Mitchell, B., & Franco, J. (2018). HOLC “redlining” maps: The persistent structure of segregation and economic inequality. National Community Reinvestment Coalition.
- Nardone, A., Casey, J. A., Morello-Frosch, R., Mujahid, M., Balmes, J. R., & Thakur, N. (2020). Associations between historical residential redlining and current age-adjusted rates of emergency department visits due to asthma across eight cities in California: An ecological study. *The Lancet Planetary Health*, 4(1), e24–e31. [https://doi.org/10.1016/S2542-5196\(19\)30241-4](https://doi.org/10.1016/S2542-5196(19)30241-4)
- National Community Reinvestment Coalition. (2022). Redlining and neighborhood health. National Community Reinvestment Coalition.
- Raymond, E. L., Miller, B., McKinney, M., & Braun, J. (2021). Gentrifying Atlanta: Investor purchases of rental housing, evictions, and the displacement of Black residents. *Housing Policy Debate*, 31(3–5), 818–834. <https://doi.org/10.1080/10511482.2021.1887318>
- Rothstein, R. (2017). *The color of law: A forgotten history of how our government segregated America*. Liveright.
- Sampson, R. J. (2012). *Great American city: Chicago and the enduring neighborhood effect*. University of Chicago Press.
- Sharkey, P. (2013). *Stuck in place: Urban neighborhoods and the end of progress toward racial equality*. University of Chicago Press.
- Sills, S. J. (2021). Enduring impact of racialized planning in Southern cities (EPU Policy Brief). UNCG Center for Housing and Community Studies. <https://chcs.uncg.edu/wp-content/uploads/2022/01/EPU-White-Paper-Dr.-Sills-FINAL-1.pdf>
- Sills, S. J. (2023). From disparity to parity: Unveiling the impact of racial planning and segregation on High Point’s Black community and the roadmap to resolution. Innovative Research Insights, LLC.
- Sills, S. J. (2026a). Concentrated disadvantage and the geography of opportunity: Framing document for the Guilford County Data Atlas. Innovative Research Insights, LLC.
- Sills, S. J. (2026b). Neighborhood indicators and indices: A methodological literature review for the Guilford County Data Atlas. Innovative Research Insights, LLC.
- Singh, G. K. (2003). Area deprivation and widening inequalities in US mortality, 1969–1998. *American Journal of Public Health*, 93(7), 1137–1143. <https://doi.org/10.2105/AJPH.93.7.1137>
- Smith, T., Noble, M., Noble, S., Wright, G., McLennan, D., & Plunkett, E. (2015). *The English indices of deprivation 2015*. Department for Communities and Local Government.
- Taylor, K.-Y. (2019). *Race for profit: How banks and the real estate industry undermined Black homeownership*. University of North Carolina Press.

Townsend, P., Phillimore, P., & Beattie, A. (1988). *Health and deprivation: Inequality and the North*. Croom Helm.

Wilson, W. J. (1987). *The truly disadvantaged: The inner city, the underclass, and public policy*. University of Chicago Press.

ARCHIVAL & LEGAL SOURCES

Buchanan v. Warley, 245 U.S. 60 (1917).

City Planning and Zoning Commission, City of High Point. (1935). Building zone map of the City of High Point, Guilford County, North Carolina (Technical Advisory Corporation, consulting engineers; E. S. Draper, collaborating; hand-colored by H. V. Koonts, January 1935) [Map].

Home Owners' Loan Corporation, Division of Research and Statistics. (1937). Residential security map of Greensboro, North Carolina (base map April 1936; grades issued June 2, 1937) [Map]. digital scan via Mapping Inequality, University of Richmond Digital Scholarship Lab, <https://dsl.richmond.edu/panorama/redlining>

Shelley v. Kraemer, 334 U.S. 1 (1948).

Village of Euclid v. Ambler Realty Co., 272 U.S. 365 (1926).

DATA SOURCES & TECHNICAL DOCUMENTATION

Centers for Disease Control and Prevention. (2024). *PLACES: Local data for better health, 2024 release* [Data set and methodology]. U.S. Department of Health and Human Services. <https://www.cdc.gov/places>

Centers for Disease Control and Prevention / Agency for Toxic Substances and Disease Registry. (2022). *CDC/ATSDR Social Vulnerability Index 2022* [Data set and documentation]. <https://www.atsdr.cdc.gov/placeandhealth/svi>

Economic Innovation Group. (2024). *Distressed Communities Index, 2019–2023* [Data set and methodology]. <https://eig.org/dci>

Guilford County Health Department. (2026). *Tract-level birth outcomes (2022–2023) and period life expectancy (2009–2020)*, Guilford County [Administrative data delivery from North Carolina State Center for Health Statistics vital records].

Noelke, C., McArdle, N., DeVoe, B., Leonardos, M., Lu, Y., Li, M., Baek, M., & Acevedo-Garcia, D. (2024). *Child Opportunity Index 3.0 technical documentation*. [diversitydatakids.org](https://www.diversitydatakids.org), Brandeis University. <https://www.diversitydatakids.org>

North Carolina State Board of Elections. (2026). *Statewide voter registration and voter history files (ncvoter41, ncvhis41)* [Data sets]. <https://dl.ncsbe.gov>

- Opportunity Insights. (2022). The Opportunity Atlas: Neighborhood-level estimates of children's outcomes in adulthood [Data set]. <https://www.opportunityatlas.org>
- Overture Maps Foundation. (2026). Overture Places, June 2026 release [Data set]. <https://overturemaps.org>
- Public Environmental Data Partners. (2025). EPA EJScreen 2024 national block group file [Preservation copy]. Harvard Dataverse. <https://doi.org/10.7910/DVN/OES2R7>
- U.S. Census Bureau. (2024). American Community Survey 2019–2023 five-year estimates [Data set]. <https://api.census.gov>
- U.S. Census Bureau. (2021). 2020 Census self-response rate archive [Data set].
- U.S. Department of Agriculture, Economic Research Service. (2021). Food Access Research Atlas: Documentation [Data set and methodology]. <https://www.ers.usda.gov/data-products/food-access-research-atlas>
- U.S. Department of Housing and Urban Development / U.S. Postal Service. (2026). Aggregated USPS administrative data on address vacancies, Q1 2026 [Data set]. <https://www.huduser.gov>
- U.S. Department of the Treasury, CDFI Fund. (2018). Designated Qualified Opportunity Zones [Data set]. <https://www.cdfifund.gov>

APPENDIX A: COMPLETE VARIABLE CATALOG

All 218 catalog variables of the GCNVI v10.6 master file: measure, source, geography and vintage, and condensed derivation notes. Variables marked ‡ are unmapped in the companion Atlas (empty or constant).

Variable	Measure	Source · geography · vintage	Derivation notes
A1_est	Median household income	U.S. Census Bureau, ACS 5-Year 2019–2023 · 2020 BG (tract-imputed where BG-suppressed)	ACS table B19013; direct estimate or computed share
A2_est	Median HH income; county ratio	U.S. Census Bureau, ACS 5-Year 2019–2023 · 2020 BG (tract-imputed where BG-suppressed)	ACS table B19013; direct estimate or computed share
A3_est	Poverty rate (all ages)	U.S. Census Bureau, ACS 5-Year 2019–2023 · 2020 BG (tract-imputed where BG-suppressed)	ACS table B17001; direct estimate or computed share
A4_est	Deep poverty (<50% FPL)	U.S. Census Bureau, ACS 5-Year 2019–2023 · 2020 BG (tract-imputed where BG-suppressed)	ACS table C17002; direct estimate or computed share
A5_est	Child poverty rate	U.S. Census Bureau, ACS 5-Year 2019–2023 · 2020 BG (tract-imputed where BG-suppressed)	ACS table B17006; direct estimate or computed share
A6_est	SNAP receipt	U.S. Census Bureau, ACS 5-Year 2019–2023 · 2020 BG (tract-imputed where BG-suppressed)	ACS table B22003; direct estimate or computed share
A7_est	Public assistance income	U.S. Census Bureau, ACS 5-Year 2019–2023 · 2020 BG (tract-imputed where BG-suppressed)	ACS table B19057; direct estimate or computed share
A8_est	Prime-age employment ratio (25–54)	U.S. Census Bureau, ACS 5-Year	ACS table B23001; direct estimate or computed share

		2019–2023 · 2020 BG (tract-imputed where BG- suppressed)	
A9_est	Unemployment rate (16+)	U.S. Census Bureau, ACS 5-Year 2019–2023 · 2020 BG (tract-imputed where BG- suppressed)	ACS table B23025; direct estimate or computed share
A10_est	Not in labor force (25– 64)	U.S. Census Bureau, ACS 5-Year 2019–2023 · 2020 BG (tract-imputed where BG- suppressed)	ACS table B23001; direct estimate or computed share
A11_est	Single-parent HH with children	U.S. Census Bureau, ACS 5-Year 2019–2023 · 2020 BG (tract-imputed where BG- suppressed)	ACS table B11003; direct estimate or computed share
A12_est	Households with no vehicle	U.S. Census Bureau, ACS 5-Year 2019–2023 · 2020 BG (tract-imputed where BG- suppressed)	ACS table B25044; direct estimate or computed share
A13_est	Gini coefficient (income inequality)	U.S. Census Bureau, ACS 5-Year 2019–2023 · 2020 tract → BG (Gini; BG MOE too high)	ACS table B19083; direct estimate or computed share
B1_est	Less than HS attainment (25+)	U.S. Census Bureau, ACS 5-Year 2019–2023 · 2020 BG (tract-imputed where BG- suppressed)	ACS table B15003; direct estimate or computed share
B2_est	HS equivalent or higher (25+)	U.S. Census Bureau, ACS 5-Year 2019–2023 · 2020 BG (tract-imputed where BG- suppressed)	ACS table B15003; direct estimate or computed share
B3_est	Bachelor's degree or higher (25+)	U.S. Census Bureau, ACS 5-Year 2019–2023 · 2020 BG (tract-imputed where BG- suppressed)	ACS table B15003; direct estimate or computed share
B6_est	Preschool enrollment (age 3–4)	U.S. Census Bureau, ACS 5-Year 2019–2023 · 2020 BG (tract-imputed	ACS table B14003; direct estimate or computed share

		where BG-suppressed)	
B8_est	Limited-English-speaking HH	U.S. Census Bureau, ACS 5-Year 2019–2023 · 2020 BG (tract-imputed where BG-suppressed)	ACS table C16002; direct estimate or computed share
C1_est	% Black or African American	U.S. Census Bureau, ACS 5-Year 2019–2023 · 2020 BG (tract-imputed where BG-suppressed)	ACS table B02001; direct estimate or computed share
C2_est	% Hispanic or Latino	U.S. Census Bureau, ACS 5-Year 2019–2023 · 2020 BG (tract-imputed where BG-suppressed)	ACS table B03003; direct estimate or computed share
C3_est	% Foreign-born	U.S. Census Bureau, ACS 5-Year 2019–2023 · 2020 BG (tract-imputed where BG-suppressed)	ACS table B05002; direct estimate or computed share
C4_est	Age dependency ratio	U.S. Census Bureau, ACS 5-Year 2019–2023 · 2020 BG (tract-imputed where BG-suppressed)	ACS table B01001; direct estimate or computed share
C5_est	ICE (Race-Income)	U.S. Census Bureau, ACS 5-Year 2019–2023 · 2020 BG (tract-imputed where BG-suppressed)	ACS table B19001A–I; direct estimate or computed share
C6_est	% with a disability	U.S. Census Bureau, ACS 5-Year 2019–2023 · 2020 BG (tract-imputed where BG-suppressed)	ACS table B18101; direct estimate or computed share
C7_est	Population density	U.S. Census Bureau, ACS 5-Year 2019–2023 · 2020 BG (tract-imputed where BG-suppressed)	ACS table B01003 + TIGER; direct estimate or computed share
D1_est	Homeownership rate	U.S. Census Bureau, ACS 5-Year 2019–2023 · 2020 BG (tract-imputed	ACS table B25003; direct estimate or computed share

		where BG-suppressed)	
D2_est	Housing vacancy rate	U.S. Census Bureau, ACS 5-Year 2019–2023 · 2020 BG (tract-imputed where BG-suppressed)	ACS table B25002; direct estimate or computed share
D3_est	Vacant for sale (not seasonal)	U.S. Census Bureau, ACS 5-Year 2019–2023 · 2020 BG (tract-imputed where BG-suppressed)	ACS table B25004; direct estimate or computed share
D4_est	Renter-occupied share	U.S. Census Bureau, ACS 5-Year 2019–2023 · 2020 BG (tract-imputed where BG-suppressed)	ACS table B25003; direct estimate or computed share
D5_est	Median home value	U.S. Census Bureau, ACS 5-Year 2019–2023 · 2020 BG (tract-imputed where BG-suppressed)	ACS table B25077; direct estimate or computed share
D6_est	Median gross rent	U.S. Census Bureau, ACS 5-Year 2019–2023 · 2020 BG (tract-imputed where BG-suppressed)	ACS table B25064; direct estimate or computed share
D7_est	Renter cost burden ≥30%	U.S. Census Bureau, ACS 5-Year 2019–2023 · 2020 BG (tract-imputed where BG-suppressed)	ACS table B25070; direct estimate or computed share
D8_est	Severe renter cost burden ≥50%	U.S. Census Bureau, ACS 5-Year 2019–2023 · 2020 BG (tract-imputed where BG-suppressed)	ACS table B25070; direct estimate or computed share
D9_est	Owner cost burden ≥30% (SMOC)	U.S. Census Bureau, ACS 5-Year 2019–2023 · 2020 BG (tract-imputed where BG-suppressed)	ACS table B25091; direct estimate or computed share
D10_est	Overcrowding (>1.5 per room)	U.S. Census Bureau, ACS 5-Year 2019–2023 · 2020 BG (tract-imputed	ACS table B25014; direct estimate or computed share

		where BG-suppressed)	
D11_est	Pre-1950 housing stock	U.S. Census Bureau, ACS 5-Year 2019–2023 · 2020 BG (tract-imputed where BG-suppressed)	ACS table B25034; direct estimate or computed share
D12_est	Mobile-home share	U.S. Census Bureau, ACS 5-Year 2019–2023 · 2020 BG (tract-imputed where BG-suppressed)	ACS table B25024; direct estimate or computed share
D13_est	Lacking complete plumbing	U.S. Census Bureau, ACS 5-Year 2019–2023 · 2020 BG (tract-imputed where BG-suppressed)	ACS table B25049; direct estimate or computed share
D14_est	Moved in past year	U.S. Census Bureau, ACS 5-Year 2019–2023 · 2020 BG (tract-imputed where BG-suppressed)	ACS table B07003 → B25038 / B07001; direct estimate or computed share
D15_est	SF-detached / MF ratio	U.S. Census Bureau, ACS 5-Year 2019–2023 · 2020 BG (tract-imputed where BG-suppressed)	ACS table B25024; direct estimate or computed share
G6a_est ‡	No health insurance (all ages)	U.S. Census Bureau, ACS 5-Year 2019–2023 · 2020 BG (tract-imputed where BG-suppressed)	ACS table B27010; direct estimate or computed share
M1_est	Broadband subscription	U.S. Census Bureau, ACS 5-Year 2019–2023 · 2020 BG (tract-imputed where BG-suppressed)	ACS table B28002; direct estimate or computed share
M2_est	Computer ownership	U.S. Census Bureau, ACS 5-Year 2019–2023 · 2020 BG (tract-imputed where BG-suppressed)	ACS table B28001; direct estimate or computed share
M4_est	Mobile-only internet HH	U.S. Census Bureau, ACS 5-Year 2019–2023 · 2020 BG (tract-imputed	ACS table B28002 / B28003; direct estimate or computed share

		where BG-suppressed)	
C7_density_per_sqmi	Population density (per sq mi)	Census TIGERweb (land area) × ACS pop · 2020 BG	ACS population ÷ TIGER ALAND
Y1	Household income rank, children of p25 parents (kfr_pooled_po...	Opportunity Atlas (Chetty et al.) · 2010 tract → 2020 tract (OI crosswalk) → BG broadcast	Criterion; tract value broadcast to member BGs
Y1_late	OA v2 late-cohort income rank (sensitivity)	Opportunity Atlas (Chetty et al.) · 2010 tract → 2020 tract (OI crosswalk) → BG broadcast	Criterion; tract value broadcast to member BGs
Y2	P(top-quintile income), p25 parents (kfr_top20_pooled_pool ed_...	Opportunity Atlas (Chetty et al.) · 2010 tract → 2020 tract (OI crosswalk) → BG broadcast	Criterion; tract value broadcast to member BGs
Y3	Employment rate at 30+, p25 parents (working_pooled_poole d_p2...	Opportunity Atlas (Chetty et al.) · 2010 tract → 2020 tract (OI crosswalk) → BG broadcast	Criterion; tract value broadcast to member BGs
Y4	Child Opportunity Index 3.0 composite (national z-score, 2023)	diversitydatakids.org COI 3.0 · 2020 tract → BG broadcast	Criterion; z_COI_nat year=2023
PLACES_ACCESS2	No health insurance, adults 18–64 (crude prev., %)	CDC PLACES (BRFSS small-area estimates) · 2020 tract → BG broadcast	Direct tract join
PLACES_ARTHRITIS	Arthritis prevalence (crude prev., %)	CDC PLACES (BRFSS small-area estimates) · 2020 tract → BG broadcast	Direct tract join
PLACES_BINGE	Binge drinking (crude prev., %)	CDC PLACES (BRFSS small-area estimates) · 2020 tract → BG broadcast	Direct tract join
PLACES_BPHIGH	High blood pressure (crude prev., %)	CDC PLACES (BRFSS small-area estimates) · 2020 tract → BG broadcast	Direct tract join
PLACES_BPMED	BP medication adherence (crude prev., %)	CDC PLACES (BRFSS small-area estimates) · 2020 tract → BG broadcast	Direct tract join

PLACES_CANCER	Cancer (non-skin) prevalence (crude prev., %)	CDC PLACES (BRFSS small-area estimates) · 2020 tract → BG broadcast	Direct tract join
PLACES_CASTHMA	Current asthma (crude prev., %)	CDC PLACES (BRFSS small-area estimates) · 2020 tract → BG broadcast	Direct tract join
PLACES_CHD	Coronary heart disease (crude prev., %)	CDC PLACES (BRFSS small-area estimates) · 2020 tract → BG broadcast	Direct tract join
PLACES_CHECKUP	Routine checkup, past year (crude prev., %)	CDC PLACES (BRFSS small-area estimates) · 2020 tract → BG broadcast	Direct tract join
PLACES_CHOLSCREEN	Cholesterol screening (crude prev., %)	CDC PLACES (BRFSS small-area estimates) · 2020 tract → BG broadcast	Direct tract join
PLACES_COGNITION	Cognitive disability (crude prev., %)	CDC PLACES (BRFSS small-area estimates) · 2020 tract → BG broadcast	Direct tract join
PLACES_COLON_SCREEN	Colorectal cancer screening (crude prev., %)	CDC PLACES (BRFSS small-area estimates) · 2020 tract → BG broadcast	Direct tract join
PLACES_COPD	COPD prevalence (crude prev., %)	CDC PLACES (BRFSS small-area estimates) · 2020 tract → BG broadcast	Direct tract join
PLACES_CSMOKING	Current smoking (crude prev., %)	CDC PLACES (BRFSS small-area estimates) · 2020 tract → BG broadcast	Direct tract join
PLACES_DENTAL	Dental visit, past year (crude prev., %)	CDC PLACES (BRFSS small-area estimates) · 2020 tract → BG broadcast	Direct tract join
PLACES_DEPRESSION	Depression prevalence (crude prev., %)	CDC PLACES (BRFSS small-area estimates) · 2020	Direct tract join

		tract → BG broadcast	
PLACES_DIABETES	Diagnosed diabetes (crude prev., %)	CDC PLACES (BRFSS small-area estimates) · 2020 tract → BG broadcast	Direct tract join
PLACES_DISABILITY	Any disability (crude prev., %)	CDC PLACES (BRFSS small-area estimates) · 2020 tract → BG broadcast	Direct tract join
PLACES_EMOTIONSPT	Lacking social/emotional support (crude prev., %)	CDC PLACES (BRFSS small-area estimates) · 2020 tract → BG broadcast	Direct tract join
PLACES_FOODINSECU	Food insecurity, past 12 mo (crude prev., %)	CDC PLACES (BRFSS small-area estimates) · 2020 tract → BG broadcast	Direct tract join
PLACES_FOODSTAMP	Received food stamps, past 12 mo (crude prev., %)	CDC PLACES (BRFSS small-area estimates) · 2020 tract → BG broadcast	Direct tract join
PLACES_GHLTH	Fair or poor self-rated health (crude prev., %)	CDC PLACES (BRFSS small-area estimates) · 2020 tract → BG broadcast	Direct tract join
PLACES_HEARING	Hearing disability (crude prev., %)	CDC PLACES (BRFSS small-area estimates) · 2020 tract → BG broadcast	Direct tract join
PLACES_HIGHCHOL	High cholesterol (crude prev., %)	CDC PLACES (BRFSS small-area estimates) · 2020 tract → BG broadcast	Direct tract join
PLACES_HOUSINSECU	Housing insecurity, past 12 mo (crude prev., %)	CDC PLACES (BRFSS small-area estimates) · 2020 tract → BG broadcast	Direct tract join
PLACES_INDEPLIVE	Independent-living disability (crude prev., %)	CDC PLACES (BRFSS small-area estimates) · 2020 tract → BG broadcast	Direct tract join

PLACES_LACKTRPT	Lacked reliable transportation (crude prev., %)	CDC PLACES (BRFSS small-area estimates) · 2020 tract → BG broadcast	Direct tract join
PLACES_LONELINESS	Feelings of loneliness (crude prev., %)	CDC PLACES (BRFSS small-area estimates) · 2020 tract → BG broadcast	Direct tract join
PLACES_LPA	No leisure-time physical activity (crude prev., %)	CDC PLACES (BRFSS small-area estimates) · 2020 tract → BG broadcast	Direct tract join
PLACES_MAMMOUSE	Mammography use, women 50–74 (crude prev., %)	CDC PLACES (BRFSS small-area estimates) · 2020 tract → BG broadcast	Direct tract join
PLACES_MHLTH	Frequent mental distress (crude prev., %)	CDC PLACES (BRFSS small-area estimates) · 2020 tract → BG broadcast	Direct tract join
PLACES_MOBILITY	Mobility disability (crude prev., %)	CDC PLACES (BRFSS small-area estimates) · 2020 tract → BG broadcast	Direct tract join
PLACES_OBESITY	Obesity prevalence (crude prev., %)	CDC PLACES (BRFSS small-area estimates) · 2020 tract → BG broadcast	Direct tract join
PLACES_PHLTH	Frequent physical distress (crude prev., %)	CDC PLACES (BRFSS small-area estimates) · 2020 tract → BG broadcast	Direct tract join
PLACES_SELF CARE	Self-care disability (crude prev., %)	CDC PLACES (BRFSS small-area estimates) · 2020 tract → BG broadcast	Direct tract join
PLACES_SHUTUTILITY	Utility shutoff threat, past 12 mo (crude prev., %)	CDC PLACES (BRFSS small-area estimates) · 2020 tract → BG broadcast	Direct tract join
PLACES_SLEEP	Short sleep duration (crude prev., %)	CDC PLACES (BRFSS small-area estimates) · 2020	Direct tract join

		tract → BG broadcast	
PLACES_STROKE	Stroke prevalence (crude prev., %)	CDC PLACES (BRFSS small-area estimates) · 2020 tract → BG broadcast	Direct tract join
PLACES_TEETHLOST	All teeth lost, adults 65+ (crude prev., %)	CDC PLACES (BRFSS small-area estimates) · 2020 tract → BG broadcast	Direct tract join
PLACES_VISION	Vision disability (crude prev., %)	CDC PLACES (BRFSS small-area estimates) · 2020 tract → BG broadcast	Direct tract join
SVI_RPL_THEMES	SVI overall percentile	CDC/ATSDR SVI 2022 · 2020 tract → BG broadcast	Direct tract join
SVI_RPL_THEME1	SVI Theme 1: Socioeconomic status	CDC/ATSDR SVI 2022 · 2020 tract → BG broadcast	Direct tract join
SVI_RPL_THEME2	SVI Theme 2: Household characteristics	CDC/ATSDR SVI 2022 · 2020 tract → BG broadcast	Direct tract join
SVI_RPL_THEME3	SVI Theme 3: Racial & ethnic minority status	CDC/ATSDR SVI 2022 · 2020 tract → BG broadcast	Direct tract join
SVI_RPL_THEME4	SVI Theme 4: Housing type & transportation	CDC/ATSDR SVI 2022 · 2020 tract → BG broadcast	Direct tract join
HMDA_K7_mean_orig_2019_2024	Mean annual mortgage originations, 2019–2024	FFIEC HMDA · 2010- vintage tracts (2018–21) + 2020- vintage (2022–25)...	Vintage-aware multi-year mean per tract → BG
SLD_NatWalkInd	National Walkability Index	EPA Smart Location Database v3 · 2010 BG → 2020 (direct 262 + tract- crosswalk 76)	Direct-BG join + crosswalk-mean fallback
SLD_D1A	Gross residential density	EPA Smart Location Database v3 · 2010 BG → 2020 (direct 262 + tract- crosswalk 76)	Direct-BG join + crosswalk-mean fallback
SLD_D1B	Gross population density	EPA Smart Location Database v3 · 2010 BG → 2020 (direct 262 + tract- crosswalk 76)	Direct-BG join + crosswalk-mean fallback
SLD_D1C	Gross employment density	EPA Smart Location Database v3 · 2010 BG → 2020 (direct	Direct-BG join + crosswalk-mean fallback

		262 + tract-crosswalk 76)	
SLD_D2A_JPHH	Jobs per household	EPA Smart Location Database v3 · 2010 BG → 2020 (direct 262 + tract-crosswalk 76)	Direct-BG join + crosswalk-mean fallback
SLD_D2B_E8MIXA	8-tier employment mix (entropy)	EPA Smart Location Database v3 · 2010 BG → 2020 (direct 262 + tract-crosswalk 76)	Direct-BG join + crosswalk-mean fallback
SLD_D3A	Total road network density	EPA Smart Location Database v3 · 2010 BG → 2020 (direct 262 + tract-crosswalk 76)	Direct-BG join + crosswalk-mean fallback
SLD_D3B	Pedestrian-oriented intersection density	EPA Smart Location Database v3 · 2010 BG → 2020 (direct 262 + tract-crosswalk 76)	Direct-BG join + crosswalk-mean fallback
SLD_D3BMM4	Multi-modal intersection density	EPA Smart Location Database v3 · 2010 BG → 2020 (direct 262 + tract-crosswalk 76)	Direct-BG join + crosswalk-mean fallback
SLD_D4A	Distance to nearest transit stop	EPA Smart Location Database v3 · 2010 BG → 2020 (direct 262 + tract-crosswalk 76)	Direct-BG join + crosswalk-mean fallback
SLD_D4C	Aggregate transit service frequency	EPA Smart Location Database v3 · 2010 BG → 2020 (direct 262 + tract-crosswalk 76)	Direct-BG join + crosswalk-mean fallback
SLD_D4D	Transit frequency per sq mi	EPA Smart Location Database v3 · 2010 BG → 2020 (direct 262 + tract-crosswalk 76)	Direct-BG join + crosswalk-mean fallback
SLD_D5AR	Jobs within 45-min auto commute	EPA Smart Location Database v3 · 2010 BG → 2020 (direct 262 + tract-crosswalk 76)	Direct-BG join + crosswalk-mean fallback
SLD_D5BR	Jobs within 45-min transit commute	EPA Smart Location Database v3 · 2010 BG → 2020 (direct 262 + tract-crosswalk 76)	Direct-BG join + crosswalk-mean fallback
SLD_D5CR	Regional centrality (auto)	EPA Smart Location Database v3 · 2010	Direct-BG join + crosswalk-mean fallback

		BG → 2020 (direct 262 + tract-crosswalk 76)	
SLD_D5CRI	Regional centrality index	EPA Smart Location Database v3 · 2010 BG → 2020 (direct 262 + tract-crosswalk 76)	Direct-BG join + crosswalk-mean fallback
SLD_TotPop	Total population (SLD base)	EPA Smart Location Database v3 · 2010 BG → 2020 (direct 262 + tract-crosswalk 76)	Direct-BG join + crosswalk-mean fallback
SLD_CountHU	Housing units (SLD base)	EPA Smart Location Database v3 · 2010 BG → 2020 (direct 262 + tract-crosswalk 76)	Direct-BG join + crosswalk-mean fallback
SLD_TotEmp	Total employment (SLD base)	EPA Smart Location Database v3 · 2010 BG → 2020 (direct 262 + tract-crosswalk 76)	Direct-BG join + crosswalk-mean fallback
SLD_Ac_Land	Land area (acres)	EPA Smart Location Database v3 · 2010 BG → 2020 (direct 262 + tract-crosswalk 76)	Direct-BG join + crosswalk-mean fallback
SLD_Ac_Unpr	Unprotected land area (acres)	EPA Smart Location Database v3 · 2010 BG → 2020 (direct 262 + tract-crosswalk 76)	Direct-BG join + crosswalk-mean fallback
FARA_LILATracts_1And10	Low-income & low-access flag (1 mi/10 mi)	USDA ERS Food Access Research Atlas 2019 · 2010 tract → 2020 (crosswalk)	Weighted tract crosswalk
FARA_LILATracts_halfAnd10	Low-income & low-access flag (1/2 mi/10 mi)	USDA ERS Food Access Research Atlas 2019 · 2010 tract → 2020 (crosswalk)	Weighted tract crosswalk
FARA_LowIncomeTracts	Low-income tract flag	USDA ERS Food Access Research Atlas 2019 · 2010 tract → 2020 (crosswalk)	Weighted tract crosswalk
FARA_lapophalfshare	Share of pop low access at 1/2 mi	USDA ERS Food Access Research Atlas 2019 · 2010 tract → 2020 (crosswalk)	Weighted tract crosswalk

FARA_lapop1share	Share of pop low access at 1 mi	USDA ERS Food Access Research Atlas 2019 · 2010 tract → 2020 (crosswalk)	Weighted tract crosswalk
FARA_lalowihalfshare	Low-income pop low access at 1/2 mi	USDA ERS Food Access Research Atlas 2019 · 2010 tract → 2020 (crosswalk)	Weighted tract crosswalk
FARA_lahunvhalfshare	HU without vehicle, low access at 1/2 mi	USDA ERS Food Access Research Atlas 2019 · 2010 tract → 2020 (crosswalk)	Weighted tract crosswalk
FARA_PovertyRate	Tract poverty rate (FARA)	USDA ERS Food Access Research Atlas 2019 · 2010 tract → 2020 (crosswalk)	Weighted tract crosswalk
N1_adi_natrank	Area Deprivation Index; national rank (v4.0.1, 2024)	Neighborhood Atlas (UW-Madison) · 2020 BG direct	Direct join; GQ/PH-suppressed BGs unranked
N1_adi_staternk	Area Deprivation Index; state rank	Neighborhood Atlas · 2020 BG direct	Direct join
N1b_adistd_ntl	Standardized ADI; national (income-standardized, 2022)	NaNDA (ICPSR) · 2020 BG direct	Petterson-critique variant; carry alongside N1 in Phase B
N1b_adistd_state	Standardized ADI; state	NaNDA (ICPSR) · 2020 BG direct	Direct join
N2_sdi	Social Deprivation Index (2012–2019)	Robert Graham Center · 2010 tract → 2020 (crosswalk)	Weighted crosswalk
N4_dci_distress	Distressed Communities Index score (2019–2023)	Economic Innovation Group · ZIP → ZCTA → tract (tab20 AREALAND_PART) → BG	AREA-WEIGHTED APPROXIMATION; EIG tract edition preferred for Phase B
N7_upjohn_distress_ord	Upjohn distressed-communities ordinal (1 best – 5 worst)	W.E. Upjohn Institute · 2010 tract → 2020 (crosswalk)	Categorical → ordinal mapping
N7_upjohn_pae_rate	Upjohn prime-age employment rate	W.E. Upjohn Institute · 2010 tract → 2020 (crosswalk)	Weighted crosswalk
G1_life_expectancy	Life expectancy at birth (USALEEP, NC File A)	CDC NCHS USALEEP · 2010 tract → 2020 (crosswalk)	109/119 Guilford 2010 tracts published; remainder genuinely unpublished
I1_uncg_violent_rate	Violent crime per 1,000 residents (3-agency, 2014–2018)	UNCG GC Substance Use build (GPD/HPPD/Sheriff)	GCSU all_vioPC1; provenance closed 7/8/26; current vintage pending

		· 2010 BG → 2020 (direct 262 + pop- weighted tract crossw...	
I2_uncg_property_rate	Property crime per 1,000 residents (2014– 2018)	UNCG GC Substance Use build · 2010 BG → 2020 (direct + pop- wtd crosswalk)	GCSU all_proP1
I3_uncg_cfs_rate	Police calls-for-service per 1,000 residents (2014–2018)	UNCG GC Substance Use build · 2010 BG → 2020 (direct + pop- wtd crosswalk)	GCSU ps_totP1
I4_uncg_weapons_rate	Weapons calls per 1,000 residents (2014– 2018)	UNCG GC Substance Use build · 2010 BG → 2020 (direct + pop- wtd crosswalk)	GCSU ps_wepP1
I5_uncg_allcrime_rate	All crime per 1,000 residents (2014–2018)	UNCG GC Substance Use build · 2010 BG → 2020 (direct + pop- wtd crosswalk)	GCSU all_crimeP1
I6a_ags_total_idx	AGS CrimeRisk total index (MODELED)	Applied Geographic Solutions · 2020 BG	Modeled vendor estimates (index, national avg=100). Concurrent validity vs verified I1-I5: $r=.38-.68$ (I6b x I1 violent=.68). Decision rule: enters Phase B screening; if retained in the published comp...
I6b_ags_personal_idx	AGS CrimeRisk personal-crime index (MODELED)	AGS · 2020 BG	Modeled vendor estimates (index, national avg=100). Concurrent validity vs verified I1-I5: $r=.38-.68$ (I6b x I1 violent=.68). Decision rule: enters Phase B screening; if retained in the published comp...
I6c_ags_property_idx	AGS CrimeRisk property-crime index (MODELED)	AGS · 2020 BG	Modeled vendor estimates (index, national avg=100). Concurrent validity vs verified I1-I5: $r=.38-.68$ (I6b x I1 violent=.68). Decision rule: enters Phase B screening; if retained in the published comp...
J5_econ_connectedness	Economic connectedness	Social Capital Atlas (Chetty et al.) · ZIP → BG allocation	SCA measure
J6a_clustering	Network clustering coefficient	Social Capital Atlas · ZIP → BG allocation	SCA cohesiveness
J6b_support_ratio	Support ratio	Social Capital Atlas · ZIP → BG allocation	SCA cohesiveness
J7a_volunteering	Volunteering rate	Social Capital Atlas · ZIP → BG allocation	SCA civic engagement
J7b_civic_orgs	Civic organizations	Social Capital Atlas · ZIP → BG allocation	SCA civic engagement

J8_self_response	2020 Census self-response rate	U.S. Census Bureau · 2020 tract → BG broadcast	Direct tract join
J8b_internet_response	2020 Census internet self-response rate	U.S. Census Bureau · 2020 tract → BG broadcast	Direct tract join
I7_oa_incarceration	Incarceration rate, children of p25 parents	Opportunity Atlas (jail_pooled_pooled_p25) · 2010 tract → 2020 → BG broadcast	OA outcome
J1_nonprofit_per10k	Nonprofits per 10,000 residents	Candid 990 (PolicyMap license) · geocoded points → 2020 BG	1,858 orgs; 1,447 geocoded (78%; non-matches mostly PO boxes)
J1a_arts_per10k	Arts/culture nonprofits per 10K (NTEE A)	Candid 990 · points → 2020 BG	NTEE class A
J1b_humansvc_per10k	Human-services nonprofits per 10K (NTEE P)	Candid 990 · points → 2020 BG	NTEE class P
J1c_youth_per10k	Youth-development nonprofits per 10K (NTEE O)	Candid 990 · points → 2020 BG	NTEE class O
J1d_civic_per10k	Civic/advocacy nonprofits per 10K	Candid 990 · points → 2020 BG	NTEE civic classes
J1e_np_expense_pc	Nonprofit total functional expenses per capita (\$)	Candid 990 · points → 2020 BG	Real 990 Total Functional Expenses
J2_voter_reg_rate ‡	Voter registration rate	NC State Board of Elections nc voter41 · geocoded addresses → 2020 BG	UNPOPULATED BY DESIGN: persistent cache accumulating toward 95% address threshold
E10_filing_rate	Eviction filing rate (mean of 3 most recent valid years)	Eviction Lab (Princeton), legacy validated series · 2010 BG (~2000–2016) → 2020 (direct 262 + crosswalk 76)	NEW in v9; vintage caveat for methodology; quality flags carried
E10_eviction_rate	Eviction (judgment) rate, same window	Eviction Lab · 2010 BG → 2020	Companion to filing rate
E10_filings_3yr	Eviction filings, 3-yr sum	Eviction Lab · 2010 BG → 2020	Count over the 3 valid years used
E10_last_year ‡	Most recent valid data year used	Eviction Lab · 2010 BG → 2020	Vintage tracker
E10_quality_flag ‡	Quality flag (low/imputed/subbed in used years)	Eviction Lab · 2010 BG → 2020	Max flag across used years
J10_library_dist_mi	Distance to nearest public library outlet (mi)	IMLS Public Libraries Survey FY2024 outlet file · outlet lat/lon vs	NEW in v9; centroid approximation; block-pop-weighted 1/2-mi share deferred (Klinenberg 2018)

		2020 BG internal point (TIGERweb)	
J10_library_halfmi	Library within 1/2 mile of BG internal point (0/1)	IMLS PLS FY2024 · same as J10 distance	Centroid approximation
K14_hub_cert_per10k	HUB-certified businesses per 10,000 residents	NC HUB certification; Guilford extract via Shaunne Thomas (7/3/26) · geocoded points -> 2020 BG (421/480 matched, 88%)	Voluntary, procurement-oriented certification; NOT an ownership-demographic census; sparse by construction. Source-status note: S.L. 2026-41 abolished the state HUB Office and repealed G.S. 143-128.2 through 143-128.4, so this series and its K14a/K14b sub-measures will not refresh; Phase B screening candidates only, never scored components.
K14a_hub_black_per10k	Black-owner HUB-certified businesses per 10K (n=338 countywid...	NC HUB certification (HUBCategory=Black) · geocoded points -> 2020 BG	Small-cell caveat for Phase B screening
K14b_hub_female_per10k	Female-owner HUB-certified businesses per 10K (n=98 countywid...	NC HUB certification (HUBCategory=Female) · geocoded points -> 2020 BG	Small-cell caveat; only 58 BGs > 0
K15_sbe_cert_per10k	NC SBE-certified small businesses per 10,000 residents	NC Small Business Enterprise certification; extract via Shaunne Thomas (7/3/26) · geocoded points -> 2020 BG (144/160 matched, 90%)	119/160 also HUB-certified; collinearity check with K14 in Phase B VIF
K8_sba_loans_per10k	SBA 7(a)+504 loans per 10K residents, FY2023–2025 (interim de...	SBA 7(a)/504 FOIA loan files (12/31/25 vintage) · borrower addresses geocoded -> 2020 BG (235/261 loans,...	Per-1,000-establishments denominator upgrade when K1 Overture lands
K8b_sba_amount_pc	SBA gross approval dollars per capita, FY2023–2025 (\$196.2M c...	same files · geocoded -> 2020 BG	Dollar-volume companion to K8
E1_code_violations_3yr	Code-enforcement violations, 3-yr count	Cities of Greensboro (Open Gate City) + High Point (Community Development & Hou... · 2020 BG; 3-yr window to 7/2026	GSO: REID-joined to GC_Parcels. HP: EPSG:2264->WGS84 + PIP (99%+ coords) with Census-geocode fallback; additive merge; identical 3-yr window. All enforcement types both cities; classification schemes...
E2_pct_absentee_owner	Absentee-owner share of residential parcels	Guilford County GIS (GC_Parcels FeatureServer) ·	OWNER_MAIL_STATE != 'NC' proxy over residential parcels; PIP to BG. PROXY pending NC SOS entity match for true investor-LLC flag.

		2020 BG; current assessment vintage	
E4_median_home_value	Median assessed value, residential parcels	Guilford County GIS (GC_Parcels FeatureServer) · 2020 BG; current assessment vintage	Median TOTAL_PROP_VALUE_ASSESSED over residential parcels; PIP to BG.
E5_median_year_built	Median year built, residential parcels	Guilford County GIS (GC_Parcels FeatureServer) · 2020 BG; current assessment vintage	Median YEAR_BUILT over residential parcels; PIP to BG.
E6_foreclosure_count	Foreclosure-flagged parcels, count	Guilford County GIS Data Hub (Parcel Foreclosure, upd. 7/8/26) · 2020 BG; snapshot 7/8/2026	Active mortgage-foreclosure-flagged parcels; centroid PIP to BG.
E6_foreclosure_rate_per1k_res	Foreclosure rate per 1,000 residential parcels	Guilford County GIS Data Hub (Parcel Foreclosure, upd. 7/8/26) · 2020 BG; snapshot 7/8/2026	E6 count / residential parcel count x 1,000 (denominator = E_parcel_count_residential).
E7_pct_vacant_land	Vacant-land share of all parcels	Guilford County GIS (GC_Parcels FeatureServer) · 2020 BG; current assessment vintage	Share of parcels with LAND_CLASS='VACANT'; PIP to BG.
E8_pct_recent_turnover_3_yr	Residential parcels sold in last 3 yr, share	Guilford County GIS (GC_Parcels FeatureServer) · 2020 BG; sale year >= 2023	Share of residential parcels with PKG_SALE_DATE year >= 2023; PIP to BG.
E9_building_permits_12mo	Building permits issued, 12-mo count	City of Greensboro (Open Gate City) Building Permits · 2020 BG; 12-mo window to 7/2026	Permits reprojected (NC State Plane EPSG:2264 -> WGS84), PIP to BG. COVERAGE CAVEAT: Greensboro city limits only.
E_parcel_count_total	Total parcels per BG (denominator)	Guilford County GIS (GC_Parcels FeatureServer) · 2020 BG; current assessment vintage	Count of all parcels PIP-matched to BG (222,815 countywide).
E_parcel_count_residential	Residential parcels per BG (denominator)	Guilford County GIS (GC_Parcels FeatureServer) · 2020 BG; current assessment vintage	Count of residential-class parcels PIP-matched to BG; denominator for E6 rate.
H2_supermarket_count	Supermarket / grocery / superstore count	USDA FNS SNAP Authorized Retailer file (Guilford extract) · 2020 BG; current SNAP authorization	Store_Type in {Supermarket, Grocery Store, Super Store}; PIP to BG. Store_Type SUBSTITUTES for Data Axle NAICS points (per-10K in Phase B).

H3_convenience_count	Convenience-store count	USDA FNS SNAP Authorized Retailer file (Guilford extract) · 2020 BG; current SNAP authorization	Store_Type='Convenience Store'; PIP to BG. Store_Type substitutes for Data Axle NAICS points.
H4_snap_retailer_count	All SNAP-authorized retailers, count	USDA FNS SNAP Authorized Retailer file (Guilford extract) · 2020 BG; current SNAP authorization	All SNAP-authorized retailers; PIP to BG (492/499 matched).
M3_TotalBSL	Broadband Serviceable Locations, count (denominator)	NC OneMap / NC Broadband Infrastructure Office (FCC BDC v6) · 2020 BG; FCC BDC v6 (2025 aggregation)	Level-8 hexbins areally interpolated to BG (BSL reconciled 182,492 exact).
M3_pct_served_100_20	Locations served at >=100/20 Mbps, share	NC OneMap / NC Broadband Infrastructure Office (FCC BDC v6) · 2020 BG; FCC BDC v6 (2025 aggregation)	1 - (unserved+underserved)/TotalBSL; areal interpolation. Primary M3 measure (upgraded from 'max advertised speed').
M3_pct_unserved	Unserved locations, share	NC OneMap / NC Broadband Infrastructure Office (FCC BDC v6) · 2020 BG; FCC BDC v6 (2025 aggregation)	UnservedLocations/TotalBSL; areal interpolation.
M3_pct_underserved ‡	Underserved (25-100 Mbps) locations, share	NC OneMap / NC Broadband Infrastructure Office (FCC BDC v6) · 2020 BG; FCC BDC v6 (2025 aggregation)	UnderservedLocations/TotalBSL. ZERO-VARIANCE in Guilford (no 25-100 tier in BDC v6); will drop in Phase B.
M3_pct_fiber	Locations with fiber reported, share	NC OneMap / NC Broadband Infrastructure Office (FCC BDC v6) · 2020 BG; FCC BDC v6 (2025 aggregation)	Count_Fiber/TotalBSL; areal interpolation. Technology-quality signal.
M3_pct_cable	Locations with cable reported, share	NC OneMap / NC Broadband Infrastructure Office (FCC BDC v6) · 2020 BG; FCC BDC v6 (2025 aggregation)	Count_Cable/TotalBSL; areal interpolation.

K13_bus_vac_rate	Business-address vacancy rate (%)	HUD / USPS Vacancy Data (registered access; manual quarterly download) · 2020 tract, broadcast to BG; Q1 2026 snapshot	bus_vac / ams_bus x 100 per tract, broadcast to all BGs in tract. No-stat addresses excluded per HUD convention. CAVEAT: no within-tract BG variation.
K13b_bus_vac_longterm_share	Share of vacant business addresses vacant 12+ months (%)	HUD / USPS Vacancy Data (registered access; manual quarterly download) · 2020 tract, broadcast to BG; Q1 2026 snapshot	(vac_12_24b + vac_24_36b + vac_36_bus) / bus_vac x 100 per tract, broadcast. NaN where tract has zero vacant business addresses.
K13_ams_bus_tract	Active business addresses in tract (denominator context)	HUD / USPS Vacancy Data (registered access; manual quarterly download) · 2020 tract, broadcast to BG; Q1 2026 snapshot	ams_bus broadcast; context/weighting field, not an index candidate.
K2_smallfirm_emp_share	Small-firm employment share (%; firms <50 employees)	Census LEHD LODES8 WAC 2023 · 2020 block -> BG (w_geocode[:12]); CY2023	(CFS01 + CFS02) / C000 on JT02 (All Private). NaN where BG has no private jobs. CAVEAT: workplace-area jobs; firm size at the FIRM (EIN) level, not establishment.
K5_youngfirm_emp_share	Young-firm employment share (%; firm age 0-3 yrs)	Census LEHD LODES8 WAC 2023 · 2020 block -> BG (w_geocode[:12]); CY2023	(CFA01 + CFA02) / C000 on JT02 (All Private). NaN where BG has no private jobs.
K2_wac_private_emp	Private workplace employment, count	Census LEHD LODES8 WAC 2023 · 2020 block -> BG (w_geocode[:12]); CY2023	JT02 C000 summed to BG; fillna(0) = true zero (no workplace jobs).
K2_wac_total_emp	Total workplace employment, count	Census LEHD LODES8 WAC 2023 · 2020 block -> BG (w_geocode[:12]); CY2023	JT00 C000 summed to BG; fillna(0) = true zero.
L2_retail_emp_cns07	Retail-trade workplace employment (NAICS 44-45), count	Census LEHD LODES8 WAC 2023 · 2020 block -> BG (w_geocode[:12]); CY2023	JT00 CNS07 summed to BG. Supply side of the Phase-B L2 retail-leakage model; demand side (CES x B19001) remains internal Phase B.
K1_places_count	Business/POI establishments, count	Overture Maps Foundation, Places theme · Point -> 2020 BG (PIP); release 2026-06-17.0	All places at confidence >= 0.5, PIP to BG (24,569 in-county). Per-10K conversion in Phase B. CAVEAT: Overture category taxonomy, not NAICS; confidence filter drops low-confidence POIs.

K1b_independent_share	Independent (non-branded) establishment share (%)	Overture Maps Foundation, Places theme · Point -> 2020 BG (PIP); release 2026-06-17.0	Share of BG establishments with no brand record. CAVEAT: brand-null is an independence PROXY.
J4_religious_count	Religious organizations, count	Overture Maps Foundation, Places theme · Point -> 2020 BG (PIP); release 2026-06-17.0	Keyword match on Overture primary category (church/mosque/synagogue/temple/religious/place_of_worship/etc.; 1,041 county total, church_cathedral=681 largest). Replaces the Data Axle NAICS 8131 spec. ...
H2_ovt_grocery_count	Grocery/supermarket establishments (Overture), count	Overture Maps Foundation, Places theme · Point -> 2020 BG (PIP); release 2026-06-17.0	categories.primary in (grocery_store, supermarket); n=175. COMPANION to SNAP-based H2_supermarket_count (retained); Phase B screening selects between constructs.
H3_ovt_convenience_count	Convenience-store establishments (Overture), count	Overture Maps Foundation, Places theme · Point -> 2020 BG (PIP); release 2026-06-17.0	categories.primary = convenience_store; n=194. COMPANION to SNAP-based H3_convenience_count (retained).
J2_reg_voters_count	Registered voters (all statuses), count	NC State Board of Elections (dl.ncsbe.gov): ncvoter41 + ncvhis41, Census batch ... · Registrant address -> 2020 BG; snapshot 7/2026; electi...	Residential-address geocode via complete cache (95.04% of unique addresses matched; 95.2% registrant-weighted). All voter statuses per the Stage-7 coded construct. Per-adult/CVAP rate in Phase B.
J2b_active_reg_count	Registered voters (ACTIVE status), count	NC State Board of Elections (dl.ncsbe.gov): ncvoter41 + ncvhis41, Census batch ... · Registrant address -> 2020 BG; snapshot 7/2026; electi...	ACTIVE-status subset companion; Phase B screening selects between all-status and ACTIVE constructs.
J3_voted_2024_count	Registrants who voted, 11/05/2024 general, count	NC State Board of Elections (dl.ncsbe.gov): ncvoter41 + ncvhis41, Census batch ... · Registrant address -> 2020 BG; snapshot 7/2026; electi...	ncvhis41 history join on voter_reg_num, election_lbl 11/05/2024 GENERAL.
J3_turnout_2024_rate	Turnout among registrants, 11/05/2024 general (%)	NC State Board of Elections (dl.ncsbe.gov): ncvoter41 + ncvhis41, Census	J3_voted_2024_count / J2_reg_voters_count x 100. CAVEAT: denominator is the 7/2026 registration snapshot, not the 11/2024 roll (post-

		batch ... · Registrant address -> 2020 BG; snapshot 7/2026; electi...	election removals bias rates slightly, uniformly).
G9_ejscreen_pollution_burden	EJScreen pollution burden (mean of 6 national percentiles)	EPA EJScreen 2024 (v2.32); PEDP preservation, Harvard Dataverse doi:10.7910/DVN... · Native 2020 BG; EJScreen 2024 release (original EPA en...	Mean of P_PM25, P_OZONE, P_DSLPM, P_RSEI_AIR, P_PTAF, P_LDPNT; >=5 of 6 required. CONSTRUCT NOTE: spec's 'air toxics cancer risk' discontinued in EJScreen 2024; P_RSEI_AIR (toxic releases to air) su...
G10_lead_paint_pre1960_pct	Lead paint risk: pre-1960 housing share (%)	EPA EJScreen 2024 (v2.32); PEDP preservation, Harvard Dataverse doi:10.7910/DVN... · Native 2020 BG; EJScreen 2024 release (original EPA en...	PRE1960PCT x 100 (EJScreen lead-paint indicator). National percentile companion (P_LDPNT) available in the Guilford extract.
K10_cra_sb_orig_count	CRA small-business loan originations, count (2024)	FFIEC CRA 2024 Aggregate Flat Files (Table A1-1, Small Business Originations) · 2020 tract (CRA 2022+ vintage), broadcast to BG; activ...	Sum of three loan-size bands per tract, broadcast to BGs. County reconciliation: 12,652 tract-assignable of 12,689 (99.7%; remainder 'tract not known'). CAVEATS: CRA reporters only (large banks; ~78%...
K10_cra_sb_orig_amt_000s	CRA small-business originations, \$ thousands (2024)	FFIEC CRA 2024 Aggregate Flat Files (Table A1-1, Small Business Originations) · 2020 tract (CRA 2022+ vintage), broadcast to BG; activ...	Sum of three size-band amounts per tract, broadcast. County total \$448.8M.
K10b_cra_sb_rev_lt1m_count	CRA originations to businesses with revenues < \$1M, count (20...	FFIEC CRA 2024 Aggregate Flat Files (Table A1-1, Small Business Originations) · 2020 tract (CRA 2022+ vintage), broadcast to BG; activ...	Field 19 of A1-1; the small-firm-targeted lending subset. K10b <= K10 verified everywhere.
E1_hp_code_violations_3yr	High Point component of E1 (audit column)	City of High Point Community Development & Housing (T. Wilson), delivered 7/16/... · 2020 BG via EPSG:2264 PIP; 3-yr window to 7/2026	In-window enforcement cases placed by reprojection+PIP (99.4% coords) or Census geocode; out-of-county High Point cases excluded by frame construction.
G8_lbwt_pct_2223	Low birthweight, % of births (2022-23 pooled)	Guilford County Health Department	Pooled Sum_LBW/births x100; NaN where pooled births <20 (reliability floor:

		(delivered 7/2026; NC vital records) · Native 2020 tract, broadcast to BG; CY2022+2023	4 tracts incl. 011201 at 16 births; floor held; pilot tract 112 covered via 011202) or the zero-birth airport tr...
G8b_preterm_pct_2223	Preterm births, % (2022-23 pooled)	Guilford County Health Department (delivered 7/2026; NC vital records) · same	Pooled Sum_PRETER/births x100; same floor.
G8_births_2223	Births, pooled 2022-23 count	Guilford County Health Department (delivered 7/2026; NC vital records) · same	Denominator context / Phase-B reliability weighting.
G7_life_expectancy_1620	Life expectancy at birth, 2016-2020 (yrs)	Guilford County Health Department (delivered 7/2026; NC vital records) · 2010 tract -> 2020 (population-weighted crosswalk, >=5...	CONSTRUCT SUBSTITUTION (documented): YPLL spec replaced by county-delivered period LE (inverse construct); supersedes USALEEP 2010-15. 111/126 tracts; all 15 gaps trace to 8 source-suppressed small 2...
G7b_le_change_0913_1620	Life-expectancy change, 2009-13 -> 2016-20 (yrs)	Guilford County Health Department (delivered 7/2026; NC vital records) · same	Crosswalked difference; improving-vs-declining trajectory companion.

APPENDIX B: ANALYSIS CODEBOOK

Analysis notes (read before standardization or modeling). 1. MISSING VALUES are system-missing and MEANINGFUL: every blank is a registered gap (source suppression, reliability floor, coverage boundary, or zero-population geography); never impute silently. Named exceptions: BG 370819801001 (zero-pop airport tract: G9, births); pilot-tract-112 successors 011201/011202 (life expectancy source-suppressed; 011201 also under the 20-birth G8 floor). 2. TRACT-BROADCAST variables (see Unit of Analysis column) have no within-tract variation across their BGs; cluster-aware screening advised; treat effective n as ~126 for those measures. 3. COUNTS vs RATES: several K/J/H variables are deliberately counts (per-10K or per-capita denominators deferred to the z-scoring spec; K1 establishment counts and G8 birth counts are provided as denominator/weighting candidates). 4. E1 two-city coverage: Greensboro + High Point jurisdictions; zeros in unincorporated county and small municipalities are COVERAGE ARTIFACTS, not true absence. E9 permits remain Greensboro-only. 5. I1-I5 reflect 2014-2018 conditions (2010->2020 crosswalk); I6a/b/c are MODELED vendor values cleared for internal analysis only (publication rights unresolved). M3_pct_underserved is zero-variance and will drop at screening. 6. Vintage-mixed file: each variable's Geography & Vintage column is authoritative; 2010-vintage sources were crosswalked with population weights.

Variable	Label	Metric	Unit of analysis
GEOID	Census block group GEOID (12-digit FIPS)	String identifier (A12)	2020 Census block group (native)
Tract	Census tract number (6-digit)	String identifier (A6)	2020 Census tract
A1_est	Median household income	Dollars / median value	2020 Census block group (native)
A2_est	Median HH income; county ratio	Count	2020 Census block group (native)
A3_est	Poverty rate (all ages)	Percent (0-100)	2020 Census block group (native)
A4_est	Deep poverty (<50% FPL)	Percent (0-100)	2020 Census block group (native)
A5_est	Child poverty rate	Rate	2020 Census block group (native)
A6_est	SNAP receipt	Continuous measure	2020 Census block group (native)
A7_est	Public assistance income	Continuous measure	2020 Census block group (native)
A8_est	Prime-age employment ratio (25–54)	Continuous measure	2020 Census block group (native)
A9_est	Unemployment rate (16+)	Percent (0-100)	2020 Census block group (native)
A10_est	Not in labor force (25–64)	Continuous measure	2020 Census block group (native)
A11_est	Single-parent HH with children	Continuous measure	2020 Census block group (native)

A12_est	Households with no vehicle	Continuous measure	2020 Census block group (native)
A13_est	Gini coefficient (income inequality)	Continuous measure	2020 Census block group (native)
B1_est	Less than HS attainment (25+)	Continuous measure	2020 Census block group (native)
B2_est	HS equivalent or higher (25+)	Continuous measure	2020 Census block group (native)
B3_est	Bachelor's degree or higher (25+)	Continuous measure	2020 Census block group (native)
B6_est	Preschool enrollment (age 3–4)	Continuous measure	2020 Census block group (native)
B8_est	Limited-English-speaking HH	Continuous measure	2020 Census block group (native)
C1_est	% Black or African American	Percent (0-100)	2020 Census block group (native)
C2_est	% Hispanic or Latino	Percent (0-100)	2020 Census block group (native)
C3_est	% Foreign-born	Percent (0-100)	2020 Census block group (native)
C4_est	Age dependency ratio	Continuous measure	2020 Census block group (native)
C5_est	ICE (Race-Income)	Continuous measure	2020 Census block group (native)
C6_est	% with a disability	Percent (0-100)	2020 Census block group (native)
C7_est	Population density	Density	2020 Census block group (native)
D1_est	Homeownership rate	Rate	2020 Census block group (native)
D2_est	Housing vacancy rate	Rate	2020 Census block group (native)
D3_est	Vacant for sale (not seasonal)	Continuous measure	2020 Census block group (native)
D4_est	Renter-occupied share	Percent (0-100)	2020 Census block group (native)
D5_est	Median home value	Dollars / median value	2020 Census block group (native)
D6_est	Median gross rent	Dollars / median value	2020 Census block group (native)
D7_est	Renter cost burden $\geq 30\%$	Percent (0-100)	2020 Census block group (native)
D8_est	Severe renter cost burden $\geq 50\%$	Percent (0-100)	2020 Census block group (native)
D9_est	Owner cost burden $\geq 30\%$ (SMOC)	Percent (0-100)	2020 Census block group (native)
D10_est	Overcrowding (>1.5 per room)	Continuous measure	2020 Census block group (native)

D11_est	Pre-1950 housing stock	Continuous measure	2020 Census block group (native)
D12_est	Mobile-home share	Percent (0-100)	2020 Census block group (native)
D13_est	Lacking complete plumbing	Continuous measure	2020 Census block group (native)
D14_est	Moved in past year	Continuous measure	2020 Census block group (native)
D15_est	SF-detached / MF ratio	Continuous measure	2020 Census block group (native)
G6a_est	No health insurance (all ages)	Continuous measure	2020 Census block group (native)
M1_est	Broadband subscription	Continuous measure	2020 Census block group (native)
M2_est	Computer ownership	Continuous measure	2020 Census block group (native)
M4_est	Mobile-only internet HH	Continuous measure	2020 Census block group (native)
C7_density_per_sqmi	Population density (per sq mi)	Density	2020 Census block group (native)
Y1	Household income rank, children of p25 parents (kfr_pooled_pooled_p25)	Continuous measure	Tract-level measure BROADCAST to block ...
Y1_late	OA v2 late-cohort income rank (sensitivity)	Continuous measure	Tract-level measure BROADCAST to block ...
Y2	P(top-quintile income), p25 parents (kfr_top20_pooled_pooled_p25)	Continuous measure	Tract-level measure BROADCAST to block ...
Y3	Employment rate at 30+, p25 parents (working_pooled_pooled_p25)	Rate	Tract-level measure BROADCAST to block ...
Y4	Child Opportunity Index 3.0 composite (national z-score, 2023)	Index score	Tract-level measure BROADCAST to block ...
PLACES_ACCESS2	No health insurance, adults 18–64 (crude prev., %)	Percent (0-100)	Tract-level measure BROADCAST to block ...
PLACES_ARTHRITIS	Arthritis prevalence (crude prev., %)	Percent (0-100)	Tract-level measure BROADCAST to block ...
PLACES_BINGE	Binge drinking (crude prev., %)	Percent (0-100)	Tract-level measure BROADCAST to block ...

PLACES_BPHIGH	High blood pressure (crude prev., %)	Percent (0-100)	Tract-level measure BROADCAST to block ...
PLACES_BPMED	BP medication adherence (crude prev., %)	Percent (0-100)	Tract-level measure BROADCAST to block ...
PLACES_CANCER	Cancer (non-skin) prevalence (crude prev., %)	Percent (0-100)	Tract-level measure BROADCAST to block ...
PLACES_CASTHMA	Current asthma (crude prev., %)	Percent (0-100)	Tract-level measure BROADCAST to block ...
PLACES_CHD	Coronary heart disease (crude prev., %)	Percent (0-100)	Tract-level measure BROADCAST to block ...
PLACES_CHECKUP	Routine checkup, past year (crude prev., %)	Percent (0-100)	Tract-level measure BROADCAST to block ...
PLACES_CHOLSCREEN	Cholesterol screening (crude prev., %)	Percent (0-100)	Tract-level measure BROADCAST to block ...
PLACES_COGNITION	Cognitive disability (crude prev., %)	Percent (0-100)	Tract-level measure BROADCAST to block ...
PLACES_COLON_SCREEN	Colorectal cancer screening (crude prev., %)	Percent (0-100)	Tract-level measure BROADCAST to block ...
PLACES_COPD	COPD prevalence (crude prev., %)	Percent (0-100)	Tract-level measure BROADCAST to block ...
PLACES_CSMOKING	Current smoking (crude prev., %)	Percent (0-100)	Tract-level measure BROADCAST to block ...
PLACES_DENTAL	Dental visit, past year (crude prev., %)	Percent (0-100)	Tract-level measure BROADCAST to block ...
PLACES_DEPRESSION	Depression prevalence (crude prev., %)	Percent (0-100)	Tract-level measure BROADCAST to block ...

PLACES_DIABETES	Diagnosed diabetes (crude prev., %)	Percent (0-100)	Tract-level measure BROADCAST to block ...
PLACES_DISABILITY	Any disability (crude prev., %)	Percent (0-100)	Tract-level measure BROADCAST to block ...
PLACES_EMOTIONSPT	Lacking social/emotional support (crude prev., %)	Percent (0-100)	Tract-level measure BROADCAST to block ...
PLACES_FOODINSECU	Food insecurity, past 12 mo (crude prev., %)	Percent (0-100)	Tract-level measure BROADCAST to block ...
PLACES_FOODSTAMP	Received food stamps, past 12 mo (crude prev., %)	Percent (0-100)	Tract-level measure BROADCAST to block ...
PLACES_GHLTH	Fair or poor self-rated health (crude prev., %)	Percent (0-100)	Tract-level measure BROADCAST to block ...
PLACES_HEARING	Hearing disability (crude prev., %)	Percent (0-100)	Tract-level measure BROADCAST to block ...
PLACES_HIGHCHOL	High cholesterol (crude prev., %)	Percent (0-100)	Tract-level measure BROADCAST to block ...
PLACES_HOUSINSECU	Housing insecurity, past 12 mo (crude prev., %)	Percent (0-100)	Tract-level measure BROADCAST to block ...
PLACES_INDEPLIVE	Independent-living disability (crude prev., %)	Percent (0-100)	Tract-level measure BROADCAST to block ...
PLACES_LACKTRPT	Lacked reliable transportation (crude prev., %)	Percent (0-100)	Tract-level measure BROADCAST to block ...
PLACES_LONELINESS	Feelings of loneliness (crude prev., %)	Percent (0-100)	Tract-level measure BROADCAST to block ...
PLACES_LPA	No leisure-time physical activity (crude prev., %)	Percent (0-100)	Tract-level measure BROADCAST to block ...

PLACES_MAMMOUSE	Mammography use, women 50–74 (crude prev., %)	Percent (0-100)	Tract-level measure BROADCAST to block ...
PLACES_MHLTH	Frequent mental distress (crude prev., %)	Percent (0-100)	Tract-level measure BROADCAST to block ...
PLACES_MOBILITY	Mobility disability (crude prev., %)	Percent (0-100)	Tract-level measure BROADCAST to block ...
PLACES_OBESITY	Obesity prevalence (crude prev., %)	Percent (0-100)	Tract-level measure BROADCAST to block ...
PLACES_PHLTH	Frequent physical distress (crude prev., %)	Percent (0-100)	Tract-level measure BROADCAST to block ...
PLACES_SELF CARE	Self-care disability (crude prev., %)	Percent (0-100)	Tract-level measure BROADCAST to block ...
PLACES_SHUTUTILITY	Utility shutoff threat, past 12 mo (crude prev., %)	Percent (0-100)	Tract-level measure BROADCAST to block ...
PLACES_SLEEP	Short sleep duration (crude prev., %)	Percent (0-100)	Tract-level measure BROADCAST to block ...
PLACES_STROKE	Stroke prevalence (crude prev., %)	Percent (0-100)	Tract-level measure BROADCAST to block ...
PLACES_TEETHLOST	All teeth lost, adults 65+ (crude prev., %)	Percent (0-100)	Tract-level measure BROADCAST to block ...
PLACES_VISION	Vision disability (crude prev., %)	Percent (0-100)	Tract-level measure BROADCAST to block ...
SVI_RPL_THEMES	SVI overall percentile	Continuous measure	Tract-level measure BROADCAST to block ...
SVI_RPL_THEME1	SVI Theme 1: Socioeconomic status	Continuous measure	Tract-level measure BROADCAST to block ...

SVI_RPL_THEME2	SVI Theme 2: Household characteristics	Continuous measure	Tract-level measure BROADCAST to block ...
SVI_RPL_THEME3	SVI Theme 3: Racial & ethnic minority status	Continuous measure	Tract-level measure BROADCAST to block ...
SVI_RPL_THEME4	SVI Theme 4: Housing type & transportation	Continuous measure	Tract-level measure BROADCAST to block ...
HMDA_K7_mean_orig_2019_2024	Mean annual mortgage originations, 2019–2024	Continuous measure	2020 Census block group (native)
SLD_NatWalkInd	National Walkability Index	Index score	2020 Census block group (native)
SLD_D1A	Gross residential density	Density	2020 Census block group (native)
SLD_D1B	Gross population density	Density	2020 Census block group (native)
SLD_D1C	Gross employment density	Density	2020 Census block group (native)
SLD_D2A_JPHH	Jobs per household	Continuous measure	2020 Census block group (native)
SLD_D2B_E8MIXA	8-tier employment mix (entropy)	Continuous measure	2020 Census block group (native)
SLD_D3A	Total road network density	Density	2020 Census block group (native)
SLD_D3B	Pedestrian-oriented intersection density	Density	2020 Census block group (native)
SLD_D3BMM4	Multi-modal intersection density	Density	2020 Census block group (native)
SLD_D4A	Distance to nearest transit stop	Continuous measure	2020 Census block group (native)
SLD_D4C	Aggregate transit service frequency	Continuous measure	2020 Census block group (native)
SLD_D4D	Transit frequency per sq mi	Continuous measure	2020 Census block group (native)
SLD_D5AR	Jobs within 45-min auto commute	Continuous measure	2020 Census block group (native)
SLD_D5BR	Jobs within 45-min transit commute	Continuous measure	2020 Census block group (native)
SLD_D5CR	Regional centrality (auto)	Continuous measure	2020 Census block group (native)
SLD_D5CRI	Regional centrality index	Index score	2020 Census block group (native)
SLD_TotPop	Total population (SLD base)	Continuous measure	2020 Census block group (native)
SLD_CountHU	Housing units (SLD base)	Continuous measure	2020 Census block group (native)

SLD_TotEmp	Total employment (SLD base)	Continuous measure	2020 Census block group (native)
SLD_Ac_Land	Land area (acres)	Continuous measure	2020 Census block group (native)
SLD_Ac_Unpr	Unprotected land area (acres)	Continuous measure	2020 Census block group (native)
FARA_LILATracts_1And10	Low-income & low-access flag (1 mi/10 mi)	Continuous measure	2020 Census block group (native)
FARA_LILATracts_halfAnd10	Low-income & low-access flag (1/2 mi/10 mi)	Continuous measure	2020 Census block group (native)
FARA_LowIncomeTracts	Low-income tract flag	Continuous measure	2020 Census block group (native)
FARA_lapophalfshare	Share of pop low access at 1/2 mi	Percent (0-100)	2020 Census block group (native)
FARA_lapop1share	Share of pop low access at 1 mi	Percent (0-100)	2020 Census block group (native)
FARA_lalowihalfshare	Low-income pop low access at 1/2 mi	Continuous measure	2020 Census block group (native)
FARA_lahunvhalfshare	HU without vehicle, low access at 1/2 mi	Continuous measure	2020 Census block group (native)
FARA_PovertyRate	Tract poverty rate (FARA)	Percent (0-100)	2020 Census block group (native)
N1_adi_natrank	Area Deprivation Index; national rank (v4.0.1, 2024)	Index score	2020 Census block group (native)
N1_adi_staternk	Area Deprivation Index; state rank	Index score	2020 Census block group (native)
N1b_adistd_ntl	Standardized ADI; national (income-standardized, 2022)	Continuous measure	2020 Census block group (native)
N1b_adistd_state	Standardized ADI; state	Continuous measure	2020 Census block group (native)
N2_sdi	Social Deprivation Index (2012–2019)	Index score	2020 Census block group (native)
N4_dci_distress	Distressed Communities Index score (2019–2023)	Index score	ZIP-level measure allocated to block gr...
N7_upjohn_distress_ord	Upjohn distressed-communities ordinal (1 best – 5 worst)	Continuous measure	2020 Census block group (native)
N7_upjohn_pae_rate	Upjohn prime-age employment rate	Rate	2020 Census block group (native)
G1_life_expectancy	Life expectancy at birth (USALEEP, NC File A)	Years	2020 Census block group (native)
I1_uncg_violent_rate	Violent crime per 1,000 residents (3-agency, 2014–2018)	Continuous measure	2020 Census block group (native)
I2_uncg_property_rate	Property crime per 1,000 residents (2014–2018)	Continuous measure	2020 Census block group (native)
I3_uncg_cfs_rate	Police calls-for-service per 1,000 residents (2014–2018)	Continuous measure	2020 Census block group (native)
I4_uncg_weapons_rate	Weapons calls per 1,000 residents (2014–2018)	Continuous measure	2020 Census block group (native)

I5_uncg_allcrime_rate	All crime per 1,000 residents (2014–2018)	Continuous measure	2020 Census block group (native)
I6a_ags_total_idx	AGS CrimeRisk total index (MODELED)	Index score	2020 Census block group (native)
I6b_ags_personal_idx	AGS CrimeRisk personal-crime index (MODELED)	Index score	2020 Census block group (native)
I6c_ags_property_idx	AGS CrimeRisk property-crime index (MODELED)	Index score	2020 Census block group (native)
J5_econ_connectedness	Economic connectedness	Continuous measure	ZIP-level measure allocated to block gr...
J6a_clustering	Network clustering coefficient	Continuous measure	ZIP-level measure allocated to block gr...
J6b_support_ratio	Support ratio	Continuous measure	ZIP-level measure allocated to block gr...
J7a_volunteering	Volunteering rate	Rate	ZIP-level measure allocated to block gr...
J7b_civic_orgs	Civic organizations	Continuous measure	ZIP-level measure allocated to block gr...
J8_self_response	2020 Census self-response rate	Rate	Tract-level measure BROADCAST to block ...
J8b_internet_response	2020 Census internet self-response rate	Rate	Tract-level measure BROADCAST to block ...
I7_oa_incarceration	Incarceration rate, children of p25 parents	Rate	Tract-level measure BROADCAST to block ...
J1_nonprofit_per10k	Nonprofits per 10,000 residents	Rate per 10,000	2020 Census block group (native)
J1a_arts_per10k	Arts/culture nonprofits per 10K (NTEE A)	Rate per 10,000	2020 Census block group (native)
J1b_humansvc_per10k	Human-services nonprofits per 10K (NTEE P)	Rate per 10,000	2020 Census block group (native)
J1c_youth_per10k	Youth-development nonprofits per 10K (NTEE O)	Rate per 10,000	2020 Census block group (native)
J1d_civic_per10k	Civic/advocacy nonprofits per 10K	Rate per 10,000	2020 Census block group (native)
J1e_np_expense_pc	Nonprofit total functional expenses per capita (\$)	Dollars, thousands	2020 Census block group (native)
J2_voter_reg_rate	Voter registration rate	Rate	2020 Census block group (native)
E10_filing_rate	Eviction filing rate (mean of 3 most recent valid years)	Percent (0-100)	2020 Census block group (native)

E10_eviction_rate	Eviction (judgment) rate, same window	Rate	2020 Census block group (native)
E10_filings_3yr	Eviction filings, 3-yr sum	Continuous measure	2020 Census block group (native)
E10_last_year	Most recent valid data year used	Continuous measure	2020 Census block group (native)
E10_quality_flag	Quality flag (low/imputed/subbed in used years)	Years	2020 Census block group (native)
J10_library_dist_mi	Distance to nearest public library outlet (mi)	Continuous measure	2020 Census block group (native)
J10_library_halfmi	Library within 1/2 mile of BG internal point (0/1)	Continuous measure	2020 Census block group (native)
K14_hub_cert_per10k	HUB-certified businesses per 10,000 residents	Rate per 10,000	2020 Census block group (native)
K14a_hub_black_per10k	Black-owner HUB-certified businesses per 10K (n=338 countywide)	Count	2020 Census block group (native)
K14b_hub_female_per10k	Female-owner HUB-certified businesses per 10K (n=98 countywide)	Count	2020 Census block group (native)
K15_sbe_cert_per10k	NC SBE-certified small businesses per 10,000 residents	Rate per 10,000	2020 Census block group (native)
K8_sba_loans_per10k	SBA 7(a)+504 loans per 10K residents, FY2023–2025 (interim denominato...	Rate per 10,000	2020 Census block group (native)
K8b_sba_amount_pc	SBA gross approval dollars per capita, FY2023–2025 (\$196.2M countywid...	Dollars, thousands	2020 Census block group (native)
E2_pct_absentee_owner	Absentee-owner share of residential parcels	Percent (0-100)	2020 Census block group (native)
E4_median_home_value	Median assessed value, residential parcels	Dollars / median value	2020 Census block group (native)
E5_median_year_built	Median year built, residential parcels	Dollars / median value	2020 Census block group (native)
E7_pct_vacant_land	Vacant-land share of all parcels	Percent (0-100)	2020 Census block group (native)
E8_pct_recent_turnover_3yr	Residential parcels sold in last 3 yr, share	Percent (0-100)	2020 Census block group (native)
E_parcel_count_total	Total parcels per BG (denominator)	Continuous measure	2020 Census block group (native)
E_parcel_count_residential	Residential parcels per BG (denominator)	Continuous measure	2020 Census block group (native)
E6_foreclosure_count	Foreclosure-flagged parcels, count	Count	2020 Census block group (native)
E6_foreclosure_rate_per1k_res	Foreclosure rate per 1,000 residential parcels	Rate	2020 Census block group (native)
E1_code_violations_3yr	Code-enforcement violations, 3-yr count	Count	2020 Census block group (native)
E9_building_permits_12mo	Building permits issued, 12-mo count	Count	2020 Census block group (native)
H2_supermarket_count	Supermarket / grocery / superstore count	Count	2020 Census block group (native)

H3_convenience_count	Convenience-store count	Count	2020 Census block group (native)
H4_snap_retailer_count	All SNAP-authorized retailers, count	Count	2020 Census block group (native)
M3_TotalBSL	Broadband Serviceable Locations, count (denominator)	Count	2020 Census block group (native)
M3_pct_served_100_20	Locations served at $\geq 100/20$ Mbps, share	Percent (0-100)	2020 Census block group (native)
M3_pct_unserved	Unserved locations, share	Percent (0-100)	2020 Census block group (native)
M3_pct_underserved	Underserved (25-100 Mbps) locations, share	Percent (0-100)	2020 Census block group (native)
M3_pct_fiber	Locations with fiber reported, share	Percent (0-100)	2020 Census block group (native)
M3_pct_cable	Locations with cable reported, share	Percent (0-100)	2020 Census block group (native)
K13_bus_vac_rate	Business-address vacancy rate (%)	Percent (0-100)	Tract-level measure BROADCAST to block ...
K13b_bus_vac_longterm_share	Share of vacant business addresses vacant 12+ months (%)	Percent (0-100)	Tract-level measure BROADCAST to block ...
K13_ams_bus_tract	Active business addresses in tract (denominator context)	Continuous measure	Tract-level measure BROADCAST to block ...
K2_smallfirm_emp_share	Small-firm employment share (% , firms <50 employees)	Percent (0-100)	2020 Census block group (native)
K5_youngfirm_emp_share	Young-firm employment share (% , firm age 0-3 yrs)	Percent (0-100)	2020 Census block group (native)
K2_wac_private_emp	Private workplace employment, count	Count	2020 Census block group (native)
K2_wac_total_emp	Total workplace employment, count	Count	2020 Census block group (native)
L2_retail_emp_cns07	Retail-trade workplace employment (NAICS 44-45), count	Count	2020 Census block group (native)
K1_places_count	Business/POI establishments, count	Count	2020 Census block group (native)
K1b_independent_share	Independent (non-branded) establishment share (%)	Percent (0-100)	2020 Census block group (native)
J4_religious_count	Religious organizations, count	Count	2020 Census block group (native)
H2_ovt_grocery_count	Grocery/supermarket establishments (Overture), count	Count	2020 Census block group (native)
H3_ovt_convenience_count	Convenience-store establishments (Overture), count	Count	2020 Census block group (native)
J2_reg_voters_count	Registered voters (all statuses), count	Count	2020 Census block group (native)

J2b_active_reg_count	Registered voters (ACTIVE status), count	Count	2020 Census block group (native)
J3_voted_2024_count	Registrants who voted, 11/05/2024 general, count	Count	2020 Census block group (native)
J3_turnout_2024_rate	Turnout among registrants, 11/05/2024 general (%)	Percent (0-100)	2020 Census block group (native)
G9_ejscreen_pollution_burden	EJScreen pollution burden (mean of 6 national percentiles)	Continuous measure	2020 Census block group (native)
G10_lead_paint_pre1960_pct	Lead paint risk: pre-1960 housing share (%)	Percent (0-100)	2020 Census block group (native)
K10_cra_sb_orig_count	CRA small-business loan originations, count (2024)	Count	Tract-level measure BROADCAST to block ...
K10_cra_sb_orig_amt_000s	CRA small-business originations, \$ thousands (2024)	Dollars, thousands	Tract-level measure BROADCAST to block ...
K10b_cra_sb_rev_lt1m_count	CRA originations to businesses with revenues < \$1M, count (2024)	Dollars, thousands	Tract-level measure BROADCAST to block ...
E1_hp_code_violations_3yr	High Point component of E1 (audit column)	Continuous measure	2020 Census block group (native)
G8_lbw_pct_2223	Low birthweight, % of births (2022-23 pooled)	Percent (0-100)	Tract-level measure BROADCAST to block ...
G8b_preterm_pct_2223	Preterm births, % (2022-23 pooled)	Percent (0-100)	2020 Census block group (native)
G8_births_2223	Births, pooled 2022-23 count	Count	2020 Census block group (native)
G7_life_expectancy_1620	Life expectancy at birth, 2016-2020 (yrs)	Years	Tract-level measure BROADCAST to block ...
G7b_le_change_0913_1620	Life-expectancy change, 2009-13 -> 2016-20 (yrs)	Years	2020 Census block group (native)

APPENDIX C: CORRELATION SCREEN RESULTS

The 57 candidate measures significantly associated with all four criterion variables at $p < .05$ (Pearson, pairwise-complete $n = 337$ to 338), from 208 candidates and 832 tests. Sorted by mean absolute correlation with the three aligned criteria. Y3 coefficients are reported for completeness; see Appendix J. Complete file: GCNVI_PhaseB_Correlation_Screen.xlsx (SPSS replication: exact agreement).

Variable	Measure	r Y1	p Y1	r Y2	p Y2	r Y3	p Y3	r Y4	p Y4
PLACES_EMOTIONSPT	Lacking social/emotional support (crude prev)	-0.819	<.001	-0.7	<.001	0.176	.001	-0.938	<.001
PLACES_SLEEP	Short sleep duration (crude prev., %)	-0.815	<.001	-0.706	<.001	0.183	<.001	-0.916	<.001
C5_est	ICE (Race-Income)	0.753	<.001	0.618	<.001	-0.294	<.001	0.906	<.001
PLACES_HOUSINSECU	Housing insecurity, past 12 mo (crude prev., %)	-0.803	<.001	-0.668	<.001	0.144	.008	-0.904	<.001
PLACES_CASTHMA	Current asthma (crude prev., %)	-0.797	<.001	-0.669	<.001	0.173	.001	-0.859	<.001
PLACES_OBESITY	Obesity prevalence (crude prev., %)	-0.781	<.001	-0.68	<.001	0.141	.010	-0.886	<.001
PLACES_FOODINSECU	Food insecurity, past 12 mo (crude prev., %)	-0.794	<.001	-0.655	<.001	0.119	.029	-0.893	<.001
SVI_RPL_THEME3	SVI Theme 3: Racial & ethnic minority status	-0.7	<.001	-0.604	<.001	0.312	<.001	-0.843	<.001
SVI_RPL_THEMES	SVI overall percentile	-0.711	<.001	-0.651	<.001	0.17	.002	-0.906	<.001
PLACES_SHUTUTILITY	Utility shutoff threat, past 12 mo (crude prev....	-0.782	<.001	-0.642	<.001	0.136	.012	-0.869	<.001
PLACES_LACKTRPT	Lacked reliable transportation (crude prev., %)	-0.785	<.001	-0.639	<.001	0.128	.018	-0.867	<.001
PLACES_LONELINESS	Feelings of loneliness (crude prev., %)	-0.753	<.001	-0.628	<.001	0.195	<.001	-0.84	<.001
SVI_RPL_THEME1	SVI Theme 1: Socioeconomic status	-0.716	<.001	-0.642	<.001	0.151	.005	-0.893	<.001
PLACES_FOODSTAMP	Received food stamps, past 12 mo (crude prev., ...	-0.778	<.001	-0.635	<.001	0.11	.043	-0.866	<.001
N2_sdi	Social Deprivation Index (2012–2019)	-0.736	<.001	-0.603	<.001	0.155	.004	-0.872	<.001
PLACES_MHLTH	Frequent mental distress (crude prev., %)	-0.753	<.001	-0.631	<.001	0.13	.017	-0.818	<.001
J8_self_response	2020 Census self-response rate	0.722	<.001	0.566	<.001	-0.166	.002	0.819	<.001

E4_median_home_value	Median assessed value, residential parcels	0.634	<.001	0.58	<.001	-0.253	<.001	0.793	<.001
N1b_adistd_ntl	Standardized ADI; national (income-standardized...	-0.675	<.001	-0.599	<.001	0.141	.011	-0.843	<.001
N1_adi_natrank	Area Deprivation Index; national rank (v4.0.1, ...	-0.627	<.001	-0.557	<.001	0.212	<.001	-0.845	<.001
N1b_adistd_state	Standardized ADI; state	-0.673	<.001	-0.592	<.001	0.132	.018	-0.838	<.001
N1_adi_staternk	Area Deprivation Index; state rank	-0.627	<.001	-0.555	<.001	0.182	.001	-0.839	<.001
I6b_ags_personal_idx	AGS CrimeRisk personal-crime index (MODELED)	-0.717	<.001	-0.574	<.001	0.165	.002	-0.723	<.001
PLACES_CANCER	Cancer (non-skin) prevalence (crude prev., %)	0.685	<.001	0.572	<.001	-0.214	<.001	0.701	<.001
C1_est	% Black or African American	-0.621	<.001	-0.524	<.001	0.268	<.001	-0.68	<.001
J3_turnout_2024_rate	Turnout among registrants, 11/05/2024 general (...)	0.666	<.001	0.526	<.001	-0.17	.002	0.73	<.001
D5_est	Median home value	0.531	<.001	0.498	<.001	-0.29	<.001	0.745	<.001
SVI_RPL_THEME4	SVI Theme 4: Housing type & transportation	-0.564	<.001	-0.511	<.001	0.212	<.001	-0.738	<.001
A1_est	Median household income	0.52	<.001	0.461	<.001	-0.317	<.001	0.724	<.001
A2_est	Median HH income; county ratio	0.52	<.001	0.461	<.001	-0.317	<.001	0.724	<.001
FARA_LILATracts_halfAnd10	Low-income & low-access flag (1/2 mi/10 mi)	-0.601	<.001	-0.487	<.001	0.139	.011	-0.73	<.001
FARA_LowIncomeTracts	Low-income tract flag	-0.601	<.001	-0.487	<.001	0.139	.011	-0.73	<.001
PLACES_CHOLSCREEN	Cholesterol screening (crude prev., %)	0.635	<.001	0.51	<.001	-0.108	.048	0.685	<.001
PLACES_BINGE	Binge drinking (crude prev., %)	0.559	<.001	0.484	<.001	-0.127	.020	0.718	<.001
E10_eviction_rate	Eviction (judgment) rate, same window	-0.582	<.001	-0.506	<.001	0.127	.020	-0.583	<.001
I6a_ags_total_idx	AGS CrimeRisk total index (MODELED)	-0.562	<.001	-0.44	<.001	0.139	.011	-0.619	<.001
G8_lbwt_pct_2223	Low birthweight, % of births (2022-23 pooled)	-0.485	<.001	-0.401	<.001	0.165	.003	-0.54	<.001
I6c_ags_property_idx	AGS CrimeRisk property-crime index (MODELED)	-0.492	<.001	-0.381	<.001	0.125	.022	-0.562	<.001

D1_est	Homeownership rate	0.437	<.001	0.341	<.001	- 0.221	<.001	0.549	<.001
D4_est	Renter-occupied share	-0.437	<.001	-0.341	<.001	0.221	<.001	-0.549	<.001
E10_filing_rate	Eviction filing rate (mean of 3 most recent val...	-0.462	<.001	-0.404	<.001	0.181	<.001	-0.483	<.001
A12_est	Households with no vehicle	-0.437	<.001	-0.322	<.001	0.121	.026	-0.48	<.001
A11_est	Single-parent HH with children	-0.377	<.001	-0.336	<.001	0.125	.024	-0.496	<.001
K10b_cra_sb_rev_lt1m_count	CRA originations to businesses with revenues < ...	0.356	<.001	0.25	<.001	- 0.188	<.001	0.476	<.001
E10_filings_3yr	Eviction filings, 3-yr sum	-0.308	<.001	-0.28	<.001	0.271	<.001	-0.404	<.001
HMDA_K7_mean_orig_2019_20...	Mean annual mortgage originations, 2019–2024	0.325	<.001	0.198	<.001	- 0.151	.005	0.534	<.001
B6_est	Preschool enrollment (age 3–4)	0.248	<.001	0.24	<.001	- 0.276	<.001	0.429	<.001
G8_births_2223	Births, pooled 2022-23 count	-0.289	<.001	-0.371	<.001	0.174	.001	-0.319	<.001
E2_pct_absentee_owner	Absentee-owner share of residential parcels	-0.316	<.001	-0.242	<.001	0.164	.002	-0.394	<.001
K10_cra_sb_orig_count	CRA small-business loan originations, count (20...	0.325	<.001	0.234	<.001	- 0.138	.011	0.41	<.001
K13_bus_vac_rate	Business-address vacancy rate (%)	-0.246	<.001	-0.178	.001	0.145	.008	-0.306	<.001
A9_est	Unemployment rate (16+)	-0.258	<.001	-0.198	<.001	0.111	.043	-0.257	<.001
C7_density_per_sqmi	Population density (per sq mi)	-0.221	<.001	-0.145	.008	0.186	<.001	-0.225	<.001
PLACES_HIGHCHOL	High cholesterol (crude prev., %)	0.234	<.001	0.173	.001	- 0.196	<.001	0.173	.001
G10_lead_paint_pre1960_pct	Lead paint risk: pre-1960 housing share (%)	-0.24	<.001	-0.158	.004	- 0.124	.022	-0.187	<.001
D15_est	SF-detached / MF ratio	0.143	.027	0.13	.044	- 0.205	.001	0.227	<.001
SLD_CountHU	Housing units (SLD base)	0.182	<.001	0.112	.040	0.147	.007	0.223	<.001

APPENDIX D: HIERARCHICAL OLS AND COLLINEARITY DIAGNOSTICS

Phase B joint models on the 50-predictor design (survivors less missingness and alias exclusions; listwise $n = 312$). D.1 reports increments to R-squared by domain block and criterion; D.2 reports each predictor's variance-inflation factor and standardized coefficients in the full models. Complete file: GCNVI_PhaseB_Hierarchical_OLS.xlsx.

D.1 Increments to R-squared by domain block.

Domain block (entry order)	Y1	Y2	Y3	Y4
A. Household Economic Position	0.237	0.154	0.034	0.351
C. Demographic Composition	0.338	0.239	0.080	0.493
D. Housing Conditions	0.001	0.001	0.001	0.000
E. Housing/Parcel Administrative Records	0.054	0.068	0.064	0.029
F. Built Environment & Accessibility	0.005	0.006	0.028	0.000
G. Health & Environment	0.150	0.197	0.162	0.085
H. Food Environment	0.004	0.006	0.000	0.000
I. Crime & Safety	0.027	0.020	0.019	0.001
J. Civic & Social Capital	0.004	0.007	0.003	0.001
K. Economic Activity	0.010	0.004	0.017	0.001
N. External Composites (Triangulation)	0.012	0.021	0.044	0.004
Full model R-squared	.842	.721	.452	.966
Full model adjusted R-squared	.812	.668	.347	.959

D.2 Full-model VIF and standardized coefficients. Exact linear aliases excluded before estimation: the low-income-only food-access flag (alias of the LILA designation) and the renter share (complement of homeownership). Five candidates excluded for margin-of-error missingness are listed beneath the table.

Variable	Dom	VIF	β Y1	p	β Y2	p	β Y3	p	β Y4	p
A12_est	A	2.1	0.021	.554	0.029	.529	0.033	0.617	0.013	.419
A11_est	A	2.0	0.037	.281	0.007	.885	0.0	0.999	-0.013	.407
A9_est	A	1.3	-0.022	.430	-0.005	.898	0.027	0.6	-0.008	.528
C5_est	C	23.8	-0.079	.510	-0.504	.002	-0.635	0.005	0.162	.004
C1_est	C	5.3	0.018	.747	-0.024	.751	-0.052	0.624	-0.012	.657
C7_density_per_sqmi	C	2.2	0.012	.752	0.009	.861	0.032	0.638	0.018	.299
D1_est	D	5.1	0.075	.177	0.145	.051	-0.037	0.721	0.018	.477
E4_median_home_value	E	5.9	0.152	.011	0.323	<.001	-0.146	0.191	0.102	<.001
E10_eviction_rate	E	8.7	-0.1	.169	-0.149	.124	-0.164	0.227	-0.036	.294
E10_filing_rate	E	8.1	-0.074	.293	-0.001	.994	0.154	0.24	0.028	.386
E10_filings_3yr	E	5.0	0.183	<.001	0.117	.108	0.013	0.902	-0.037	.151
E2_pct_absentee_owner	E	1.7	0.04	.213	0.075	.083	-0.065	0.281	-0.009	.567
SLD_CountHU	F	2.9	-0.115	.006	-0.071	.205	0.14	0.075	0.023	.241
PLACES_EMOTIONSPT	G	1132.8	-4.514	<.001	-3.356	.003	-4.242	0.006	-0.641	.098
PLACES_SLEEP	G	124.1	0.726	.009	0.469	.198	-0.162	0.75	0.116	.365
PLACES_HOUSINSECU	G	7049.2	-0.99	.632	4.928	.074	1.369	0.722	-3.349	<.001
PLACES_CASTHMA	G	255.9	-0.934	.018	0.406	.438	-0.886	0.228	0.865	<.001
PLACES_OBESITY	G	201.1	-1.43	<.001	-2.233	<.001	-0.005	0.994	-0.594	<.001
PLACES_FOODINSECU	G	21789.1	7.052	.053	-2.841	.556	0.464	0.945	7.24	<.001
PLACES_SHUTUTILITY	G	5117.1	2.49	.159	-0.083	.972	-0.373	0.909	2.851	<.001
PLACES_LACKTRPT	G	2613.5	-0.415	.742	-0.675	.687	-0.689	0.769	-1.737	.003
PLACES_LONELINESS	G	1189.1	3.502	<.001	2.536	.025	3.825	0.016	-0.18	.649
PLACES_FOODSTAMP	G	9509.0	-5.351	.027	0.711	.824	0.537	0.904	-4.22	<.001
PLACES_MHLTH	G	597.6	1.941	.001	-0.15	.851	-0.148	0.895	-0.53	.060
PLACES_CANCER	G	159.2	-0.408	.190	-0.664	.108	-0.767	0.186	0.099	.494
PLACES_CHOLSCREEN	G	602.0	3.026	<.001	0.744	.354	1.345	0.233	0.207	.463
PLACES_BINGE	G	78.7	0.14	.521	-0.011	.969	-0.604	0.138	0.278	.007
G8_lbw_pct_2223	G	2.4	-0.088	.023	-0.063	.219	-0.062	0.384	-0.002	.892
G8_births_2223	G	3.6	0.067	.155	-0.126	.044	0.095	0.276	-0.074	<.001
PLACES_HIGHCHOL	G	126.7	0.611	.028	1.097	.003	0.225	0.662	-0.297	.022
G10_lead_paint_pre1960_pct	G	2.7	0.008	.851	-0.04	.449	-0.007	0.928	-0.007	.709
FARA_LowIncomeTracts	H	5.5	0.182	.002	0.151	.050	0.151	0.163	0.045	.095
I6b_ags_personal_idx	I	1855.0	0.133	.901	0.385	.784	-1.119	0.571	-1.203	.016
I6a_ags_total_idx	I	32325.6	-2.757	.534	-3.645	.535	5.709	0.489	5.535	.008
I6c_ags_property_idx	I	20252.7	2.383	.497	3.033	.515	-4.969	0.447	-4.452	.007
J8_self_response	J	13.4	0.095	.293	-0.243	.043	-0.143	0.394	0.117	.006
J3_turnout_2024_rate	J	10.2	0.058	.464	-0.107	.306	0.152	0.302	-0.024	.512

K10b_cra_sb_rev_lt1m_count	K	19.4	-0.069	.525	-0.016	.913	-0.399	0.049	0.015	.767
HMDA_K7_mean_orig_2019_20...	K	5.9	-0.23	<.001	-0.151	.058	0.144	0.196	-0.045	.108
K10_cra_sb_orig_count	K	18.1	0.04	.704	0.051	.713	0.405	0.039	0.026	.596
K13_bus_vac_rate	K	2.1	0.034	.343	0.041	.394	0.072	0.287	-0.004	.799
SVI_RPL_THEME3	N	18.0	0.162	.121	0.093	.504	0.399	0.041	0.079	.106
SVI_RPL_THEMES	N	52.5	-0.23	.198	-0.187	.430	-0.509	0.126	-0.062	.453
SVI_RPL_THEME1	N	35.2	0.086	.558	-0.309	.112	0.504	0.065	-0.145	.034
N2_sdi	N	15.0	-0.283	.003	-0.118	.352	-0.434	0.015	-0.016	.727
N1b_adistd_ntl	N	95.0	0.135	.575	0.257	.420	0.0	1.0	0.152	.175
N1_adi_natrank	N	97.5	0.118	.628	0.168	.602	1.139	0.012	-0.004	.975
N1b_adistd_state	N	85.6	-0.018	.938	-0.123	.683	0.002	0.997	-0.105	.323
N1_adi_staternk	N	86.5	-0.028	.903	-0.035	.908	-1.02	0.017	-0.026	.806
SVI_RPL_THEME4	N	9.9	0.039	.619	-0.033	.745	0.013	0.927	-0.023	.523

Missingness exclusions (screen survivors not entered): median household income and its county ratio (27 suppressed block groups), median-value ACS duplicate (31), preschool enrollment (17), plus the two aliases above.

APPENDIX E: COMPOSITE MODEL DOCUMENTATION

The complete v1.0 model record. Complete file: GCNVI_v1_Model_Documentation.xlsx.

E.1 Model summary.

Element	Specification
Construction	Each component standardized on the county distribution (z), sign-aligned so higher = greater vitality, averaged with eq...
Selection basis	Correlation screen vs Y1/Y2/Y4 (Y3 excluded from selection per criterion finding), hierarchical-OLS unique contribution...
Reliability	Cronbach's alpha = .922; McDonald's omega = .935; all single-factor loadings .62-.91 (n=326 complete).
Collinearity	All component VIFs <= 6.1 (max: social-support measure); pairwise /r/ max .78.
Criterion validity	r with Y1 income mobility = +.762; Y2 top-quintile = +.622; Y4 Child Opportunity = +.873 (all p<.001). Y3 employment re...
Benchmark comparison	Outperforms ADI (/r_Y1/ .68) and SDI (.74) on income mobility with 10 transparent local measures; within .03 of SVI/ICE...
Classification	Five categories at county mean $\pm$ 0.5 / 1.5 SD: Very Low 24 · Low 94 · Moderate 92 · High 105 · Very High 20.
Pilot-tract vali...	All nine Coalition pilot tracts fall in Very Low/Low modal categories (mean scores -0.90 to -1.65; tract 139 lowest)...
Caveats	I6b is a MODELED vendor index cleared for internal analysis; resolve publication rights or substitute the 2014-18 measu...

E.2 Per-domain selection matrix.

Domain	Selection	Score	Decision & rationale
A. Household Economic	A12 no-vehicle households	0.413	SELECTED. A1/A2 (median income) scored higher (.57) but carry 27 MOE-suppressed BGs and were O...
B. Education	NA	-	EXCLUDED: sole qualifier B6 (preschool enrollment) has 17 missing and the weakest score (.31);...
C. Demographic	NA	-	EXCLUDED BY DESIGN. C5/ICE is a quasi-composite (race-income polarization; VIF 10.5, r=.93 wit...
D. Housing Conditions	D1 homeownership rate	0.442	SELECTED. D5 median value duplicates E4 with 31 missing; D4 renter share is D1's exact complem...
E. Parcel Administrat...	E4 median home value + E10 ev...	0.669 /...	TWO SELECTED (largest domain; both complete administrative measures). E4 was the only variable...
F. Built Environment	NA	,	EXCLUDED: sole qualifier (housing-unit count) is conceptually thin as vitality (.17).
G. Health & Environme...	PLACES social-support + G8 lo...	0.819 /	TWO SELECTED. PLACES capped at ONE component by design (modeled; partial Y4 circularity): top ...
H. Food Environment	FARA LILA flag	0.606	SELECTED over its aliased twin (LowIncome alone): LILA is the proper food-desert construct (in...
I. Crime & Safety	I6b personal-crime index	0.671	SELECTED WITH FLAG: modeled vendor index; strongest I-domain qualifier. Publication-rights res...
J. Civic & Social Cap...	J8 census self-response	0.703	SELECTED over J3 turnout (r=.72 between them; J8 stronger and captures broader civic engagemen...
K. Economic Activity	K10b small-firm lending	0.361	SELECTED: capital access to <\$1M-revenue firms is the Newkirknomics-core construct; strongest ...
N. External Composites	NA	-	EXCLUDED BY DESIGN: triangulation domain; including national composites would make an index of...

E.3 The ten components.

Variable	Component	Polarity	Loading	VIF	Coverage /338
A12_est	No-vehicle households (%)	–	0.62	1.53	335
D1_est	Homeownership rate (%)	+	0.74	2.15	335
E4_median_home_value	Median home value (\$)	+	0.78	2.45	338
E10_eviction_rate	Eviction judgment rate	–	0.68	1.72	338
PLACES_EMOTIONSPT	Lacking social/emotional support (%)	–	0.91	6.14	337
G8_lbw_pct_2223	Low birthweight (%; 2022-23)	–	0.65	1.59	328
FARA_LILATracts_halfAnd10	Food desert flag (LILA)	–	0.84	3.17	337
I6b_agr_personal_idx	Personal-crime risk index (modeled)	–	0.87	3.46	338
J8_self_response	Census self-response rate (%)	+	0.91	5.11	337
K10b_cra_sb_rev_lt1m_count	Small-business lending to <\$1M firms	+	0.64	1.54	338

APPENDIX F: SCORED DATASET DESCRIPTION AND ACCESS

The scored dataset is delivered in three synchronized formats: GCNVI_v1_Scored_Dataset.sav (SPSS, with variable labels), .xlsx (branded workbook), and .csv (plain interchange). Each carries one row per block group (n = 338) and thirteen fields: the block-group GEOID and tract identifier; the ten vitality-aligned component z-scores (prefixed z_, county-standardized, signed so that higher always indicates greater vitality); the composite score (GCNVI_score, the equal-weighted mean of available components, computed where at least nine of ten are present, 335 of 338); and the five-category classification (GCNVI_category, cut at the county mean plus or minus 0.5 and 1.5 standard deviations). Three block groups are unscored and carried as empty: the zero-population airport block group and two data-sparse fragments. Suppressed component cells (chiefly birth outcomes under the twenty-birth floor) remain empty in every format; users computing derived statistics should apply the analysis notes of Appendix B. The dataset is county-relative by construction and supports no cross-county comparison. Redistribution accompanies the platform governance of Section 8.3; the modeled safety component's values are cleared for the analyses of this report and remain subject to the licensing condition stated there before public platform republication.

APPENDIX G: MAP SUITE INDEX

The 33-map suite accompanying the index: the composite and the ten components, each in county, Greensboro pilot-area, and High Point pilot-area views, red-to-green with green as the vitality direction, pilot tracts outlined. Delivered as GCNVI_v1_Map_Suite.zip; the companion Atlas volume carries the full 213-plate variable series.

Subject	County view	Greensboro target	High Point target
GCNVI composite score	county/00_GCNVI_composite.png	greensboro_target/00_GCNVI_composite.png	highpoint_target/00_GCNVI_composite.png
Vehicle access	county/01_A12_est.png	greensboro_target/01_A12_est.png	highpoint_target/01_A12_est.png
Homeownership	county/02_D1_est.png	greensboro_target/02_D1_est.png	highpoint_target/02_D1_est.png
Median home value	county/03_E4_median_home_value.png	greensboro_target/03_E4_median_home_value.png	highpoint_target/03_E4_median_home_value.png
Eviction judgments	county/04_E10_eviction_rate.png	greensboro_target/04_E10_eviction_rate.png	highpoint_target/04_E10_eviction_rate.png
Social and emotional support	county/05_PLACES_EMOTIONSPT.png	greensboro_target/05_PLACES_EMOTIONSPT.png	highpoint_target/05_PLACES_EMOTIONSPT.png
Birth outcomes (LBW)	county/06_G8_lbw_pct_2223.png	greensboro_target/06_G8_lbw_pct_2223.png	highpoint_target/06_G8_lbw_pct_2223.png
Food access (LILA flag)	county/07_FARA_LILATracts_halfAnd10.png	greensboro_target/07_FARA_LILATracts_halfAnd10.png	highpoint_target/07_FARA_LILATracts_halfAnd10.png
Personal-crime risk (modeled)	county/08_I6b_ags_personal_idx.png	greensboro_target/08_I6b_ags_personal_idx.png	highpoint_target/08_I6b_ags_personal_idx.png
Census self-response	county/09_J8_self_response.png	greensboro_target/09_J8_self_response.png	highpoint_target/09_J8_self_response.png
Small-business lending	county/10_K10b_cra_sb_rev_lt1m_count.png	greensboro_target/10_K10b_cra_sb_rev_lt1m_count.png	highpoint_target/10_K10b_cra_sb_rev_lt1m_count.png

APPENDIX H: DATA ACQUISITION AND PROVENANCE RECORD

The complete acquisition record is the GCNVI Data-Pull Specification (Edition 6, v10.6), incorporated here by reference; it documents, for every source, the retrieval route, endpoint or delivery, join method, verification checks, and rebuild instructions sufficient for independent replication of the entire master file. This appendix records the project's standing decision log: the material determinations, with their dates and grounds, that shaped the file.

Decision	Date	Determination
Criterion treatment of Y3	Jul 2026	Employment criterion near-orthogonal to Y1/Y2/Y4; excluded from selection and validation targets; reported separately (Appendix J).
Demographic composition	Jul 2026	Excluded from all scoring under the demographic-neutrality safeguard; retained as context and benchmarks (Section 3.6).
National composites (Domain N)	Jul 2026	Excluded from components categorically; retained as validation benchmarks.
ICE polarization measure	Aug 2026	Strongest single screener; assigned to benchmark set as quasi-compositional (Sections 3.6, 6.3).
I6 modeled crime indices	Jul 2026	Cleared for internal analysis and this report; vendor license resolution or substitution with the measured 2014-2018 series required before public platform release.
PLACES modeled estimates	Aug 2026	Capped at one component in the composite; social-support measure selected.
Life expectancy series	Aug 2026	County health department 2016-2020 period life tables adopted; USALEEP file and city-limits data-use-agreement plan superseded; eight source-suppressed parent tracts honored as suppressed (15 of 126 tracts empty).
Birth outcomes floor	Aug 2026	Twenty-birth reliability floor; 2022-2023 pooled; suppressed cells carried as suppressed.
E1 code enforcement coverage	Aug 2026	Two-city jurisdictional coverage (Greensboro + High Point deliveries merged at v10.5); zeros outside jurisdictions flagged as coverage artifacts.
Eviction measure window	Jul 2026	Three-year judgment window per hundred renter households; court records geocoded to 2020 frame.
Business vacancy (K13)	Jul 2026	HUD/USPS Q1 2026 snapshot at tract level; retained for mapping, not composite.
Multidimensional Deprivation Index (N5)	Jul 2026	Published at county scale and above only; resolved unavailable at index geography; not approximated.
NC Secretary of State registry	Standing	In agreement negotiation; first item of the v2 measurement agenda; establishment measures rest on Overture pending delivery.
Legacy voter columns	Aug 2026	Empty legacy registration-rate column retired at v10.6 cleanup; geocoded registrant build (95.0 percent match) is the measure of record.
2010-vintage sources	Jul 2026	Population-weighted tract crosswalk with minimum-weight rule (at least half of 2020 weight covered); gaps traced to source suppression, never imputed.

APPENDIX I: VERIFICATION REGIME SUMMARY

Every pipeline build concluded with named pass or fail assertions; outputs were withheld on any failure. The table summarizes the regime by build; full check-by-check results are in the versioned run logs delivered with the pipeline.

Build	Content added	Verification highlights	Named exceptions
v9	Consolidated master through certified-business and SBA layers	Frame integrity (338/126 exact); bounds and rate checks; geocode match floors	Airport block group (zero population) registered
v10	Criterion variables; ACS backbone rebuild; federal layers	Exact tract-set matches both directions; criterion coverage floors; crosswalk minimum-weight rule	2010-vintage sources carried with weight rule
v10.1	USPS vacancy; catalog hygiene	Tract-broadcast flags; vacancy bounds	Tract-level broadcast documented
v10.2	Voter registration and turnout (geocoded build)	210,862 addresses, 95.0 percent match; voters never exceed registrants; turnout bounds	Unmatched addresses excluded, logged
v10.3	EJScreen 2024 environmental layer	338/338 join; percentile bounds; preservation-copy checksum	Manual acquisition registered (S3 block)
v10.4	FFIEC CRA 2024 small-business lending	County-total reconciliation 99.7 percent; small-firm counts never exceed totals	37 unknown-tract loans registered, not allocated
v10.5	High Point code enforcement merged (two-city E1)	98.3 percent placed (State Plane reprojection + point-in-polygon; Census geocoder fallback); window harmonized	Boundary and unplaceable cases excluded, logged; jurisdiction coverage flagged
v10.6	County vital records: birth outcomes and life expectancy	Twenty-birth floor; rate bounds; crosswalk weight rule; suppression tracing (all 15 LE gaps to 8 suppressed parents)	Suppressed cells carried as suppressed

APPENDIX J: METHODS NOTE ON THE EMPLOYMENT CRITERION (Y3)

The employment criterion required, and received, separate treatment, and this note records the evidence and the decision. Y3 is the Opportunity Atlas measure of the employment rate at age 30 or later among children of 25th-percentile parents raised in each block group. In the county's data it is near-orthogonal to the other three criteria: its correlations are .055 with adult income rank, -.015 with top-quintile probability, and -.171 with the Child Opportunity Index, against .73 to .87 among those three. The pattern extends through the predictor field: of the 57 measures significant on all four criteria, only two carry sign-consistent associations across all four; the remainder reverse direction on Y3, such that block groups with worse measured conditions exhibit modestly higher measured employment among the cohort's low-income children.

The pattern is a property of the benchmark, not an artifact of the county file. The Opportunity Atlas documentation and subsequent literature describe the mechanism: in low-mobility areas, low-income young adults enter the labor force earlier and more completely, while in high-mobility areas the same cohort at the measurement age is disproportionately enrolled in higher education, which suppresses measured employment precisely where later-life income is highest. Y3 thus benchmarks early labor-force attachment, a construct related to, but distinct from and partially opposed to, the long-run opportunity the other three criteria share.

The design consequence, adopted in Section 3.2 and applied throughout: component selection scores and composite validation use Y1, Y2, and Y4; Y3 is estimated and reported in every model for completeness (its full-model R-squared of .096 in Table 6.3 confirming the orthogonality) and is never treated as disconfirming evidence. The composite's negative correlation with Y3 (-.224) is the expected signature of the benchmark's construct, and any future user of the index encountering it should read it through this note rather than as a validity failure.

APPENDIX K: GLOSSARY AND ABBREVIATIONS

Term	Meaning
ACS	American Community Survey (U.S. Census Bureau); five-year estimates 2019-2023 used throughout
ADI	Area Deprivation Index (national benchmark)
Block group (BG)	Census statistical unit of roughly 600 to 3,000 residents; 338 in Guilford County (2020 frame); the index's unit of analysis
Broadcast	Assignment of a tract-level value uniformly to the tract's constituent block groups; flagged in the codebook
COI 3.0	Child Opportunity Index 3.0 (criterion Y4)
Composite score	Equal-weighted mean of the ten vitality-aligned, county-standardized components
CRA	Community Reinvestment Act; source of small-business lending disclosures
Criterion validity	Association of a measure with an independent benchmark of the construct; here, the Opportunity Atlas outcomes and COI 3.0
Crosswalk	Population-weighted correspondence carrying 2010-vintage tract values to 2020 geography
DCI	Distressed Communities Index (benchmark)
EJScreen	EPA environmental justice screening dataset (2024 preservation copy)
FARA / LILA	USDA Food Access Research Atlas; low-income, low-access designation (the food-desert flag)
GCNVI	Guilford County Neighborhood Vitality Index
HMDA	Home Mortgage Disclosure Act data
HOLC	Home Owners' Loan Corporation; author of the 1930s residential security maps
ICE	Index of Concentration at the Extremes (race-income polarization; benchmark)
LODES	LEHD Origin-Destination Employment Statistics
MOE	Margin of error (ACS sampling uncertainty)
OLS	Ordinary least squares regression
PLACES	CDC model-based small-area health estimates
Polarity	The sign convention aligning each component so higher values indicate greater vitality
Pre-registered hypothesis	A prediction fixed before measurement; here, the nine Coalition pilot tracts, used as a proof-of-concept validation geography rather than an eligibility geography
QOZ	Qualified Opportunity Zone (2018 federal designation)
SDI	Social Deprivation Index (benchmark)
SNAP	Supplemental Nutrition Assistance Program

Suppression	A producer's withholding of small-population estimates; honored, never imputed
SVI	CDC/ATSDR Social Vulnerability Index (benchmark)
Tract	Census unit of roughly 1,200 to 8,000 residents; 126 in Guilford County
VIF	Variance-inflation factor; collinearity diagnostic (component ceiling: 10)
Vitality categories	Very Low, Low, Moderate, High, Very High; cut at the county mean plus or minus 0.5 and 1.5 SD
z-score	A value expressed in standard deviations from the county mean